Dell PowerFlex 4.5.x and PowerFlex Manager 5.1.x

Software Install Guide

August 2026
Rev. 1.1

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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1

Introduction

The *Dell PowerFlex 4.5.x and PowerFlex Manager 5.1.x Software Install Guide* describes how to deploy a new instance of Dell PowerFlex 4.5.x and PowerFlex Manager 5.1.x.

Use this guide only for PowerFlex software installations. PowerFlex rack and PowerFlex appliance customers should not use this document for installations. For PowerFlex rack documentation, go to PowerFlex rack technical documentation. For PowerFlex appliance documentation, go to PowerFlex appliance documentation.

The target audience for this document includes Dell Technologies customers and services.

PowerFlex Manager 5.1.x includes the catalog core package 4.5.5 and above.

To deploy PowerFlex Manager 5.1.x for the first time, see detailed instructions on the following topics:

- Selecting deployment type
- Configuring PowerFlex Manager
- Deploying additional storage nodes

This Dell Technologies product uses an embedded Linux distribution that is licensed to Dell. This embedded Linux is only authorized to be used to operate the Dell product and only the Dell product.

Logical Configuration Survey or LCS is now known as Enterprise Project Services (EPS) form. References in the documentation will be updated over time.

SupportAssist is now known as Dell Technologies connectivity services. References in the documentation will be updated over time.

The Dell Code and Compatibility Knowledge Hub was previously known as the Release Certification Matrix (RCM) Portal. References in the documentation will be updated over time.

For additional PowerFlex software documentation, go to Dell PowerFlex Software: Technical Documentation.

Common PowerFlex terms

The following table identifies common PowerFlex terms and associated acronyms.

Table 1. Common PowerFlex terms

Term	Acronym
management virtual machine	MVM
management data store	MDS
metadata manager	MDM
PowerFlex management cluster	PFMC
PowerFlex management platform	PFMP
storage data client	SDC
storage data server	SDS
storage virtual machine	SVM
storage data replicator	SDR
storage data target	SDT
system configuration reporter	SCR

The following table summarizes key terminology changes. References in the documentation will be updated over time.

Table 2. Terminology changes

Term	Formerly known as
PowerFlex management node	PowerFlex controller node
Dell Technologies connectivity services	SupportAssist
Enterprise Project Services (EPS) form	Logical Configuration Survey (LCS)
Dell Code and Compatibility Knowledge Hub	Release Certification Matrix (RCM) Portal

See the PowerFlex glossary for more information about PowerFlex terms and acronyms.

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Revision history

Table 3. Revisions

Date	Document revision	Description of changes
August 2026	1.1	Technical graphic updates
June 2026	1.0	Initial release

1

Install

This section describes how to install and configure PowerFlex Manager, and deploy PowerFlex storage.

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Installation overview and prerequisites

Installation workflow

Installing PowerFlex on physical servers includes the following activities:

1. Complete the installation checklist to prepare your environment for PowerFlex Manager.

   For PowerFlex custom node or VxFlex Ready Node, see the product *Hardware Configuration and Operating System Installation Guide* and *VxFlex Ready Node deployment overview* available on the Dell Technologies support site. These documents provide step-by-step installation instructions on how to prepare the server and operating system before installing PowerFlex.

2. Install and configure PowerFlex Manager.
3. Install the PowerFlex components.
4. For a two-layer system configuration, install storage data clients (SDCs) and/or configure NVMe over TCP hosts on external servers.
5. Upload the PowerFlex license.
6. Perform any necessary post-installation tasks.
7. Move the PowerFlex system to production.

Supported configurations

The following PowerFlex Manager configurations are supported:

Table 4. PowerFlex Manager supported configurations

PowerFlex Manager deployment configuration	Offered on	Description	Minimum number of VMs (or nodes) required	CPU resources required (per node)	Memory resources required (per node)	Disk resources required (per node)
Pre-existing VMware ESXi management infrastructure	• PowerFlex appliance • PowerFlex custom node • VxFlex Ready Node • PowerEdge servers • Non-Dell Technologies hardware	Customers may use their own VMware ESXi hypervisor to host the MVMs, provided that the underlying infrastructure meets the platform's minimum system requirements. PowerFlex management platform is hosted on the three MVMs. Use the installer to set up the PowerFlex management platform. The MVMs and installer VM are deployed using the MVM OVA and installer OVA resepctively. Download the OVA files from the Dell Technologies support site. **NOTE:** The installer can be deployed either on a virtual machine or a Linux bare metal server. It is required	One installer VM or physical node for installer	8 vCPUs/ 8 CPUs	8 GB	100GB
			Three MVMs for installing	14 vCPUs	32GB	600GB

Table 4. PowerFlex Manager supported configurations (continued)

PowerFlex Manager deployment configuration	Offered on	Description	Minimum number of VMs (or nodes) required	CPU resources required (per node)	Memory resources required (per node)	Disk resources required (per node)
		only during the initial setup of the PowerFlex management platform. Once the deployment is complete, the installer VM or the Linux bare metal server configuration can be safely removed.				
Integrated management (non-KVM) on PowerFlex	• PowerFlex custom node • VxFlex Ready Node • PowerEdge servers • Non-Dell Technologies hardware	This is essentially a management controller that integrates with the PowerFlex software components on the same physical servers, instead of requiring a dedicated set of management nodes. **NOTE:** The installer can be deployed either on a virtual machine or a Linux bare metal server. It is required only during the initial setup of the PowerFlex management platform. Once the deployment is complete, the installer VM or the Linux bare metal server configuration can be safely removed. The bare-metal node is repurposed for the MDM role.	Three physical nodes for installing and configuring PowerFlex management platform	14 vCPUs	32GB	600GB
			One physical node or VM for the installer	8 vCPUs / 8 CPUs	8 GB	100GB
Pre-existing Linux bare metal / Linux VM management infrastructure	• PowerFlex custom node • VxFlex Ready Node • PowerEdge servers • Non-Dell Technologies hardware	Customers may provide three bare-metal Linux servers to host the PowerFlex management platform. It is required to install the operating system and configure the network settings on each server. All hardware must comply with the minimum system requirements specified for the platform. **NOTE:** The installer can be deployed either on a virtual machine or a Linux bare metal server. It is required only during the initial setup of the PowerFlex management platform.	Three physical nodes for installing and configuring PowerFlex management platform	14 CPUs	32 GB	600GB
			One VM or physical node for installer	8 vCPUs/ 8 CPUs	8 GB	100GB

Table 4. PowerFlex Manager supported configurations (continued)

PowerFlex Manager deployment configuration	Offered on	Description	Minimum number of VMs (or nodes) required	CPU resources required (per node)	Memory resources required (per node)	Disk resources required (per node)
		Once the deployment is complete, the installer VM or the Linux bare metal server configuration can be safely removed.				
Pre-existing Linux KVM management infrastructure	• PowerFlex custom node • VxFlex Ready Node • PowerEdge servers • Non-Dell Technologies hardware	Customers may use their own Linux KVM hypervisor to host the management virtual machines, provided that the underlying infrastructure meets the platform's minimum system requirements. Download the QCOW files from the Dell Technologies support site. **NOTE:** The installer can be deployed either on a virtual machine or a Linux bare metal server. It is required only during the initial setup of the PowerFlex management platform. Once the deployment is complete, the installer VM or the Linux bare metal server configuration can be safely removed.	Three MVMs for installing and configuring PowerFlex management platform	14 vCPUs	32 GB	600GB
			One VM or physical node for installer	8 vCPUs/ 8 CPUs	8GB	100 GB

Table 5. Minimum disk requirements

Disk type	Interface/standard	Speed (Gbps)	Support status
SATA SSD	SATA III	6	Not supported
SAS SSD	SAS 3.0	12	Supported
SAS SSD	SAS 4.0	24	Supported
NVMe SSD	Data center/Enterprise	-	Recommended

Installation checklist

This section contains a checklist of items required for installing PowerFlex. Ensure that you have done everything on this list before proceeding then perform the remaining preparation tasks covered in this guide.

Table 6. PowerFlex installation checklist

Item	Checked
General	
All servers intended for hosting PowerFlex components and PowerFlex Manager must meet system requirements.	

Table 6. PowerFlex installation checklist (continued)

Item	Checked
NOTE: The Dell-supplied hardware configurations must satisfy all the hardware requirements.	
Secure boot on servers is not supported. Ensure that secure boot is disabled in the BIOS of each server.	
All nodes must have unique hostnames.	
To use secure authentication mode, ensure that the latest OpenSSL 64-bit 1.0.2 or above is installed on all servers in the system.	
Ensure that you know the following: ● Primary and secondary MDM IP addresses ● Primary MDM admin username and password ● Primary MDM operating system hostname and password ● LIA password	
VMware ESXi	
Ensure that the required network interfaces and IP addresses are configured.	
Networking	
For more information on networking requirements, see the *Dell PowerFlex Technical Overview*.	
Powerflex Manager and its installer node	
The nodes that host PowerFlex Manager must have connectivity to both data and management PowerFlex networks. All management nodes should be on the same subnet and SSH should be enabled.	
Prepare the following nodes for installing PowerFlex Manager ● PowerFlex Manager installer node: ○ If the PowerFlex Manager installer is on a bare-metal node, install Docker version 20.10.0 or higher (Podman version 4.9.4 is also supported). ○ If you are using PowerFlex Manager installer on an existing VMware ESXi management infrastructure, deploy the provided installer OVA (SUSE Linux-based). ● If you are using the PowerFlex Manager installer on existing KVM management infrastructure, deploy the provided installer QCOW2 (SUSE Linux-based). ● Three management nodes: ○ These nodes must have IP addresses and hostnames that are unique and are on the same subnet. ○ Hostname labels may contain only the ASCII letters 'a' through 'z' (in a case-insensitive manner), the digits '0' through '9', and the hyphen '-'. ○ The PowerFlex management platform can be deployed in the following modes: ■ One dedicated management node. ■ Three dedicated management nodes with VMware ESXi or bring your own Linux dedicated management nodes. ■ Co-resident with storage or Linux hyperconverged nodes.	

Networking

PowerFlex network types and definitions:

Table 7. PowerFlex network types

Network type	Definition
General-purpose LAN	Access network resources for basic networking activities
PowerFlex data (All network together)	Data traffic between all PowerFlex core components
PowerFlex data (client-traffic only)-separate client and internal storage network	Storage data clients (SDC) to storage node traffic. This network can also be used for NVMe target (SDT) to NVMe over TCP host traffic.

Table 7. PowerFlex network types (continued)

Network type	Definition
	NOTE: It can be a separate data network than the SDS-SDC network. If you are using PowerFlex file this network is also used to connect the SDS to the SDCs on the PowerFlex file controller nodes
PowerFlex data (server-traffic only)-separate client and internal storage network	Storage data server to storage data server traffic only. The same network is used for NVMe target (SDT), SDR (Replication), MDM internal traffic.
PowerFlex management	PowerFlex system management
PowerFlex replication	Storage data replicator (SDR) to storage data server (storage or storage data server) or system to system replication traffic • SDR to storage data clients (application or storage data clients) traffic • SDR to SDR traffic

Networking requirements

The following are the networking requirements for installing PowerFlex.

- Setting jumbo frames to use a maximum transmission unit (MTU) of 9000 allows for the transmission of large packets.
- When deploying PowerFlex on each node, there should be one IP address that can access PowerFlex Manager.
- If you want to separate PowerFlex management IP addresses and data networks, you can have more than one network.
- If you are deploying a server with one IP address, that IP address has to serve as both the PowerFlex management IP address and the data IP address.
- If you are using more than one network, create the network that is routed and can reach PowerFlex Manager as PowerFlex management type network, and other networks as PowerFlex datatype networks. For more information about networking, see the *Dell PowerFlex 4.6.x Technical Overview*.

NOTE: If bonding interfaces and using mode 6 `ALB` the updelay option may need to be increased above zero (which is the default) in the bond configuration file.

For example:

```
MTU='9000'
BOOTPROTO='none'
STARTMODE='auto'
ZONE='public'
BONDING_MASTER='yes'
BONDING_SLAVE0='em3'
BONDING_SLAVE1='p2p1'
BONDING_MODULE_OPTS='mode=6 miimon=100 updelay=100'
```

Deploying PowerFlex Manager

Deploy PowerFlex Manager in accordance with your environment.

PowerFlex Manager can be installed in three ways:

- VMware ESXi environment: Three VMs are running a management virtual machine cluster. PowerFlex Manager is deployed using a fourth VM acting as an installer VM.
- Integrated PowerFlex management (non-KVM): PowerFlex Manager is installed on four bare metal nodes, and then PowerFlex block storage is deployed on the same nodes.
- Linux management environment: The PowerFlex management platform is installed on the Linux operating system directly.
- Customer-provided KVM environment: Enables the setup of the PowerFlex management platform system for management purposes in a KVM environment.

Software packages

For more information about software packages, see the *Dell PowerFlex Release Notes* available on the Dell Technologies support site.

Download PowerFlex installation packages

Use the following procedure to download and extract the installation bundle if you are deploying a PowerFlex cluster outside of PowerFlex Manager.

About this task

Based on the deployment scenario, download either of the following PowerFlex installation packages:

- Use the following packages for vSphere, integrated PowerFlex management (non-KVM), and Linux deployments:
  - `Software_Only_Complete_4.5.x.x.zip`— Contains core PowerFlex software and the compliance matrix (`cm-xxxxxxxx.gpg`) file
- Use the following package for VMware vSphere or virtualized deployments:
  - `PowerFlex_Manager_5.1.x.x.zip`— Contains the OVA templates for the Installer node and for the management node
- Use the following package for deploying with a bare metal Linux installation:
  - `PFMP2-5.1.x.x-xxx.tgz`— Contains the files and scripts for PowerFlex Manager deployment (bare-metal)

Steps

1. On the Dell Technologies support site, download the installation bundle for the version of PowerFlex.
2. Copy the installation bundle to your personal computer or desktop.
3. Extract the installation bundle.

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Deploying the PowerFlex management platform

To deploy the PowerFlex management platform, use one of the following installation options:

- Deploying the PowerFlex management platforminto a VMware vSphere environment
- Deploying the PowerFlex management platforminto a Integrated PowerFlex management (non-KVM) environment
- Deploying the PowerFlex management platforminto a Linux management environment

Deploying into a VMware vSphere environment

Use this option if the PowerFlex management virtual machines reside in an existing VMware vSphere environment.

NOTE: The storage for PowerFlex management platform must be at least an SSD SAS level specification, it does not need to be shared storage, each node has its own partitions to use.

PowerFlex management platform virtual machine requirements

Before you begin, ensure that the VMs you are installing the PowerFlex management platform meet the following requirements:

The PowerFlex management platform virtual machine requirements for storage, CPU, and RAM for PowerFlex 4.5.x or later are the following:

Table 8. VM requirements

System scale	CPU cores	RAM (GB)	Storage capacity (GB)
Low scale	10	32	300
Medium scale	10	32	350
High scale	14	32	450

Deploying the PowerFlex management platform

This section describes how to install and configure the PowerFlex management platform.

This section includes the deployment and configuration of the temporary PowerFlex management platform installer VM. This PowerFlex management platform installer VM is used to deploy the containerized services required for the PowerFlex management platform.

The following diagram depicts the PowerFlex installer VM and management VM deployed in a virtual environment and highlights the interfaces that must be configured:

NOTE: For PowerFlex software, including the `oob` VLAN is optional.

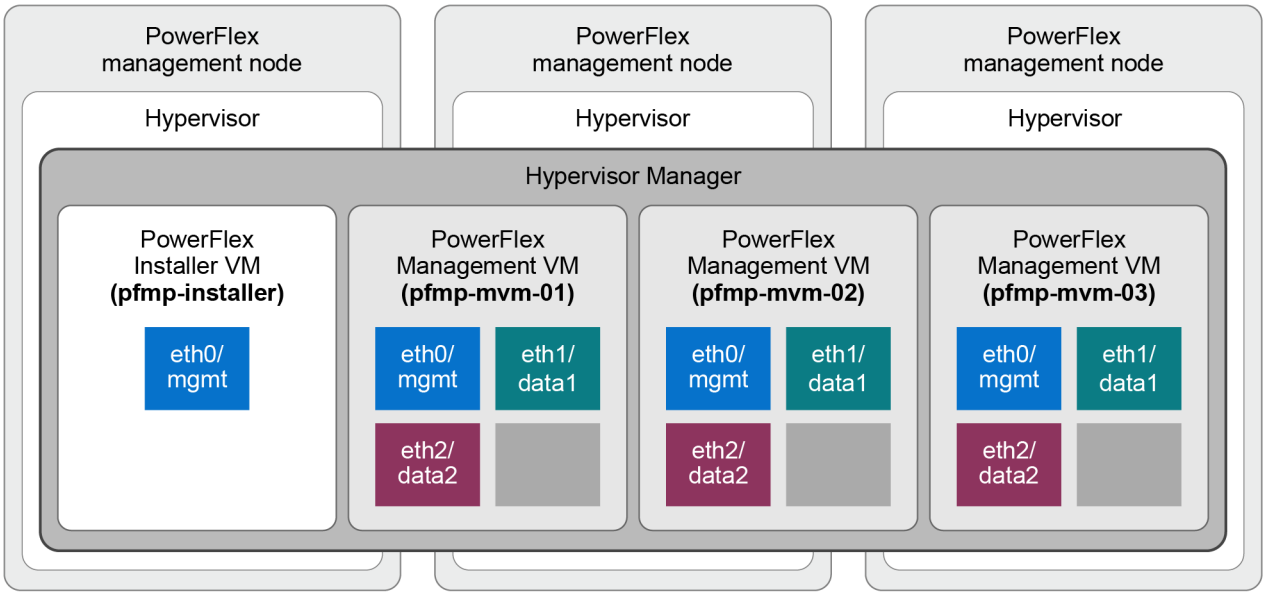


Figure 1. PowerFlex installer VM and MVMs deployed in a virtual environment

PowerFlex management platform cluster configuration parameters

The following table describes the configuration parameters for the `PFMP_Config.json` file, which is used for the configuration and deployment of the PowerFlex management platform cluster.

The following table describes the `PFMP_Config.json` and its configuration parameters:

Table 9. PFMP_Config.json configuration parameters

Parameter	Value	Description
Nodes	Hostname IP address	Minimum of three VMs/nodes The IP addresses of the nodes on which the PowerFlex management platform cluster is deployed, along with the hostnames that must be used for these nodes. Ensure that the hostname is in lowercase.
ClusterReservedIPPoolCIDR	IP address	Minimum / 23 network The minimum recommended subnet size is /23. Ensure that the private subnet does not conflict with any existing subnets in the datacenter. Users could use the same subnet for multiple PowerFlex management platform deployments. The default subnets 10.43.x.x and 10.42.x.x, as listed in the JSON file, can be used. If these defaults conflict with the your environment, they can be modified accordingly.
ServiceReservedIPPoolCIDR	IP address	Minimum / 23 network The minimum recommended subnet size is /23. Ensure that the private subnet does not conflict with any existing subnets in the datacenter.

Table 9. PFMP_Config.json configuration parameters (continued)

Parameter	Value	Description
		Users could use the same subnet for multiple PowerFlex management platform deployments. The default subnets 10.43.x.x and 10.42.x.x, as listed in the JSON file, can be used. If these defaults conflict with the your environment, they can be modified accordingly.
RoutableIPPoolCIDR	<flex-node-mgmt> <flex-data>	flex-node-mgmt and data pools require five unused IP addresses that are used by the web server, Dell Technologies connectivity services, and internal services.
PFMPHostname	FQDN	FQDN which is used to connect to PowerFlex Manager using a browser. The FQDN must be resolvable through DNS and be lower case.
PFMPHostIP	IP address	This parameter specifies the IP address for the ingress IP address. **NOTE:** The ingress IP address must be selected from one of the IP addresses specified in the flex-node-mgmt-<vlanid> IP service range.

Deploying into a VMware vSphere environment using the PowerFlex management platform deployment wizard

The PowerFlex management platform deployment wizard offers a streamlined and automated solution for installing the PowerFlex management platform and the associated VMs, significantly reducing the complexity of manual deployment for a VMware environment.

As a standalone application compatible with Windows and Linux, the wizard enforces a standardized deployment workflow, validates configuration inputs, and provides centralized monitoring with job progress. The wizard automates hundreds of manual tasks such as network configuration, SSH key setup, storage provisioning, VM deployment, and the full PowerFlex management platform installation.

Restrictions and requirements

Ensure the following before using the PowerFlex management platform deployment wizard:

- 3-node PowerFlex management node cluster
- Default MVM credentials are required during the deployment
- NTP/DNS are required

Prerequisities

- The PowerFlex management platform deployment wizard is run from local disk (not a network share location).
- Local reliable network connection to PowerFlex management node cluster.
- Correct VLAN configuration on network switches.
- Allocated IP addresses and hostnames for the installer VM, MVMs and PowerFlex management platform.
- Access to installer VM OVA and MVM OVA files.
- The OVA location requires 100GB of temporary free space.
- vCenter with compute, storage, and networking resources.

Network IP address and hostname planning

Make note of the following parameters for use later in the deployment process:

Table 10. Network IP and hostname planning

Name	FQDN	IP address	Username	Password
vCenter				
-	Primary		Secondary	
DNS				
NTP				
Domain				

Table 11. Installer VM details

Installer VM	Management
VM/hostname	
Management IP address	
Subnet mask	
Default gateway	

Table 12. PowerFlex management cluster node IP address

PowerFlex management cluster IP address	Management
Host-1 IP address	
Host-1 IP address	
Host-1 IP address	

Table 13. MVM-1

MVM-1	Management	OOB*	Data 1	Data 2	Data3	Data 4
VM/hostname						
IP address						
Subnet mask						
Default gateway						

Where OOB* is optional.

Table 14. MVM-2

MVM-2	Management	OOB*	Data 1	Data 2	Data3	Data 4
VM/hostname						
IP address						
Subnet mask						
Default gateway						

Where OOB* is optional.

Table 15. MVM-3

MVM-3	Management	OOB*	Data 1	Data 2	Data3	Data 4
VM/hostname						

Table 15. MVM-3 (continued)

MVM-3	Management	OOB*	Data 1	Data 2	Data3	Data 4
IP address						
Subnet mask						
Default gateway						

Where OOB* is optional.

Download and launch the PowerFlex management platform deployment wizard

Use this procedure to download and launch the PowerFlex management platform deployment wizard.

Steps

1. Download the PowerFlex management platform deployment wizard package, go to on the Dell Technologies support site.
2. Copy the package ZIP file to a local jump host that can access the cluster that is being installed.
   Ensure a reliable network connection for image transfers.
3. Unzip the package:

Table 16. Operating systems available

Windows	Linux
Double-click `powerflex-pfmp-deploy-WIN.bat`.	a. Open terminal and navigate to the application folder. b. Run `./powerflex-pfmp-deploy-LINUX.sh`.

PFMP deployment wizard workflow

The PFMP Deployment Wizard features an interface that, once launched, presents a main splash screen outlining each step of the process. You simply click **Configure** to input deployment data and **Deploy** to initiate the automated deployment.

For the full procedure on how to use this tool see the `Help.PDF` in the unzipped package directory. The following high level workflow is an overview of how the tool works:

1. Verify all prerequisites have been met.
2. Download and launch the wizard.
3. Select deployment type: VMware.
4. Input details for the installer VM and deploy installer VM.
5. Input details for MVMs and install MVMs.
6. Input details for PFMP and fully deploy PFMP.
7. Deploy optional VMs:
   - CloudLink
   - Jump VM
   - Secure connect gateway VM
   - Policy manager VM

Once the PowerFlex management platform is deployed, go to *Configuring PowerFlex Manager*.

Deploy the PowerFlex management platform cluster

Use the following procedure to deploy the PowerFlex management platform.

Steps

1. Using SSH, connect as a non-root user (**`delladmin`**) to the PowerFlex management platform installer VM and run the following command:
   `ssh delladmin@<pfmp installer ip>`

2. Access the `config` directory:
   `cd /opt/dell/pfmp/PFMP_Installer/config`
3. To configure the `PFMP_Config.json`, run **`sudo vi PFMP_Config.json`** and update the configuration parameters.
   For example:

```
{
"Nodes":
[
    {
     "hostname": "pfmp-mvm-01",
     "ipaddress": "10.10.105.30"
    },
    {
     "hostname": "pfmp-mvm-02",
     "ipaddress": "10.10.105.31"
    },
    {
     "hostname": "pfmp-mvm-03",
     "ipaddress": "10.10.105.32"
    }
],
  "ClusterReservedIPPoolCIDR" : "10.42.0.0/23",
  "ServiceReservedIPPoolCIDR" : "10.43.0.0/23",
  "RoutableIPPoolCIDR" : [{"flex-node-mgmt-105":"10.10.105.20-10.10.105.24"},
                          {"flex-node-oob-101":"10.10.101.20-10.10.101.24"},
                          {"flex-data1-151":"192.168.151.20-192.168.151.24"},
                          {"flex-data2-152":"192.168.152.20-192.168.152.24"}],
  "PFMPHostname" : "dellpowerflex.com",
  "PFMPHostIP" : "10.10.105.20"
}
```

NOTE:

- **`"RoutableIPPoolCIDR"`** requires a pool range of at least five IP addresses. By default, the first IP address in the range becomes the PowerFlex Manager user interface address.
- PMFPHostIP has to be in the range of **`"RoutableIPPoolCIDR" :{"flex-node-mgmt-105":"10.10.105.20-10.10.105.24"}`**.
- The **`"PFMPHostname" : "dellpowerflex.com"`** and **`"PFMPHostIP"`** are user-defined and must have the appropriate DNS and PTR records created before installation.

Routed networks must be accessible through the interfaces on the hosts or VMs. If PowerFlex Manager requires access to non-routed networks, add the required interfaces to the host or VM and add an additional line to the RoutableIPPoolCIDR section for that interface.

4. Access the scripts folder in the working directory:
   `cd /opt/dell/pfmp/PFMP_Installer/scripts`
5. Set up the installer:
   `sudo ./setup_installer.sh`

   The **`setup_installer.sh`** starts a docker container with the appropriate network settings.

6. Install the PowerFlex management platform:
   `sudo ./install_PFMP.sh`

   The **`install_PFMP.sh`** uses the docker installation to deploy PowerFlex Manager management cluster across the three virtual machines or hosts.

7. When prompted, do the following:
   - Enter **`No`** to indicate that no SSH keys are used for authentication connection to the cluster nodes. If SSH keys are used, enter **`Yes`**.
   - Enter the non-default SSH username for the nodes specified in `PFMP_Config.json`.
   - Enter **`Yes`** to indicate that the passwords are the same for all cluster nodes.
   - Enter the non-default SSH password for the nodes specified in `PFMP_Config.json`.

- Enter **No** to indicate that the nodes used for PowerFlex management platform are not integrated PowerFlex management (non-KVM) nodes.

NOTE: If you want to deploy a PowerFlex cluster using a CSV topology file, ensure that you use the single SSH key for all nodes.

View the progress of the installation in the following logs that are in the following directory:

- `/opt/dell/pfmp/PFMP_Installer/logs/`
- `/opt/dell/pfmp/atlantic/logs/`

NOTE: Installation may take more than two hours. When the deployment is complete, output similar to the following should appear:

```
UI can be accessed at: powerflex_manager.fqdn.com which needs to be resolved to
<PFMPHostIP>
```

- If the PowerFlex management platform deployment fails, collect the PowerFlex management platform support data and provide the support bundle files to Dell Technologies Support. For information about collecting PowerFlex management platform support data, see the *Dell PowerFlex and PowerFlex Manager Software Install Guide*.
- For more information about troubleshooting, see the *Dell PowerFlex and PowerFlex Manager Software Install Guide*.

(Optional) Collect PowerFlex management platform support data

Use the following procedure to collect PowerFlex management platform support data if the central log collection from the PowerFlex Manager is not available or you are troubleshooting PowerFlex management platform installation issues.

Steps

1. If required, transfer the data collection executable, `pfmp_support`, to all PowerFlex management platform cluster member hosts.

   The data collection utility file is available in the PowerFlex management platform installer and Dell Technologies support site.
   - Get the utility file from the PowerFlex management platform installer: Go to the `PFMP_Installer/scripts` directory.
   - Download the data collection utility from the Dell Technologies support site: Search for **PFMP log collection Script** file on the Dell Technologies support site.

   NOTE: Depending on the transfer method, add the execution permissions using the `chmod +x <file name>` command.

2. On one of the cluster member nodes, run the data collection utility as a user with superuser permissions, such as root user.

   For example, the utility executable exists under `/root`, as follows:

```
# /root/pfmp_support
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting shared kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

3. If the PowerFlex management platform cluster consists of more than one node, on each of the remaining nodes, run the utility as a user with superuser permissions, such as `root`, skipping shared Kubernetes data collection.

```
# /root/pfmp_support --skip-kubernetes-shared
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting server data
collecting general hardware data
```

```
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

4. From all cluster member nodes, provide the resulting support bundle files to Dell Technologies Support.
   By default, bundle files are named `/tmp/powerflex-pfmpsupport/pfmpSupport.tgz`.

Uninstall the PowerFlex management platform cluster

If deployment fails, use the following steps to reset the PowerFlex management platform cluster environment or uninstall the cluster.

Steps

1. Change the directory to the PowerFlex management platform installer scripts directory:
   `cd /opt/dell/pfmp/PFMP_Installer/scripts`
2. Run the following script to reset or uninstall the PowerFlex management platform cluster.
   `sudo ./reset_PFMP.sh`
   NOTE: If any troubleshooting issues are present, you can redeploy the cluster using *Deploy the PowerFlex management platform cluster*.

Remove the PowerFlex management platform installer VM

Use this procedure to remove the PowerFlex management platform installer VM.

Prerequisites

Before removing the PowerFlex management platform installer VM take a backup of the **`PFMP_Config.json`** file at `/opt/dell/pfmp/PFMP_Installer/config`. The **`PFMP_Config.json`** file is required when restoring PowerFlex Manager.

Steps

1. Log in to the VMware vCenter Server Appliance.
2. From the menu, click **Shortcuts** > **Hosts and Clusters**.
3. Right-click **PowerFlex management platform installer VM** > **Power** > **Power Off**.
4. Once powered off, right-click **Delete from Disk**.

Disable kdump on PowerFlex management cluster and management virtual machines

Use this procedure to disable kdump on PowerFlex management nodes.

Prerequisites

NOTE: It is recommended to back up the GRUB file before editing the GRUB file.

Steps

1. SSH to the node and log in as **`delladmin`**
2. Verify the service status, type the following: **`systemctl status kdump.service`**
3. Stop the kdump service, type the following: **`sudo systemctl stop kdump.service`**
4. Disable the kdump service, type the following: **`sudo systemctl disable kdump.service`**
5. Modify the GRUB configuration file, type the following:**`#vi /etc/default/grub`**

a. Locate the line starting with `GRUB_CMDLINE_LINUX_DEFAULT`.
b. Change the **crashkernel** parameter to **crashkernel=no**.

   NOTE: If the parameter is **crashkernel=<some_size>**, change it to **crashkernel=no**.

c. Save the file, press **ESC**+ **:wq**.

6. Update the GRUB configuration file, type the following: **sudo grub2-mkconfig -o /boot/grub2/grub.cfg**
7. Reboot the system, type the following: **sudo reboot**

Perform PowerFlex Manager initial configuration

After installation, use the following procedure to complete the initial configuration of PowerFlex Manager.

Steps

1. Open a web browser and enter **https://<pfmphostname>**, where **<pfmphostname>** is taken from the PFMP_config.json file.
2. Log in to PowerFlex Manager, see the *Dell PowerFlex Family 4.x Security Configuration Guide* for log in details.
3. At the first login, you are prompted to change the password.
4. On the **Welcome** page, read the instructions, and click **Next**.
5. If you want to enable SupportAssist, see related information, and click **Next**.
6. Select **I want to deploy a new instance of PowerFlex**.
7. Review the summary and click **Finish**.

Next steps

Deploy PowerFlex storage, for detailed procedures see related information.

Deploying the PowerFlex management platform into an integrated PowerFlex management (non-KVM) environment

Integrated PowerFlex management (non-KVM) deployment deploys the PowerFlex management platform onto the boot device of three PowerFlex storage nodes where the PowerFlex Manager and PowerFlex storage co-reside on the same node.

PowerFlex Manager is installed on a mirrored NVMe class boot device, such as one of the following:

- R660 BOSS-N1
- R650 BOSS-S2

The minimum NVMe disk size required is 480 GB.

After the operating system and PowerFlex Manager are installed, PowerFlex storage is deployed across the same management nodes through CSV.

Deployment workflow

The following table describes the main steps for deploying the PowerFlex management platform on the BOSS card in a integrated PowerFlex management (non-KVM) deployment.

Table 17. PowerFlex management platform integrated PowerFlex management (non-KVM) deployment workflow

Step	Action	Reference
1	Configure the BIOS settings and iDRAC (if applicable).	For Dell PowerFlex custom node and VxFlex Ready Node deployments, see the following guides available on the Dell Technologies support site.

Table 17. PowerFlex management platform integrated PowerFlex management (non-KVM) deployment workflow (continued)

Step	Action	Reference
2	Install the Linux operating system using vendor best practices. **NOTE:** When deploying the PowerFlex management platform in a integrated PowerFlex management (non-KVM) environment on a BOSS card, the maximum space should be allocated to the `/root` partition.	• *Dell PowerFlex Custom Node R660-R6625-R760-R7625 Hardware Configuration and Operating System Installation Guide for Linux* • *Dell PowerFlex Custom Node R650-R6525-R750 Hardware Configuration and Operating System Installation Guide for Linux*
3	Configure the network on the nodes and the PowerFlex management platform installer.	Configure the PowerFlex management platform installer node with the same parameters as the other nodes. When the PowerFlex management platform installer functionality is removed it will be used as a primary MDM.
4	Set up the YUM/Zypper repository to update the packages as needed.	See the *Adding repositories* topic in the *Managing software with command line tools | Administration Guide | SUSE Linux Enterprise Server 15 SP6*.
6	Deploy the PowerFlex management platform installer and run the PowerFlex management platform deployment.	See *Deploying the PowerFlex management platform*for more information.
7	Log in to PowerFlex Manager.	Verify that the PowerFlex Manager is deployed successfully.
8	Deploy PowerFlex with PowerFlex Manager on the integrated PowerFlex management (non-KVM) nodes using the CSV procedure.	See *Deploying a PowerFlex cluster using a CSV topology file* for more information.

Integrated PowerFlex management (non-KVM) deployment prerequisites

There are certain prerequisites that are required by PowerFlex Manager before you can deploy a Kubernetes cluster.

The following operating systems are supported:

- Red Hat Enterprise Linux 8.8, 8.9, 9.2
- SUSE Linux Enterprise Server (SLES) 15 SP3, 15 SP4, 15 SP5 and 15 SP6

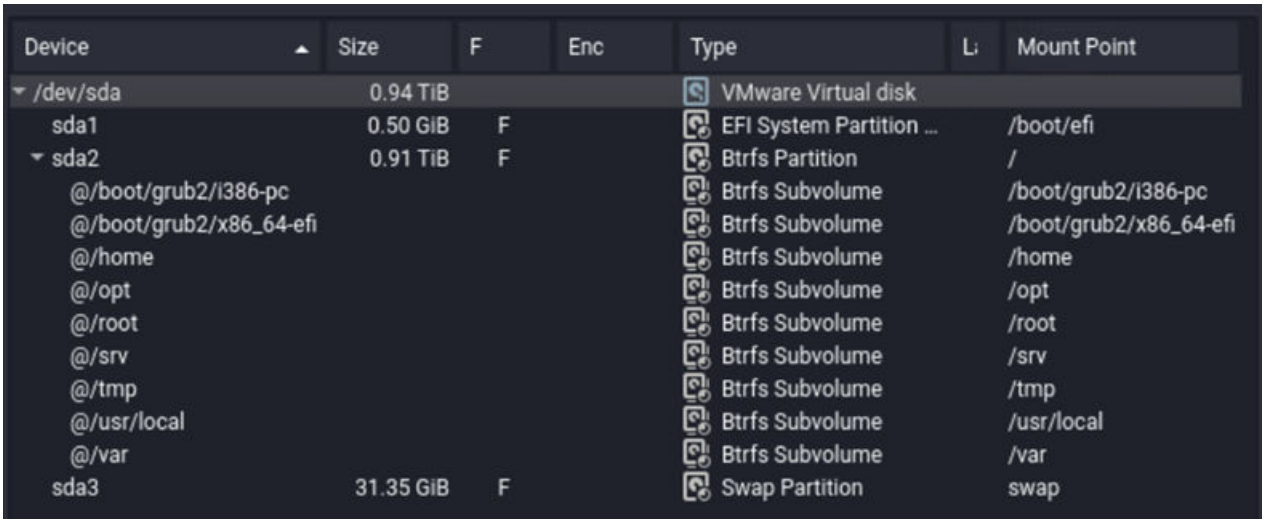
The following modules must be installed on the participating nodes before starting a Integrated PowerFlex management (non-KVM) deployment.

The following list of modules must be installed on the participating nodes for SUSE Linux Enterprise Server before starting a Integrated PowerFlex management (non-KVM) deployment. The modules are located on the install ISO.

- python3
- python3-pip
- sshpass
- java-17-openjdk-headless
- jq
- haproxy
- keepalived
- lvm2
- skopeo
- apparmor-parser
- firewalld
- iptables
- libapr1

The following list of modules must be installed on the participating nodes for Red Hat Enterprise Linux before starting a Integrated PowerFlex management (non-KVM) deployment. The modules are located on the install ISO.

- python3
- python3-pip
- httpd-tools
- python3-libselinux
- sshpass
- java-17-openjdk-headless
- jq
- haproxy
- keepalived
- lvm2
- skopeo
- firewalld
- iptables
- All nodes must have their clocks synchronized with a valid time source.
- Ensure that the default route is configured before installing PowerFlex Manager.
- When installing the operating system to the boot device, it is important to allocate maximum available space on the `/(root)` partition on all the four nodes.
  - In the following image all the remaining space (.91 TB) was allocated to the `/(root)` on partition sda1.



The following table describes the roles of the PowerFlex management platform deployment nodes.

Table 18. Roles of PowerFlex management platform deployment nodes

Number of nodes	Role	Recommended function
Node1	Installer	• Removes the installer functionality. • Primary MDM role after the PowerFlex management platform deployment is complete. • Node1 can also be a temporary VM if it is a three-node deployment. The MDM roles will shift down to node2, node3 and node 4.
Node2	PFMP1	Secondary MDM role
Node3	PFMP2	Tiebreaker role
Node4	PFMP3	Standby MDM role or only SDS or SDC node.

Deploy the PowerFlex management platform installer

Use the following procedure to download and prepare the PowerFlex management platform installer. The PowerFlex management platform installer is deployed temporarily on what will become the primary PowerFlex MDM on a co-resident system. Alternatively, you could deploy the PowerFlex management platform from a VM on a hypervisor external to the system that has network connectivity to the deployment nodes. After deployment, the PowerFlex management platform installer is removed or the installer VM can be deleted, if this is a three-node deployment.

Prerequisites

The node that is used for the PowerFlex management platform installer will require access to meet one of the following requirements:

- To the Internet and attached to a SUSE account.
- Attached to a local SUSE RMT server at the customer site.
- Attached to a local zypper (SLES) or yum (Red Hat Enterprise Linux) repository at the customer site that contains the needed packages and their dependencies such as the installer ISO.

Steps

1. Log in to the installer node, which will become the primary MDM, and install docker.

   For SLES, run the following command:

   ```
   zypper install docker python3-docker-compose
   ```

   For Red Hat Enterprise Linux, run the following command:

   ```
   yum install podman-docker podman-plugins
   ```

2. Create and go to the `pfmp` directory with the following commands:

   ```
   sudo mkdir -p /opt/dell/pfmp
   cd /opt/dell/pfmp
   ```

3. From your desktop, download the PowerFlex installation file from the Dell Technologies support site.
   a. Click **Sign In** and log in with your account.
   b. Search for PowerFlex software and select it from the list.
   c. Go to **Drivers and Download** and select **PowerFlex Manager Bundle 4.5.x Build xxx**.
   d. Download the PowerFlex Manager bundle.
4. Copy the bundle to the `/opt/dell/pfmp` directory created on the first node.
5. Extract the package with the following command:

   ```
   sudo tar -xvf PowerFlex-Manager-bundle-4.5.x-Build-xxx-tgz
   ```

PowerFlex management platform cluster configuration parameters

The following table describes the configuration parameters for the `PFMP_Config.json` file, which is used for the configuration and deployment of the PowerFlex management platform cluster.

The following table describes the `PFMP_Config.json` and its configuration parameters:

Table 19. PFMP_Config.json configuration parameters

Parameter	Value	Description
Nodes	Hostname IP address	Minimum of three VMs/nodes The IP addresses of the nodes on which the PowerFlex management platform cluster is deployed, along with the hostnames that must be used for these nodes. Ensure that the hostname is in lowercase.
ClusterReservedIPPoolCIDR	IP address	Minimum / 23 network

Table 19. PFMP_Config.json configuration parameters (continued)

Parameter	Value	Description
		The minimum recommended subnet size is /23. Ensure that the private subnet does not conflict with any existing subnets in the datacenter. Users could use the same subnet for multiple PowerFlex management platform deployments. The default subnets 10.43.x.x and 10.42.x.x, as listed in the JSON file, can be used. If these defaults conflict with the your environment, they can be modified accordingly.
ServiceReservedIPPoolCIDR	IP address	Minimum / 23 network The minimum recommended subnet size is /23. Ensure that the private subnet does not conflict with any existing subnets in the datacenter. Users could use the same subnet for multiple PowerFlex management platform deployments. The default subnets 10.43.x.x and 10.42.x.x, as listed in the JSON file, can be used. If these defaults conflict with the your environment, they can be modified accordingly.
RoutableIPPoolCIDR	<flex-node-mgmt> <flex-data>	flex-node-mgmt and data pools require five unused IP addresses that are used by the web server, Dell Technologies connectivity services, and internal services.
PFMPHostname	FQDN	FQDN which is used to connect to PowerFlex Manager using a browser. The FQDN must be resolvable through DNS and be lower case.
PFMPHostIP	IP address	This parameter specifies the IP address for the ingress IP address. **NOTE:** The ingress IP address must be selected from one of the IP addresses specified in the flex-node-mgmt-<vlanid> IP service range.

Deploy the PowerFlex management platform cluster

Use the following procedure to deploy the PowerFlex management platform cluster.

Steps

1. Go the `config` directory:
   `cd /opt/dell/pfmp/PFMP_Installer/config`
2. Open the `PFMP_Config.json` file for editing:
   `sudo vi PFMP_Config.json`
3. Update the configuration parameters in the `PFMP_Config.json` file according to the details for your deployment. Refer as necessary to the table in the *PowerFlex management platform cluster configuration parameters* topic in related information. .

CAUTION: Modify the file carefully. Removing `[]` or `{}` causes the install script to fail.

The configuration file varies according to the needs of the customer. For example, in the following example, there are one management interface and two data networks.

NOTE: Ensure the hostname is the same as that defined on the physical node.

```
{
    "Nodes":
    [
      {
        "hostname": "mvm-245",
        "ipaddress": "10.234.89.245"
      },
      {
        "hostname": "mvm-246",
        "ipaddress": "10.234.89.246"
      },
      {
        "hostname": "mvm-247",
        "ipaddress": "10.234.89.247"
      }
    {
    "hostname": "mvm-248",
    "ipaddress": "10.234.89.248"
    }
    {
    "hostname": "mvm-249",
    "ipaddress": "10.234.89.249"
    }
    ],
    "ClusterReservedIPPoolCIDR" : "10.42.0.0/23",
    "ServiceReservedIPPoolCIDR" : "10.43.0.0/23",
    "RoutableIPPoolCIDR": [
    {
      "flex-node-mgmt-<vlanid>": "10.234.89.250 - 10.234.89.254"
    },
    {
      "data1-<vlanid>": "192.168.151.20- 192.168.151.24"
    },
    {
      "data2-<vlanid>": "192.168.152.20 - 192.168.152.24"
    }
  ],
    "PFMPHostname" : "powerflex_manager.fqdn.com",
    "PFMPHostIP" : "10.234.89.250"
}
```

NOTE:

- `RoutableIPPoolCIDR` requires a pool range of at least five IP addresses. By default, the first IP address in the range becomes the PowerFlex Manager user interface address.
- `PMFPHostIP` must be in the range of **`"RoutableIPPoolCIDR" : [{"mgmt":"10.234.89.250 - 10.234.89.254"}]`**.
- `PFMPHostname` and `PFMPHostIP` are user-defined and must have the appropriate DNS and PRT records created before the installation. `PFMPHostname` must be set to a hostname with a fully qualified domain name (FQDN) where the DNS resolves to the `PFMPHostIP`.
- The `PFMPHostname` value, *powerflex_manager.fqdn.com* is used here as an example.

Routed networks must be accessible through the interfaces on the hosts or virtual machines (VMs). If PowerFlex Manager requires access to non-routed networks, add the required interfaces to the host or VM and add an additional line to the `RoutableIPPoolCIDR` section for that interface.

4. Access the scripts folder in the working directory:
   `cd /opt/dell/pfmp/PFMP_Installer/scripts`

5. Ensure that the installer host has IPv4 forwarding enabled:
   a. On the installer host, run the following command:
      sudo sysctl net.ipv4.ip_forward
      Sample output: `net.ipv4.ip_forward = 1`
   b. If the command returns `0 net.ipv4.ip_forward = 0`, add **net.ipv4.ip_forward = 1** to the `/etc/sysctl.conf` file.
   c. Run the following command **sudo sysctl -p** to reload the configuration file.
6. Run the following command to set up the installer:
   sudo ./setup_installer.sh
   The **setup_installer.sh** installs and starts a docker container.
7. Run the following command to install the PowerFlex management platform.
   sudo ./install_PFMP.sh
8. Answer the following questions in the prompt::

```
Are ssh keys used for authentication connecting to the cluster nodes?
Answer= No (unless ssh keys are used)
Please enter the ssh username for the nodes specified in the PFMP_Config.json[root]
Answer: (your ssh username)
Are passwords the same for all the cluster nodes?
Answer: Yes
Please enter the ssh password for the nodes specified in the PFMP_Config.json.
Answer: (your ssh password)
Are the nodes used for the PFMP cluster, co-res nodes?
Answer: Yes
```

   NOTE: When deploying PowerFlex storage with a CSV file using SSH keys for the nodes, ensure that you use the same keys for all nodes.

   Installation may take more than two hours. You can view the progress of the installation in the following logs, located in the directory:

   - `/opt/dell/pfmp/PFMP_Installer/logs/`
   - `/opt/dell/pfmp/atlantic/logs/`

   When the deployment is complete, output similar to the following should appear:

```
UI can be acccessed at: powerflex_manager.fqdn.com which needs to be resolved to
<PFMPHostIP>
```

9. Use to access the PowerFlex Manager login screen.
10. In case of a failure or to check the progress, see the following log files:
    - PowerFlex management platform installer logs: `/opt/dell/pfmp/PFMP_Installer/logs`
    - Platform installer logs: `/opt/dell/pfmp/atlantic/logs/bedrock.log`

    If the PowerFlex management platform deployment fails, collect the PowerFlex management platform support data and provide the support bundle files to Dell Technologies Support. For information about collecting PowerFlex management platform data, see *(Optional) Collect PowerFlex management platform support data*.

Post deployment cleanup

After you deploy the PowerFlex management platform, perform the following steps to uninstall the PowerFlex management platform installer. The PowerFlex management platform installer is not longer needed after you deploy the PowerFlex management platform cluster.

Prerequisites

NOTE: If this is a three-node deployment with an installer VM, you can now shutdown the installer VM and skip this procedure.

Steps

1. Using SSH, connect to the node that the PowerFlex management platform installer is deployed on, and disable IPv4 forwarding:
   a. Open the `sysctl.conf` for editing:
      `sudo vi /etc/sysctl.conf`
   b. Modify the following line: `net.ipv4.ip forward = 1` to **`net.ipv4.ip forward = 0`**
   c. Run the following command to reload the configuration file:
      `sudo sysctl -p`
2. Remove the PowerFlex management platform installer:
   `sudo ./reset_installer.sh`
3. Delete the install directory with the following command:
   `rm -r /opt/dell/pfmp`
4. Stop and remove the docker container:
   `sudo systemctl disable docker`

   `sudo systemctl stop docker`

   `zypper remove docker python3-docker-compose`

   Use the following command for Red Hat:

   `yum remove podman-docker podman-plugins`

(Optional) Collect PowerFlex management platform support data

Use the following procedure to collect PowerFlex management platform support data if the central log collection from the PowerFlex Manager is not available or you are troubleshooting PowerFlex management platform installation issues.

Steps

1. If required, transfer the data collection executable, `pfmp_support`, to all PowerFlex management platform cluster member hosts.

   The data collection utility file is available in the PowerFlex management platform installer and Dell Technologies support site.
   - Get the utility file from the PowerFlex management platform installer: Go to the `PFMP_Installer/scripts` directory.
   - Download the data collection utility from the Dell Technologies support site: Search for **PFMP log collection Script** file on the Dell Technologies support site.

   NOTE: Depending on the transfer method, add the execution permissions using the `chmod +x <file name>` command.

2. On one of the cluster member nodes, run the data collection utility as a user with superuser permissions, such as root user.

   For example, the utility executable exists under `/root`, as follows:

```
# /root/pfmp_support
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting shared kubernetes data
```

```
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

3. If the PowerFlex management platform cluster consists of more than one node, on each of the remaining nodes, run the utility as a user with superuser permissions, such as `root`, skipping shared Kubernetes data collection.

```
# /root/pfmp_support --skip-kubernetes-shared
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

4. From all cluster member nodes, provide the resulting support bundle files to Dell Technologies Support.
   By default, bundle files are named `/tmp/powerflex-pfmpsupport/pfmpSupport.tgz`.

Uninstall the PowerFlex management platform cluster

If deployment fails, use the following steps to reset the PowerFlex management platformcluster environment or uninstall the cluster.

Prerequisites

NOTE: If this is a three-node deployment with an installer VM, you can now delete the VM and skip this procedure.

Steps

1. Change the directory to the PowerFlex management platform installer scripts directory:
   `cd /opt/dell/pfmp/PFMP_Installer/scripts`
   If the PowerFlex management platform deployment fails, collect the PowerFlex management platform support data and provide the support bundle files to Dell Technologies Support. For information about collecting PowerFlex management platform data, see related information.
2. Run the following script to reset or uninstall a PowerFlex management platform cluster:
   `./reset_PFMP.sh`

Post deployment cleanup

After you deploy the PowerFlex management platform, perform the following steps to uninstall the PowerFlex management platform installer. The PowerFlex management platform installer is not longer needed after you deploy the PowerFlex management platform cluster.

Steps

1. Using SSH, connect to the node that the PowerFlex management platform installer is deployed on, and disable IPv4 forwarding:
   a. Open the `sysctl.conf` for editing:
      `sudo vi /etc/sysctl.conf`
   b. Modify the following line: `net.ipv4.ip forward = 1` to **`net.ipv4.ip forward = 0`**

c. Run the following command to reload the configuration file:
   sudo sysctl -p
2. Delete the install directory with the following command:
   rm -r /opt/dell/pfmp
3. Stop and remove the Docker/Podman container:
   sudo systemctl disable docker

   sudo systemctl stop docker

   zypper remove docker python3-docker-compose

   NOTE: Use the following command for Red Hat: **yum remove podman-docker podman-plugins**

Remove temporary installer VM

Once installation is complete, the temporary installer VM may be removed from the host if the deployment was handled by an external hypervisor.

About this task

NOTE: It is recommended to backup the `PFMP_config.json` file used to the jump server to deploy the production PowerFlex Manager instance.

Steps

1. Connect to the hypervisor where the temporary installer VM is deployed.
2. Shutdown the temporary installer VM.
3. Remove the temporary installer VM.

Perform PowerFlex Manager initial configuration

After installation, use the following procedure to complete the initial configuration of PowerFlex Manager.

Steps

1. Open a web browser and enter **https://<pfmphostname>**, where **<pfmphostname>** is taken from the PFMP_config.json file.
2. Log in to PowerFlex Manager, see the *Dell PowerFlex Family 4.x Security Configuration Guide* for log in details.
3. At the first login, you are prompted to change the password.
4. On the **Welcome** page, read the instructions, and click **Next**.
5. If you want to enable SupportAssist, see related information, and click **Next**.
6. Select **I want to deploy a new instance of PowerFlex**.
7. Review the summary and click **Finish**.

Next steps

Deploy PowerFlex storage, for detailed procedures see related information.

Deploying the PowerFlex management platform into a Linux management environment

In a PowerFlex bring your own Linux management configuration, the PowerFlex management platform is installed on the Linux operating system directly.

Deployment workflow into a Linux management environment

The steps to deploy into a Linux management environment are as follows:

NOTE: The storage for PowerFlex management platform must be at least an SSD SAS level specification, it does not need to be shared storage, each node has its own partitions to use.

Table 20. Deployment workflow for a Linux management environment

Step	Action	Reference
1	Configure BIOS settings and the BMC controller (if applicable).	See the *Dell PowerFlex Custom Node R660-R6625-R760-R7625-R860 Hardware Configuration and Operating System Installation Guide for Linux* or *Dell Technologies PowerFlex Custom Node R650/R750/R6525 Hardware Configuration and Operating System Installation Guide for Linux* available at Dell Technologies support site. See the *Dell VxFlex Ready Node R640-R740xd-R840 Hardware Configuration and Operating System Guide for Linux* available at Dell Technologies support site.
2	Install the Linux operating system using the vendor's best practices.	When deploying the operating system reserve 200 GB for /var and 600 GB for /opt
3	Configure the network on the nodes.	N/A
4	Set up the yum \ zypper repository to update the packages as needed.	See the *Adding repositories* topic in the *Managing software with command line tools | Administration Guide | SUSE Linux Enterprise Server 15 SP6* available at the SUSE documentation website.
5	Deploy the PowerFlex management platform installer and run PowerFlex deployment.	See the *Deploying the PowerFlex management platform* for more information.
6	Log in to PowerFlex Manager and deploy a PowerFlex system.	To deploy PowerFlex, see *Deploy PowerFlex storage*.

Linux management deployment prerequisites

Ensure that the following prerequisites are met before deploying the PowerFlex management platform Kubernetes clusters:

Supported operating systems

- Red Hat Enterprise Linux 8.8. 9.2
- SUSE Linux Enterprise Server (SLES) 15 SP3, 15 SP4

Required modules

The following modules must be installed on the participating nodes before starting the deployment:

Table 21. Deployment prerequisites for SUSE Linux Enterprise Server

Module	Location	Repository
python3	On ISO	N/A
python3-pip	On ISO	N/A
sshpass	SUSEConnect	PackageHub/15.x/x86_64
java-11-openjdk-headless	On ISO	N/A
java-11-openjdk-headless	On ISO	N/A
jq	On ISO	N/A
haproxy	SUSEConnect	sle-ha/15.x/x86_64
keepalived	SUSEConnect	sle-ha/15.x/x86_64
lvm2	On ISO	N/A
skopeo	SUSEConnect	sle-ha/15.x/x86_64
apparmor-parser	On ISO	N/A
libapr1	SUSEConnect	sle-module-basesystem/15.x/x86_64

Table 22. Deployment prerequisites for Red Hat Linux Enterprise

Module	Location	Repository
python3	On ISO	N/A
python3-pip	On Satellite	rhel-x-for-x86_64-appstream-rpms
httpd-tools	On Satellite	rhel-x-for-x86_64-appstream-rpms
python3-libselinux	On ISO	N/A
sshpass	On Satellite	rhel-x-for-x86_64-appstream-rpms
java-11-openjdk-headless	On Satellite	rhel-x-for-x86_64-appstream-rpms
java-17-openjdk-headless	On Satellite	rhel-x-for-x86_64-appstream-rpms
jq	On Satellite	rhel-x-for-x86_64-appstream-rpms
haproxy	On Satellite	rhel-x-for-x86_64-appstream-rpms
keepalived	On Satellite	rhel-x-for-x86_64-appstream-rpms
lvm2	On ISO	N/A
skopeo	On Satellite	rhel-x-for-x86_64-appstream-rpms

Volumes and mapping commands

Dedicate all unused storage to the `/(root)` partition.

Additional prerequisites

- All nodes must have their clocks synchronized with a valid time source.
- Ensure that the default route is configured before installing the PowerFlex management platform.

Deployment design for Linux management

The following details best practices in relation to component layout:

NOTE: The following steps are also applicable for a Linux VM running on top of KVM. The KVM configuration is not in this guide, and customers can manage the setup:

- Number of nodes: 3
- Node 1: Kubernetes for the PowerFlex management platform
- Node 2: Kubernetes for the PowerFlex management platform
- Node 3: Kubernetes for the PowerFlex management platform

Configuring Linux management nodes

Use the procedures in this section to configure Linux management nodes.

Setting up the PowerFlex management platform

Deploy the PowerFlex management platform installer

Use the following procedure to download and prepare the PowerFlex management platform installer. The PowerFlex management platform installer is deployed temporarily on what will become the primary PowerFlex MDM on a co-resident system. Alternatively, you could deploy the PowerFlex management platform from a VM on a hypervisor external to the system that has network connectivity to the deployment nodes. After deployment, the PowerFlex management platform installer is removed or the installer VM can be deleted, if this is a three-node deployment.

Prerequisites

The node that is used for the PowerFlex management platform installer will require access to meet one of the following requirements:

- To the Internet and attached to a SUSE account.
- Attached to a local SUSE RMT server at the customer site.
- Attached to a local zypper (SLES) or yum (Red Hat Enterprise Linux) repository at the customer site that contains the needed packages and their dependencies such as the installer ISO.

Steps

1. Log in to the installer node, which will become the primary MDM, and install docker.

   For SLES, run the following command:

   ```
   zypper install docker python3-docker-compose
   ```

   For Red Hat Enterprise Linux, run the following command:

   ```
   yum install podman-docker podman-plugins
   ```

2. Create and go to the `pfmp` directory with the following commands:

   ```
   sudo mkdir -p /opt/dell/pfmp
   cd /opt/dell/pfmp
   ```

3. From your desktop, download the PowerFlex installation file from the Dell Technologies support site.
   a. Click **Sign In** and log in with your account.
   b. Search for PowerFlex software and select it from the list.
   c. Go to **Drivers and Download** and select **PowerFlex Manager Bundle 4.5.x Build xxx**.
   d. Download the PowerFlex Manager bundle.
4. Copy the bundle to the `/opt/dell/pfmp` directory created on the first node.
5. Extract the package with the following command:

   ```
   sudo tar -xvf PowerFlex-Manager-bundle-4.5.x-Build-xxx-tgz
   ```

Deploying the PowerFlex management platform

This section describes how to install and configure the PowerFlex management platform.

This includes the deployment and configuration of the temporary PowerFlex management platform installer VM. This PowerFlex management platform installer VM is used to deploy the containerized services required for the PowerFlex management platform.

The following diagram depicts the PowerFlex installer VM and management VM deployed in a virtual environment and highlights the interfaces needed to be configured:

NOTE: For PowerFlex software, including the `oob` VLAN is optional.

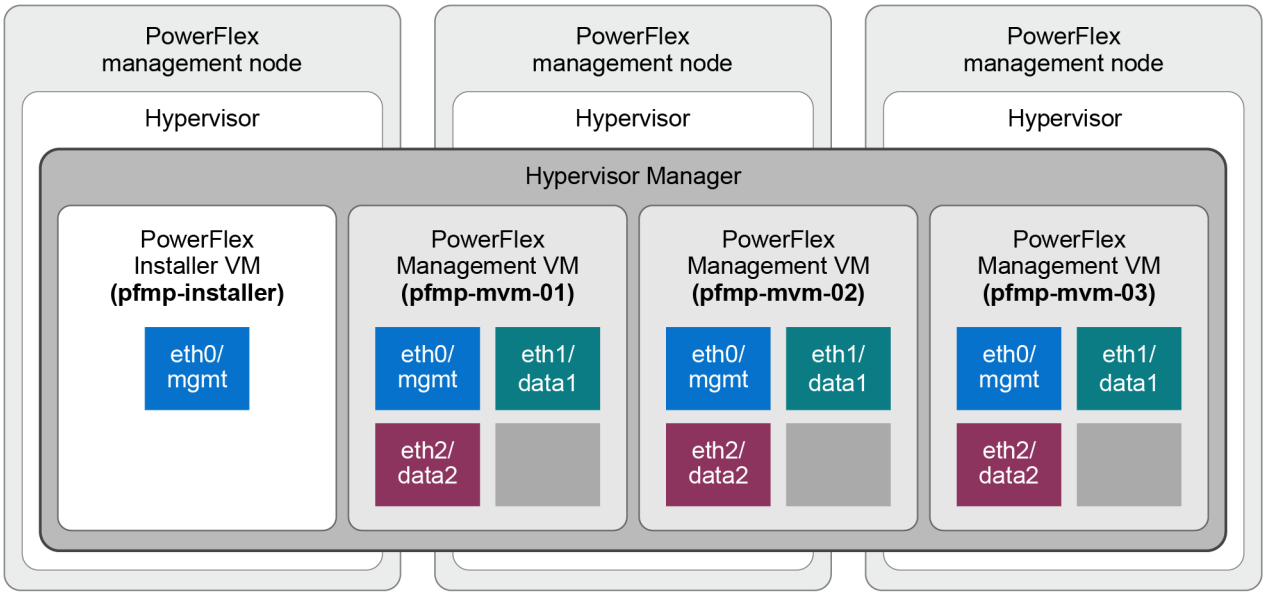


Figure 2. PowerFlex installer VM and MVMs deployed in a virtual environment

Deploy the PowerFlex management platform cluster

Use the following procedure to deploy the PowerFlex management platform.

Steps

1. Using SSH, connect as a non-root user (**delladmin**) to the PowerFlex management platform installer VM and run the following command:

   ssh delladmin@<pfmp installer ip>

2. Access the `config` directory:

   cd /opt/dell/pfmp/PFMP_Installer/config

3. To configure the `PFMP_Config.json`, run **sudo vi PFMP_Config.json** and update the configuration parameters.

   For example:

```
{
"Nodes":
[
    {
     "hostname": "pfmp-mvm-01",
     "ipaddress": "10.10.105.30"
    },
    {
     "hostname": "pfmp-mvm-02",
     "ipaddress": "10.10.105.31"
    },
    {
```

```
    "hostname": "pfmp-mvm-03",
    "ipaddress": "10.10.105.32"
   }
],
  "ClusterReservedIPPoolCIDR" : "10.42.0.0/23",
  "ServiceReservedIPPoolCIDR" : "10.43.0.0/23",
  "RoutableIPPoolCIDR" : [{"flex-node-mgmt-105":"10.10.105.20-10.10.105.24"},
                          {"flex-node-oob-101":"10.10.101.20-10.10.101.24"},
                          {"flex-data1-151":"192.168.151.20-192.168.151.24"},
                          {"flex-data2-152":"192.168.152.20-192.168.152.24"}],
  "PFMPHostname" : "dellpowerflex.com",
  "PFMPHostIP" : "10.10.105.20"
}
```

NOTE:

- **"RoutableIPPoolCIDR"** requires a pool range of at least five IP addresses. By default, the first IP address in the range becomes the PowerFlex Manager user interface address.
- PMFPHostIP has to be in the range of **"RoutableIPPoolCIDR" :{"flex-node-mgmt-105":"10.10.105.20-10.10.105.24"}**.
- The **"PFMPHostname" : "dellpowerflex.com"** and **"PFMPHostIP"** are user-defined and must have the appropriate DNS and PTR records created before installation.

Routed networks must be accessible through the interfaces on the hosts or VMs. If PowerFlex Manager requires access to non-routed networks, add the required interfaces to the host or VM and add an additional line to the RoutableIPPoolCIDR section for that interface.

4. Access the scripts folder in the working directory:
   cd /opt/dell/pfmp/PFMP_Installer/scripts
5. Set up the installer:
   sudo ./setup_installer.sh
   The **setup_installer.sh** starts a docker container with the appropriate network settings.
6. Install the PowerFlex management platform:
   sudo ./install_PFMP.sh
   The **install_PFMP.sh** uses the docker installation to deploy PowerFlex Manager management cluster across the three virtual machines or hosts.
7. When prompted, do the following:
   - Enter **No** to indicate that no SSH keys are used for authentication connection to the cluster nodes. If SSH keys are used, enter **Yes**.
   - Enter the non-default SSH username for the nodes specified in `PFMP_Config.json`.
   - Enter **Yes** to indicate that the passwords are the same for all cluster nodes.
   - Enter the non-default SSH password for the nodes specified in `PFMP_Config.json`.
   - Enter **No** to indicate that the nodes used for PowerFlex management platform are not integrated PowerFlex management (non-KVM) nodes.

   NOTE: If you want to deploy a PowerFlex cluster using a CSV topology file, ensure that you use the single SSH key for all nodes.

   View the progress of the installation in the following logs that are in the following directory:

   - `/opt/dell/pfmp/PFMP_Installer/logs/`
   - `/opt/dell/pfmp/atlantic/logs/`

   NOTE: Installation may take more than two hours. When the deployment is complete, output similar to the following should appear:

   ```
   UI can be accessed at: powerflex_manager.fqdn.com which needs to be resolved to
   <PFMPHostIP>
   ```

- If the PowerFlex management platform deployment fails, collect the PowerFlex management platform support data and provide the support bundle files to Dell Technologies Support. For information about collecting PowerFlex management platform support data, see the *Dell PowerFlex and PowerFlex Manager Software Install Guide*.
- For more information about troubleshooting, see the *Dell PowerFlex and PowerFlex Manager Software Install Guide*.

(Optional) Collect PowerFlex management platform support data

Use the following procedure to collect PowerFlex management platform support data if the central log collection from the PowerFlex Manager is not available or you are troubleshooting PowerFlex management platform installation issues.

Steps

1. If required, transfer the data collection executable, `pfmp_support`, to all PowerFlex management platform cluster member hosts.

   The data collection utility file is available in the PowerFlex management platform installer and Dell Technologies support site.
   - Get the utility file from the PowerFlex management platform installer: Go to the `PFMP_Installer/scripts` directory.
   - Download the data collection utility from the Dell Technologies support site: Search for **PFMP log collection Script** file on the Dell Technologies support site.

   NOTE: Depending on the transfer method, add the execution permissions using the `chmod +x <file name>` command.

2. On one of the cluster member nodes, run the data collection utility as a user with superuser permissions, such as root user.

   For example, the utility executable exists under `/root`, as follows:

```
# /root/pfmp_support
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting shared kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

3. If the PowerFlex management platform cluster consists of more than one node, on each of the remaining nodes, run the utility as a user with superuser permissions, such as `root`, skipping shared Kubernetes data collection.

```
# /root/pfmp_support --skip-kubernetes-shared
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

4. From all cluster member nodes, provide the resulting support bundle files to Dell Technologies Support.
   By default, bundle files are named `/tmp/powerflex-pfmpsupport/pfmpSupport.tgz`.

Uninstall the PowerFlex management platform cluster

If deployment fails, use the following steps to reset the PowerFlex management platform cluster environment or uninstall the cluster.

Steps

1. Change the directory to the PowerFlex management platform installer scripts directory:
   cd /opt/dell/pfmp/PFMP_Installer/scripts
2. Run the following script to reset or uninstall the PowerFlex management platform cluster.
   sudo ./reset_PFMP.sh
   NOTE: If any troubleshooting issues are present, you can redeploy the cluster using *Deploy the PowerFlex management platform cluster*.

Post deployment cleanup

After you deploy the PowerFlex management platform, perform the following steps to uninstall the PowerFlex management platform installer. The PowerFlex management platform installer is not longer needed after you deploy the PowerFlex management platform cluster.

Prerequisites

NOTE: If this is a three-node deployment with an installer VM, you can now shutdown the installer VM and skip this procedure.

Steps

1. Using SSH, connect to the node that the PowerFlex management platform installer is deployed on, and disable IPv4 forwarding:
   a. Open the `sysctl.conf` for editing:
      sudo vi /etc/sysctl.conf
   b. Modify the following line: `net.ipv4.ip forward = 1` to **net.ipv4.ip forward = 0**
   c. Run the following command to reload the configuration file:
      sudo sysctl -p
2. Remove the PowerFlex management platform installer:
   sudo ./reset_installer.sh
3. Delete the install directory with the following command:
   rm -r /opt/dell/pfmp
4. Stop and remove the docker container:
   sudo systemctl disable docker

   sudo systemctl stop docker

   zypper remove docker python3-docker-compose

   Use the following command for Red Hat:

   yum remove podman-docker podman-plugins

Remove temporary installer VM

Once installation is complete, the temporary installer VM may be removed from the host.

About this task

NOTE: It is recommended to back up the `PFMP_config.json` file used to the jump server to deploy the production PowerFlex Manager instance.

Steps

1. Connect to the temporary installer VM with the command: **sudo virsh console tempinstaller**.
2. Press **Enter** to get the login prompt.
3. Log in as **delladmin** with the password **delladmin**.
4. Issue a command to perform a graceful shutdown of the VM: **sudo virsh shutdown tempinstaller**
5. Undefine the VM and remove the associated storage: **sudo virsh undefine tempinstaller --remove-all-storage**
6. Verify the installer is no longer defined: **sudo virsh list --all**

Perform PowerFlex Manager initial configuration

After installation, use the following procedure to complete the initial configuration of PowerFlex Manager.

Steps

1. Open a web browser and enter **https://<pfmphostname>**, where **<pfmphostname>** is taken from the PFMP_config.json file.
2. Log in to PowerFlex Manager, see the *Dell PowerFlex Family 4.x Security Configuration Guide* for log in details.
3. At the first login, you are prompted to change the password.
4. On the **Welcome** page, read the instructions, and click **Next**.
5. If you want to enable SupportAssist, see related information, and click **Next**.
6. Select **I want to deploy a new instance of PowerFlex**.
7. Review the summary and click **Finish**.

Next steps

Deploy PowerFlex storage, for detailed procedures see related information.

Deploying the PowerFlex management platform into a customer-provided KVM environment

Use this section to deploy the Powerflex management platform in a customer-provided KVM environment.

Deploy and configure production MVM images

Use this procedure to deploy and configure production MVM images.

Steps

1. Log in to the first host as **delladmin**.
2. Download the QCOW image from the Dell Support site and copy the `pfmp-k8s-xxx-<date>.qcow2` image to all the three nodes on `/var/lib/libvirt/images/mvm-name`.
3. Deploy the MVM images using the following commands:

```
$ sudo virt-install --name mvm-name1 --ram 32768 --disk /var/lib/libvirt/
images/mvm-name --import --vcpus=14 --network bridge=bridge-<mgmt-VLAN>,model=virtio
--network bridge=bridge-<oob VLAN>,model=virtio --network bridge=bridge-<data1
VLAN>,model=virtio --network bridge=bridge-<data2 VLAN>,model=virtio --osinfo sle15 --
noautoconsole

$ sudo virt-install --name mvm-name2 --ram 32768 --disk /var/lib/libvirt/
images/mvm-name --import --vcpus=14 --network bridge=bridge-<mgmt-VLAN>,model=virtio
--network bridge=bridge-<oob VLAN>,model=virtio --network bridge=bridge-<data1
VLAN>,model=virtio --network bridge=bridge-<data2 VLAN>,model=virtio --osinfo sle15 --
noautoconsole
```

```
$ sudo virt-install --name mvm-name3 --ram 32768 --disk /var/lib/libvirt/
images/mvm-name --import --vcpus=14 --network bridge=bridge-<mgmt-VLAN>,model=virtio
--network bridge=bridge-<oob VLAN>,model=virtio --network bridge=bridge-<data1
VLAN>,model=virtio --network bridge=bridge-<data2 VLAN>,model=virtio --osinfo sle15 --
noautoconsole
```

4. Start the VMs:

   This should be run on all the 3 nodes where MVM is deployed:

```
$ sudo virsh start <mvm-name1>
$ sudo virsh start <mvm-name2>
$ sudo virsh start <mvm-name3>
```

Configure production MVMs

Use this procedure to configure the MVMs.

Steps

1. On the individual hosts, run the following command: **sudo virsh console <mvm-name>**.
2. Press **Enter** to get the login prompt.
3. Log in as **delladmin** with the password **delladmin**.
4. Change the hostname, type **sudo hostnamectl set-hostname <mvm hostname>**.
5. Configure the flex-stor-mgmt-<vlanid> eth0 interface as follows:
   a. Type: **sudo vi /etc/sysconfig/network/ifcfg-eth0**

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
MTU=<mtu>
```

   For example:

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.150.10
NETMASK=255.255.255.0
MTU=1500
```

   b. Configure the default route and run the following command: **sudo vi /etc/sysconfig/network/routes. default <gateway ip> - <interface>**.

   For example: **default 10.10.150.1 - eth0**.

   c. Configure the DNS search and DNS servers. Run the following command: **sudo vi /etc/sysconfig/network/config** and modify the following:

   NETCONFIG_DNS_STATIC_SEARCHLIST="<search domain>"

   NETCONFIG_DNS_STATIC_SERVERS="< dns_ip1 dns_ip2>"

   For example:

   NETCONFIG_DNS_STATIC_SEARCHLIST="example.com"

   NETCONFIG_DNS_STATIC_SERVERS=" 10.10.105.240 10.10.105.241"

6. Configure the eth1 interface with flex-oob-mgmt-<vlanid> (if OOB is active) or flex-data1-<vlanid>interface as follows:
   a. Type: **sudo vi /etc/sysconfig/network/ifcfg-eth1**

```
DEVICE=eth1
NAME=eth1
```

```
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
MTU=<mtu> # 1500 for OOB or 9000 for data network
```

For example:

```
DEVICE=eth1
NAME=eth1
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.150.10
NETMASK=255.255.255.0
MTU=9000
```

7. Configure the eth2 interface with flex-data1-<vlanid> (if OOB is active) or flex-data2-<vlanid>interface as follows:
   a. Type: **sudo vi /etc/sysconfig/network/ifcfg-eth2**

```
DEVICE=eth2
NAME=eth2
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
MTU=<mtu>
```

For example:

```
DEVICE=eth2
NAME=eth2
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.150.10
NETMASK=255.255.255.0
MTU=9000
```

8. Configure the eth3 interface with flex-data2-<vlanid> (if OOB is active) or flex-data3-<vlanid> interface as follows:
   a. Type: **sudo vi /etc/sysconfig/network/ifcfg-eth3**

```
DEVICE=eth3
NAME=eth3
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
MTU=<mtu>
```

For example:

```
DEVICE=eth3
NAME=eth3
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.150.10
NETMASK=255.255.255.0
MTU=9000
```

9. Configure the eth4 interface with flex-data3-<vlanid> (if OOB is active) or flex-data4-<vlanid>interface as follows:
   a. Type: **sudo vi /etc/sysconfig/network/ifcfg-eth4**

```
DEVICE=eth4
NAME=eth4
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
```

```
NETMASK=<mask>
MTU=<mtu>
```

For example:

```
DEVICE=eth4
NAME=eth4
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.150.10
NETMASK=255.255.255.0
MTU=9000
```

10. Configure the eth5 interface with flex-data4-<vlanid> (if OOB is active) interface as follows:
    a. Type: **sudo vi /etc/sysconfig/network/ifcfg-eth5**

```
DEVICE=eth5
NAME=eth5
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
MTU=<mtu>
```

For example:

```
DEVICE=eth5
NAME=eth5
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.150.10
NETMASK=255.255.255.0
MTU=9000
```

11. To restart the IP addresses and networks, type **sudo systemctl restart network**.
12. Exit the console, close the window.
13. Repeat the above for the other production MVMs on the hosts theya re running on, gathered from step 1.

Configure the NTP service

Use this procedure to configure the NTP service.

Steps

1. SSH to the first MVM as **delladmin**.
2. To configure the NTP server, type **sudo vi /etc/chrony.conf** and add the following: **server <ntp ip address> iburst**.

   For example: **server 10.10.105.240 iburst**
3. To enable chronyd, type **sudo systemctl enable chronyd**.
4. To reboot, type **sudo reboot**.
5. Log in as **delladmin**
6. To check the server is synced to NTP, type **chronyc tracking**.

   NOTE: This may take a few minutes to sync. Examples of chronyc tracking status are as follows:

Table 23. Chronyc tracking status

Successful chronyc tracking status	Unsuccessful chronyc tracking status
A configured NTP server should be listed under `Reference ID`, `Stratum` should be a value greater than 0 and	Stratum 0 means not synced and Leap status says `Not synchronized`:

Table 23. Chronyc tracking status

Successful chronyc tracking status	Unsuccessful chronyc tracking status

`System Time` should report much less than 1 second time difference:
`node1:~ # chronyc tracking`
`Reference ID : 64400012 (0.pool.ntp.org)`
`Stratum : 4`
`Ref time (UTC) : Mon Mar 07 17:15:39 2022`
`System time : 0.000001647 seconds slow of NTP time`
`Last offset : -0.000006101 seconds`
`RMS offset : 0.000020258 seconds`
`Frequency : 33.247 ppm slow`
`Residual freq : -0.000 ppm`
`Skew : 0.016 ppm`
`Root delay : 0.040418178 seconds`
`Root dispersion : 0.033837218 seconds`
`Update interval : 1024.6 seconds`
`Leap status : Normal`
	`node1:~ # chronyc tracking`
	`Reference ID : 00000000 ()`
	`Stratum : 0`
	`Ref time (UTC) : Thu Jan 01 00:00:00 1970`
	`System time : 0.000000000 seconds fast of NTP time`
	`Last offset : +0.000000000 seconds`
	`RMS offset : 0.000000000 seconds`
	`Frequency : 0.000 ppm slow`
	`Residual freq : +0.000 ppm`
	`Skew : 0.000 ppm`
	`Root delay : 1.000000000 seconds`
	`Root dispersion : 1.000000000 seconds`
	`Update interval : 0.0 seconds`
	`Leap status : Not synchronized`

7. Repeat these steps for all MVMs.

Deploy the temporary installer VM

A temporary installer VM is required to install the second PowerFlex Manager instance.

About this task

The VM is placed on local storage of one of the existing hosts and removed once the installation is complete.

Prerequisites

Ensure the QCOW2 image for the PowerFlex management platform installer VM must be accessible to the system.

Steps

1. SSH to one of the PowerFlex Manager controller hosts.
2. Download and copy the QCOW2 image for the installer to `/var/lib/libvirt/images/`.
3. Deploy the virtual machine, type:

```
$ sudo virt-install --name tempinstaller --ram 8192 --vcpus=8  --disk /var/lib/
libvirt/images/<img name>.qcow2   --network bridge=bridge-<mgmt-VLAN>,model=virtio --
osinfo sle15 --import --noautoconsole
```

Configure temporary installer

Steps

1. Connect to the temporary installer VM with the command **`sudo virsh console tempinstaller`**.
2. Press **Enter** to get the login prompt.
3. Log in as **`delladmin`** with the password **`delladmin`**.
4. Configure the flex-stor-mgmt-<vlanid> eth0 interface:
   a. Type: **`sudo vi /etc/sysconfig/network/ifcfg-eth0`**

```
DEVICE=eth0
NAME=eth0
```

```
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
```

For example:

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.105.10
NETMASK=255.255.255.0
```

b. Configure the default route. Type:

```
sudo vi /etc/sysconfig/network/routes
default<gateway ip> - <interface>
```

For example: **default 10.10.105.1 - eth0**.

c. Configure DNS search and DNS servers. Type: **sudo vi /etc/sysconfig/network/config** and modify the following:

NETCONFIG_DNS_STATIC_SEARCHLIST="<search domain>"

NETCONFIG_DNS_STATIC_SERVERS="< dns_ip1 dns_ip2>"

For example:

NETCONFIG_DNS_STATIC_SEARCHLIST="example.com"

NETCONFIG_DNS_STATIC_SERVERS=" 10.10.105.240 10.10.105.241"

5. To restart the IP addresses and network, type: **sudo systemctl restart network**.

Deploy the PowerFlex management platform cluster

Use this task to complete the steps needed to configure the `PFMP_Config.json` file which is used for the configuration and deployment of the PowerFlex management platform cluster.

Prerequisites

The following locations contain log files for troubleshooting:

- PowerFlex management platform installer logs: `/opt/dell/pfmp/PFMP_Installer/logs`
- Platform installer logs: `/opt/dell/pfmp/atlantic/logs/bedrock.log`

The following table describes the `PFMP_Config.json` and its configuration parameters.

NOTE: When configuring the RoutableIPPoolCIDR parameter in `PFMP_Config.json`, the subset of the IP addresses must be contiguous. It is five contiguous IP addresses for each flex-node-mgmt-<vlanid>, flex-oob-mgmt-<vlanid>, flex-data1-<vlanid>, and flex-data2-<vlanid>.

Table 24. PFMP_config.JSON and configuration parameters

Key	Value	Description
Nodes (minimum three VMs / nodes)	hostname IP address	The IP addresses of the nodes on which the PowerFlex management platform cluster will be deployed, along with the hostnames that need to be used for these nodes. Ensure that the hostname is in lowercase.
ClusterReservedIPPoolCIDR (minimum / 23 network)	IP address	The minimum recommended subnet size is /23. Ensure that the private subnet does not conflict with any existing subnets in the datacenter. Every PowerFlex management platform instance must use unique subnets.

Table 24. PFMP_config.JSON and configuration parameters (continued)

Key	Value	Description
		For example, PFMP-PROD should use different subnets than the PFMP-MGMT instance as well as all subsequent PowerFlex management platform instances.
ServiceReservedIPPoolCIDR (minimum / 23 network)	IP address	The minimum recommended subnet size is /23. Ensure that the private subnet does not conflict with any existing subnets in the datacenter. Every PowerFlex management platform instance must use unique subnets. For example, PFMP-PROD should use different subnets than the PFMP-MGMT instance as well as all subsequent PowerFlex management platform instances.
RoutableIPPoolCIDR	flex-stor-mgmt-150 flex-oob-mgmt-<vlanid> flex-data1-<vlanid> flex-data2-<vlanid> flex-data3-<vlanid> (if required) flex-data4-<vlanid> (if required)	IP requirements (number of IP addresses): Five contiguous IP addresses ● flex-stor-mgmt-150 (5) ● flex-oob-mgmt-<vlanid> (5) ● flex-data1-<vlanid> (5) ● flex-data2-<vlanid> (5) ● flex-data3-<vlanid> (5) (if required) ● flex-data4-<vlanid> (5) (if required) IP addresses are for the following services: ● Ingress (web server) IP address ● HTTP IP address ● SNMP IP address (receiving SNMP traps) ● SupportAssist remote ● Additional IP for HTTP services (will be used only during upgrade)
PFMPHostname	FQDN	This is the FQDN or hostname which a user can connect to the PowerFlex management platform UI using a browser. The FQDN/hostname needs to be resolvable through DNS or through a host file (where the browser is running).
PFMPHostIP	IP address	This parameter specifics the IP address for the ingress IP address. **NOTE:** The ingress IP address must be selected from one of the IP addresses specified in the flex-node-mgmt-<vlanid> IP service range.

Steps

1. To SSH as non-root user to the PowerFlex management platform Installer, run the following command: **`ssh delladmin@<pfmp installer ip>`**.
2. To navigate to the config directory, run the following command: **`cd /opt/dell/pfmp/PFMP_Installer/config`**
3. To configure the `PFMP_Config.json`, run the following command: **`sudo vi PFMP_Config.json`** and update the configuration parameters.

For example:

```
{
"Nodes":
[
    {
     "hostname": "pfmp-mvm-01",
     "ipaddress": "10.10.150.30"
    },
    {
     "hostname": "pfmp-mvm-02",
     "ipaddress": "10.10.150.31"
    },
    {
     "hostname": "pfmp-mvm-03",
     "ipaddress": "10.10.150.32"
    }
],
  "ClusterReservedIPPoolCIDR" : "10.42.0.0/23",
  "ServiceReservedIPPoolCIDR" : "10.43.0.0/23",
  "RoutableIPPoolCIDR" : [{"flex-stor-mgmt-150":"10.10.150.20-10.10.105.24"},
                          {"flex-node-oob-101":"10.10.101.20-10.10.101.24"},
                          {"flex-data1-151":"192.168.151.20-192.168.151.24"},
                          {"flex-data2-152":"192.168.152.20-192.168.152.24"}],
  "PFMPHostname" : "dellpowerflex.com",
  "PFMPHostIP" : "10.10.150.20"
}
```

NOTE: PMFPHostIP has to be in the range of **"RoutableIPPoolCIDR" : {"flex-stor-mgmt-<vlanid>":"10.10.150.20-10.10.150.24"}**.

NOTE: If the customer is planning for four data networks, add flex-data3-<vlanid> and flex-data4-<vlanid> also in PFMP_Config.json file.

4. To set up the installer, run the following command: **sudo ./setup_installer.sh**.
5. Navigate to the scripts folder in the working directory, type: **cd /opt/dell/pfmp/PFMP_Installer/scripts**

   NOTE: If the PowerFlex management platform has already been deployed on the management VMs, and you need to redeploy the PowerFlex management platform, the previous installation must be removed and cleaned from the VMs, type the following command: **sudo ./reset_installer.sh**.

   The reset_installer.sh script will provide the same prompts as the setup_installer.sh script, but removes the deployed cluster.

6. To install PowerFlex management platform, run the following command: **sudo ./install_PFMP.sh** (enter sudo user and password for target nodes when prompted, when prompted, answer the questions provided.
7. When prompted, do the following:
   - Enter **No** to indicate that no SSH keys are used for authentication connection to the cluster nodes. If SSH keys are used, enter **Yes**.
   - Enter the SSH username of the non-root user for the nodes specified in `PFMP_Config.json`. The default user is delladmin.
   - Enter **Yes** to indicate that passwords are the same for all cluster nodes.
   - Enter the SSH password of the non-root user for the nodes specified in `PFMP_Config.json`.
   - Enter **No** to indicate that the nodes used for PowerFlex management platform are not co-resident nodes.
8. Once the installer is complete, use **powerflex_manager.fqdn.com** to log in to PowerFlex Manager using username and password.

   If a failure has occurred with PowerFlex management platform installer, the relevant errors are located at: `/opt/dell/pfmp/atlantic/logs/bedrock.log`. For more information on troubleshooting, see *Troubleshooting a PowerFlex management platform deployment*.

Remove temporary installer VM

Once installation is complete, the temporary installer VM may be removed from the host.

About this task

NOTE: It is recommended to back up the `PFMP_config.json` file used to the jump server to deploy the production PowerFlex Manager instance.

Steps

1. Connect to the temporary installer VM with the command: **`sudo virsh console tempinstaller`**.
2. Press **Enter** to get the login prompt.
3. Log in as **`delladmin`** with the password **`delladmin`**.
4. Issue a command to perform a graceful shutdown of the VM: **`sudo virsh shutdown tempinstaller`**
5. Undefine the VM and remove the associated storage: **`sudo virsh undefine tempinstaller --remove-all-storage`**
6. Verify the installer is no longer defined: **`sudo virsh list --all`**

Disable kdump on PowerFlex management cluster (KVM) and management virtual machines

Use this procedure to disable kdump on PowerFlex management nodes (KVM).

Prerequisites

NOTE: It is recommended to back up the GRUB file before editing the GRUB file.

Steps

1. SSH to the node and login as **`delladmin`**.
2. Verify the service status, type the following: **`systemctl status kdump.service`**
3. Stop the kdump service, type the following: **`sudo systemctl stop kdump.service`**
4. Disable the kdump service, type the following: **`sudo systemctl disable kdump.service`**
5. Modify the GRUB configuration file, type the following:**`sudo vi /etc/default/grub`**
   a. Locate the line starting with `GRUB_CMDLINE_LINUX_DEFAULT`.
   b. Change the **`crashkernel`** parameter to **`crashkernel=no`**.

      NOTE: If the parameter is **`crashkernel=<some_size>`**, change it to **`crashkernel=no`**.

   c. Save the file, press **ESC**+ **:wq**.
6. Update the GRUB configuration file, type the following: **`sudo grub2-mkconfig -o /boot/grub2/grub.cfg`**
7. Reboot the system, type the following: **`sudo reboot`**

Configure firewall on PowerFlex Manager (optional)

PowerFlex Manager uses Kubernetes, which is incompatible with firewalld. Firewalld can interfere with the dynamic requirements of Kubernetes, so firewalld must be disabled. Use this optional task to disable firewalld and achieve the same capabilities using a Calico network policy update.

About this task

In PowerFlex Manager, firewalld is disabled automatically. This task describes how to configure the Calico firewall policy on PowerFlex Manager if required. Use this task to configure custom ports beyond the default custom ports in PowerFlex Manager.

NOTE: Configuring the firewall on PowerFlex Manager is optional.

Steps

1. On the management virtual machine, run the following command:

```
kubectl patch hr gnwpolicy -n powerflex --type merge --patch '{"spec":{"values":
{"customtcpports":[xx,yy],"iscores":"false"}}}'
```

Where `[xx,yy]` are the ports.

For example, the following command adds TCP ports 90 and 80:

```
kubectl patch hr gnwpolicy -n powerflex --type merge --patch '{"spec":{"values":
{"customtcpports":[90,80],"iscores":"false"}}}'
```

Where `iscores=false` or `iscores=true` is based on whether PowerFlex management platform is running on a separate management cluster.

You must open the complete list of ports that must be opened on `customtcpports` and `customudpports`. Additional ports are opened after providing the complete list of ports including the ports opened earlier.

For example, if you have opened `customtcpports` 80 and 90, and at a later point you open ports 8000 and 9000, add the new and existing ports as shown in the following example. Also, ensure to set the *iscores* flag appropriately.

```
kubectl patch hr gnwpolicy -n powerflex --type merge --patch '{"spec":{"values":
{"customtcpports":[90,80,9000,8000],"iscores":"false"}}}'
```

2. Run the following command to verify the newly added ports:

```
kubectl get globalnetworkpolicy -o yaml
```

5

Troubleshooting a PowerFlex management platform deployment

This troubleshooting section contains verification steps and methods that can be used to identify and address misconfigurations to ensure a successful PowerFlex management platform deployment.

The PowerFlex management platform runs on Kubernetes across three virtual machines that reside on controller nodes or a customer-provided virtual environment. The PowerFlex management platform deployment process can fail for several reasons. To ensure the expected state of the PowerFlex installer and management virtual machines (MVMs) before initiating the deployment.

Verifying the network time protocol setup

Use this section to verify the network time protocol (NTP) servers are synchronized before the install is initiated.

Time synchronization across all hosts and virtual machines is required. Functioning NTP servers are required for deployment. All PowerFlex nodes and virtual machines must point to a functioning NTP server and time must be verified as synchronized before the install is attempted.

Verify the NTP setup on a vCenter Server Appliance

Use this procedure to verify the NTP is setup properly on a vCenter Server Appliance.

Steps

1. Access the vCenter Console at `https://<vcenterIP>:5480`
2. Log in as **`root`**.
3. Select **Time**.
4. Navigate to **Time Synchronization**.
5. Verify at least one of the NTP servers is set and has a green check.

Verify the NTP setup on a VMware ESXi-based PowerFlex node

Use the following procedure to verify the NTP is set up properly on a VMware ESXi-based PowerFlex node.

Steps

1. In the VMware vSphere Client home page, go to **Home** > **Hosts and Clusters**.
2. Select a host.
3. On the **Configure** tab, select **System** > **Time Configuration**.
4. Click **Test services**.
5. Verify that the check passes and time are accurate.

Verify the NTP setup on a Linux-based PowerFlex node

Use the following procedure to verify the NTP setup on a Linux-based PowerFlex node.

Steps

1. SSH to the Linux-based PowerFlex node.

2. Type the following: **chronyc tracking**.

   Output should have a valid reference ID and time should be accurate. The following gives an example output:

```
pfmp-mvm-01:~ # chronyc tracking
Reference ID    : 0AEA7C8B (dns.example.com)
Stratum         : 4
Ref time (UTC)  : Tue Jan 30 16:09:53 2024
System time     : 0.000041504 seconds slow of NTP time
Last offset     : -0.000069052 seconds
RMS offset      : 0.000188316 seconds
Frequency       : 16.336 ppm slow
Residual freq   : -0.011 ppm
Skew            : 0.077 ppm
Root delay      : 0.001050830 secondsRoot dispersion : 0.005389712 seconds
Update interval : 1034.1 seconds
Leap status     : Normal
pfmp-mvm-01:~ #
```

   If chronyc tracking returns invalid data, see the NTP setup in *Configure the NTP service*.

Verifying the DNS

PowerFlex uses mTLS and SSL certificates across all services, hosts, and virtual machines. To get proper host resolution valid DNS A-records and PTR records are required. Use this section to verify the DNS is functioning as required.

Verify DNS on vCenter Server Appliance

Use this procedure to verify DNS on vCenter Server Appliance.

Steps

1. SSH to the vCenter Appliance.
2. At the command prompt, type: **shell**.
3. Type the following: **nslookup <FQDN>**.

   Example output:

```
[root@pfmp-vcenter:~] nslookup pfmp-vcenter.example.com
Server:         10.10.105.240
Address:        10.10.105.240:53

Name:   pfmp-vcenter.example.com
Address: 10.10.105.2

[root@pfmp-vcenter:~]
```

4. Type the following: **nslookup <IP address>**.

   Example output:

```
[root@ pfmp-vcenter:~] nslookup 10.10.105.2
Server:         10.10.105.240
Address:        10.10.105.240:53

123.123.123.45.in-addr.arpa      name = pfmp-vcenter.example.com

[root@pfmp-vcenter:~]
```

Verify DNS on a VMware ESXi-based PowerFlex node

Use the following procedure to verify DNS on a VMware ESXi-based PowerFlex node.

Steps

1. SSH to the VMware ESXi-based PowerFlex node.
2. Type the following: **`nslookup <FQDN>`**.

   Example output:

```
[root@pfmp-r650-01:~] nslookup pfmp-r650-01.example.com
Server:         10.10.105.240
Address:        10.10.105.240:53

Name:   pfmp-r650-01.example.com
Address: 10.10.105.3

[root@pfmp-r650-01:~]
```

3. Type the following: **`nslookup <IP address>`**.

   Example output:

```
[root@pfmp-r650-01:~] nslookup 10.10.105.3
Server:         10.10.105.240
Address:        10.10.105.240:53

123.123.123.45.in-addr.arpa      name = pfmp-r650-01.example.com

[root@pfmp-r650-01:~]
```

Verify DNS on a Linux-based PowerFlex node

Use the procedure to modify or verify DNS on a Linux-based PowerFlex node.

Steps

1. SSH to the Linux-based PowerFlex node.
2. To update or modify the DNS entries:
   a. Go the file path, run the following command: `sudo vi /etc/sysconfig/network/config`
   b. Add or modify the line:
      - **`NETCONFIG_DNS_STATIC_SEARCHLIST=”xxx.com”`**, where **`xxx.com`** is the domain name.
      - **`NETCONFIG_DNS_STATIC_SERVERS="x.x.x.x y.y.y.y"`**, where **`x.x.x.x`** and **`y.y.y.y`** with your DNS server IP address.
   c. Save the file, press **Esc**+ **`:wq`**.
   d. Refresh and update the configuration, run the following command: **`sudo netconfig update -f`**.

Check DNS on a Linux-based PowerFlex node

Use this procedure to check DNS on a Linux-based PowerFlex node.

Steps

1. SSH to the Linux-based PowerFlex node.
2. Type the following: **`nslookup <Linux IP>`**.

   Example output:

```
pfmp-mvm-01:~ nslookup 10.10.105.30
Server:         10.10.105.240
```

```
Address:        10.10.105.240:53

123.123.123.45.in-addr.arpa      name = pfmp-mvm-01.example.com

pfmp-mvm-01:~
```

Verifying the network

Use this section to verify the network. The PowerFlex management platform requires both internal and external network communication. See the *Dell PowerFlex Appliance and PowerFlex Rack Cabling and Connectivity Guide* for more information.

Verify each of the three PowerFlex management virtual machines has the required interfaces:

- flex-node-mgmt-<vlanid>
- flex-oob-mgmt-<vlanid> (if required)
- flex-data1-<vlanid>
- flex-data2-<vlanid>
- flex-data3-<vlanid> (if required)
- flex-data4-<vlanid> (if required)

Verify the management virtual machine network communication

Use the following procedure to verify the management virtual machine network communication using a ping command.

Steps

1. SSH to each PowerFlex management virtual machine node.
2. For flex-node-mgmt-<vlanid> and flex-oob-mgmt-<vlanid>, type the following: **`ping <peer IP address>`**.
3. For flex-data1-<vlanid>, flex-data2-<vlanid>, flex-data3-<vlanid> (if required), and flex-data4-<vlanid> (if required), type the following: **`ping <peer IP address>`**.

   Using the following examples, verify all interfaces within the same subnet are reachable:

```
pfmp-mvm-01 mgmt/eth0 <-> pfmp-mvm-02 mgmt/eth0
pfmp-mvm-01 mgmt/eth0 <-> pfmp-mvm-03 mgmt/eth0

pfmp-mvm-01 oob/eth1 <-> pfmp-mvm-02 oob/eth1
pfmp-mvm-01 oob/eth1 <-> pfmp-mvm-03 oob/eth1

pfmp-mvm-01 data1/eth2 <-> pfmp-mvm-02 data1/eth2
pfmp-mvm-01 data1/eth2 <-> pfmp-mvm-03 data1/eth2

pfmp-mvm-01 data2/eth3 <-> pfmp-mvm-02 data2/eth3
pfmp-mvm-01 data2/eth3 <-> pfmp-mvm-03 data2/eth3

pfmp-mvm-02 mgmt/eth0 <-> pfmp-mvm-03 mgmt/eth0
pfmp-mvm-02 oob/eth1 <-> pfmp-mvm-03 oob/eth1

pfmp-mvm-02 data1/eth2 <-> pfmp-mvm-03 data1/eth2
pfmp-mvm-02 data2/eth3 <-> pfmp-mvm-03 data2/eth3
```

   Example output:

```
pfmp-mvm-01:~ # ping 10.10.105.31
PING pfmp-mvm-02 (10.10.105.31) 56(84) bytes of data.
64 bytes from pfmp-mvm-02 (10.10.105.31): icmp_seq=1 ttl=64 time=0.112 ms
64 bytes from pfmp-mvm-02 (10.10.105.31): icmp_seq=2 ttl=64 time=0.077 ms
64 bytes from pfmp-mvm-02 (10.10.105.31): icmp_seq=3 ttl=64 time=0.086 ms

--- from pfmp-mvm-02 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3063ms
rtt min/avg/max/mdev = 0.077/0.378/1.240/0.497 ms
pfmp-mvm-01:~
```

Verify routing

Use the following procedure to verify the routing of networks.

About this task

Verify the routing for the following networks:

- flex-node-mgmt-<vlanid> route to flex-stor-mgmt-<vlanid>
- flex-node-mgmt-<vlanid> route to pfmc-sds-mgmt-<vlanid>
- pfmc-vm-mgmt-<vlanid> route to pfmc-sds-mgmt-<vlanid> (optional, used for multi-VLAN configurations)

Steps

1. SSH to each PowerFlex management virtual machine on flex-node-mgmt-<vlanid>.
2. Type the following: **`ping <pfmc-sds-mgmt-<vlandid> target ip address>`**.
3. Type the following: **`ping <flex-stor-mgmt-<vlanid> target ip address>`**.

   Example output:

```
pfmp-mvm-01:~ # ping 10.10.150.10
PING 10.10.150.10 (10.10.150.10) 56(84) bytes of data.
64 bytes from 10.10.150.10: icmp_seq=1 ttl=64 time=0.112 ms
64 bytes from 10.10.150.10: icmp_seq=2 ttl=64 time=0.077 ms
64 bytes from 10.10.150.10: icmp_seq=3 ttl=64 time=0.086 ms

--- from 10.10.150.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3063ms
rtt min/avg/max/mdev = 0.077/0.378/1.240/0.497 ms
pfmp-mvm-01:~


pfmp-mvm-01:~ # ping 10.10.140.10
PING 10.10.140.10 (10.10.140.10) 56(84) bytes of data.
64 bytes from 10.10.140.10: icmp_seq=1 ttl=64 time=0.112 ms
64 bytes from 10.10.140.10: icmp_seq=2 ttl=64 time=0.077 ms
64 bytes from 10.10.140.10: icmp_seq=3 ttl=64 time=0.086 ms

--- from 10.10.140.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3063ms
rtt min/avg/max/mdev = 0.077/0.378/1.240/0.497 ms
pfmp-mvm-01:~
```

Verifying the PFMP_config.json file

Use this section to verify the `PFMP_config.json` file.

The PowerFlex management platform deployment process requires a configuration file to be filled out with customer network information. The `install_pfmp.sh` script uses the data in PFMP_config.json to deploy all the required packages, install Kubernetes, and deploy all the pods that are required to run PowerFlex Manager.

The template file is found on the installer VM under: `/opt/dell/pfmp/PFMP_Installer/config/PFMP_Config.json`.

Verify the PFMP_config.json key input restrictions

Use this procedure to verify the key input restrictions for the PFMP_config.json file.

Steps

Verify the following `PFMP_config.json` key input restrictions:

- Ensure that all hostnames and network names are in lowercase.
- Management virtual machine hostname values must NOT be fully qualified (FQDN).
- ClusterReserved and ServiceReserved Pools are left at default if not used.

- RoutableIPPoolCIDR requires five unused contiguous IP addresses in each range.
- The PFMPHostname value must be fully qualified (FQDN).
- PFMPHostIP value must be in the flex-node-mgmt-<vlanid> range.

6

Configuring PowerFlex Manager

After installing the PowerFlex management platform, perform the tasks in this section to complete the initial PowerFlex Manager configuration procedures. PowerFlex Manager is the web user interface for the PowerFlex management platform.

After you configure PowerFlex Manager, the remaining PowerFlex installation and configuration tasks are performed using PowerFlex Manager.

Log in to PowerFlex Manager

Perform the following steps to log in to PowerFlex Manager for the first time in order to continue configuration activities and deploy storage resources.

Prerequisites

- Ensure that you have access to a web browser that has network connectivity with PowerFlex Manager.
- Ensure that you know the address that was configured for accessing PowerFlex Manager. This address was configured for `PFMPHostname` in the JSON file during PowerFlex Manager installation.
- Prepare a new password for accessing PowerFlex Manager. The default password cannot be considered as a new password. For more information about default credentials, see the *Dell PowerFlex Family 4.x Security Configuration Guide*.

  Password rules are as follows:

  - Fewer than 32 characters
  - Only alphanumeric and punctuation characters

NOTE: If you are using **Windows Active Directory (AD)** as an **LDAP server**, you must NOT use hostnames (NetBIOS names) longer than **15** characters. Hostnames exceeding this limit may cause issues during deployment and integration.

Steps

1. Log in to PowerFlex Manager using the default credentials. The default credentials are provided in the *Dell PowerFlex Family 4.x Security Configuration Guide*.
2. Change the default password.

   Make a note of the password for future use. This password is for the SuperUser who is performing the initial configuration activities. Additional users and passwords can be configured later in the process.
3. Click **Submit**.
   You are now logged into the system. The initial configuration wizard is displayed. On the **Welcome** page, read the instructions, and click **Next** to proceed with initial configuration activities.

Perform the initial setup

The first time you log in to PowerFlex Manager, you are prompted with an **Initial Configuration** wizard, which prompts you to configure the basic settings that are required to start using PowerFlex Manager.

Before you begin, have the following information available:

- Dell Technologies connectivity services configuration details
- Information about whether you intend to use a compliance file
- Information about the type of installation you want to perform, including details about your existing PowerFlex instance, if you intend to import from another PowerFlex instance

To configure the basic settings:

1. On the **Welcome** page, read the instructions and click **Next**.
2. On the **SupportAssist** page, optionally enable Dell Technologies connectivity services and specify Dell Technologies connectivity services connection settings, and click **Next**.

NOTE: If you want to configure secure connect gateway and policy manager, complete *Enabling Dell Technologies connectivity services* before proceeding as enabling Dell Technologies connectivity services is an optional procedure and only applicable if you opt for it within your service agreement.

3. On the **Installation Type** page, specify whether you want to deploy a new instance of PowerFlex or import an existing instance, and click **Next**.
4. On the **Summary** page, verify all settings for Dell Technologies connectivity services, compliance, and installation type. Click **Finish** to complete the initial setup.

After completing the initial setup, you can begin to configure PowerFlex Manager and deploy resources from the **Getting Started** page.

Deploying and configuring Dell Technologies connectivity services for the customer-provided KVM environment

There are two options to configure events and alerts:

- Connect directly
- Connect using secure connect gateway
- If you connect directly, only the call home option is available
- If you connect through secure connect gateway, all options through secure connect gateway are enabled

Deploy and configure secure connect gateway and policy manager using QCOW2 images

Use this procedure to deploy and configure secure connect gateway and policy manager using QCOW2 images.

Steps

1. Download the policy manager and secure connect gateway Hyper-V VHD files from the Dell Technologies support site.
2. Log in to the KVM host as **delladmin**.
3. Download the VHD files to the KVM host (to `/tmp`).
4. Unzip the secure connect gateway VHD image, type: **unzip <filename>**
5. Convert the VHD image to a QCOW2 image, type: **qemu-img convert -f vpc -O qcow2 <filename> scgvm.qcow2**
6. Unzip the POLICYMGR VHD image, type: **unzip <filename>**
7. Convert the VHD image to a QCOW2 image, type: **qemu-img convert -f vpc -O qcow2 <filename> policymgrvm.qcow2**
8. Move the resulting QCOW2 images to `/mnt/general`, type: **mv *.qcow2 /mnt/general**
9. Create VMs, type: **virt-install**:

```
sudo virt-install --name scgvm --ram 16384 --disk /mnt/general/scgvm.qcow2 --import
--vcpus=4 --network bridge=bridge-<vlan>,model=virtio --osinfo sle15 --noautoconsole
--print-xml --graphics vnc,listen=<PMC host mgmt IP>,port=5901 > /tmp/scgvm.xml

sudo virt-install --name policymgr --ram 16384 --disk /mnt/general/policymgrvm.qcow2
--import --vcpus=4 --network bridge=bridge-<vlan>,model=virtio --osinfo sle15 --
noautoconsole --print-xml --graphics vnc,listen=<PMC host mgmt IP>,port=5902 > /tmp/
scgvm.xml
```

10. Add crm configuration to instantiate both VMs as cluster objects, type:

```
sudo crm configure primitive scgvm VirtualDomain params config="/mnt/general/
scgvm.xml" remoteuri="qemu+ssh://hacluster@%n/system" start_resource=true meta allow-
migrate=true target-role=Stopped op monitor interval=10s timeout=30s
```

```
sudo crm configure colocation co-general-scgvm info: scgvm cl-sharedfs-general

sudo crm configure order or-general-scgvm Mandatory: cl-sharedfs-general scgvm

sudo crm configure primitive policymgrvm VirtualDomain params config="/mnt/general/
policymgrvm.xml" remoteuri="qemu+ssh://hacluster@%n/system" start_resource=true meta
allow-migrate=true target-role=Stopped op monitor interval=10s timeout=30s

sudo crm configure colocation co-general-policymgrvm info: policymgrvm cl-sharedfs-
general

sudo crm configure order or-general-policymgrvm Mandatory: cl-sharedfs-general
policymgrvm
```

11. Start the resources, type: **sudo crm resource start scgvm policymgrvm**
12. Temporarily add the firewall rule to host to allow the VNC communication, type: **sudo firewall-cmd --add-port=5901/tcp --add-port=5902/tcp**
13. Use the VNCviewer from the jump server (or laptop connected to service port) to connect to the console of scgvm, see Secure Connect Gateway 5.x — Virtual Edition Deployment Guide for the secure connect gateway and policy manager deployment procedures.
14. Clean up the firewall, type: **sudo firewall-cmd --remove-port=5901/tcp --remove-port=5902/tcp**
15. Clean up the downloaded files (`zip` and `VHD` files), type: **rm ~/POLICYMGR_* ; rm ~/SCG_***

Configure secure connect gateway and policy manager

Use this procedure if you need to configure secure connect gateway and policy manager.

Steps

1. Connect to jump server and launch a console.
2. In a terminal on the jump server, type the following: **virt-viewer -a -c qemu+ssh://delladmin@<KVM Host>/system**.

   A window displays listing the VMs running on the host.
3. Select the secure connect gateway / policy manager VM to launch the console.
4. Agree terms, type **ALT** + **A** and type **ALT** + **N** to go next.
5. Edit the network settings, type **ALT** + **I**.
   a. Provide the IP address and subnet mask, type **ALT** + **N**.
   b. Provide the hostname, DNS and domain search, type **ALT** + **S**, and type **ALT** + **N** to go next.
   c. Provide routing, type **ALT** + **U**, and to add route, type **ALT** + **D**.
   d. Provide the gateway and select device as eth0, **ALT** + **O** and type **ALT** + **N** to go next.
   e. Select time zone and type **ALT** + **N** to go next.
   f. Provide the root password and type **ALT** + **N** to go next.
   g. Maintain SSL as default, and type **ALT** + **N** to go next.

      NOTE: This step is only for Policy Manager (PM).

   h. Maintain the user admin as default and type **ALT** + **N** to go next.
6. Wait for the configuration to complete. Once the configuration is complete access web UI for further configuration.
7. Repeat steps 3 to 6 to configure policy manager.

Register and log in to the secure connect gateway user interface

Use this procedure to log in to the secure connect gateway user interface.

Steps

To log in to the secure connect gateway, see *Registering and signing in to Secure Connect Gateway* in the Secure Connect Gateway 5.x — Virtual Edition Deployment Guide.

Ensure you complete the following procedures:

- *Sign in to Secure Connect Gateway*
- *Register Secure Connect Gateway*
- *Sign in using administrator account or network credentials*

Configure policy manager settings on secure connect gateway

Policy manager for secure connect gateway is a device access management technology that is delivered as a virtual appliance. Policy manager for secure connect gateway can be deployed on a hypervisor.

About this task

You can configure policy manager for secure connect gateway to perform the following tasks:

- Control remote access to your devices.
- Maintain an audit log of remote connections and file transfers.
- Access administration actions performed on policy manager for secure connect gateway.

NOTE: Policy manager is not managed by PowerFlex Manager.

Steps

1. Sign in to secure connect gateway.
2. Go to **Settings** > **Environment configuration** > **Connectivity details** > **Policy Manager**.
3. Click **Enable remote Policy manager**.
4. Enter the hostname or IP address, port number, username, and password of the policy manager server.

   NOTE: If the port is SSL secured, the port number must be **8443**. If the port is not SSL secured, the port number must be **8888**.

5. Click **Enable SSL** if policy manager for secure connect gateway is installed on a server that is secured by SSL.
6. If the policy manager server connects to the Internet through a proxy server, perform the following steps in the **Customer proxy server** section:
   a. Click **Enable proxy server for Policy manager only**.
   b. Enter the hostname or IP address and port number.
   c. Click **Proxy requires authentication** and enter the username and password to access the proxy server.
7. Test the connection and click **Apply**.

Enabling Dell Technologies connectivity services for VMware ESXi

There are two options to configure events and alerts:

- Connect directly
- Connect using secure connect gateway
- If you connect directly, only the call home option is available
- If you connect through secure connect gateway, all options through secure connect gateway are enabled

Deploy and configure secure connect gateway

Secure connect gateway is an enterprise monitoring technology that monitors your devices and proactively detects hardware issues that may occur.

Prerequisites

Ensure your system complies to the minimum requirements, for more information, see the *Secure Connect Gateway 5.x - Virtual Edition Deployment Guide*.

Steps

Use the PowerFlex management platform deployment wizard. See the help document in the deployment wizard for more information.

Register and log in to the secure connect gateway user interface

Use this procedure to log in to the secure connect gateway user interface.

Steps

To log in to the secure connect gateway, see *Registering and signing in to Secure Connect Gateway* in the Secure Connect Gateway 5.x — Virtual Edition Deployment Guide.

Ensure you complete the following procedures:

- *Sign in to Secure Connect Gateway*
- *Register Secure Connect Gateway*
- *Sign in using administrator account or network credentials*

Deploy policy manager for secure connect gateway

Use this procedure to deploy policy manager for secure connect gateway.

About this task

NOTE: Policy manager for secure connect gateway is optional.

PowerFlex Manager does not manage policies or policy manager for secure connect gateway.

This chapter does not cover procedures to create and manage policies. For information on creating and managing policies, see *Policy Manager for Secure Connect Gateway 5.00.00.xy User's Guide*.

Prerequisites

Before deploying policy manager for secure connect gateway, ensure the following:

- Upgrade secure connect gateway
- Download the latest version of policy manager for secure connect gateway from Dell Technologies support site.

Steps

1. Use the PowerFlex management platform deployment wizard to deploy policy manager. See the help document in the deployment wizard for more information.
2. Configure policy manager using the following steps:
   a. Go to **`https://<hostname(FQDN) or IP address>:8443`**
   b. Enter admin as the user ID.
   c. If you are logging in for the first time, enter the default password.
   d. After logging in for the first time, enter the new administrator account password when prompted and click **Save**.
   e. Enter the new password and click **Log In**.

Configure policy manager settings on secure connect gateway

Policy manager for secure connect gateway is a device access management technology that is delivered as a virtual appliance. Policy manager for secure connect gateway can be deployed on a hypervisor.

About this task

You can configure policy manager for secure connect gateway to perform the following tasks:

- Control remote access to your devices.
- Maintain an audit log of remote connections and file transfers.
- Access administration actions performed on policy manager for secure connect gateway.

NOTE: Policy manager is not managed by PowerFlex Manager.

Steps

1. Sign in to secure connect gateway.
2. Go to **Settings** > **Environment configuration** > **Connectivity details** > **Policy Manager**.
3. Click **Enable remote Policy manager**.
4. Enter the hostname or IP address, port number, username, and password of the policy manager server.

   NOTE: If the port is SSL secured, the port number must be **8443**. If the port is not SSL secured, the port number must be **8888**.

5. Click **Enable SSL** if policy manager for secure connect gateway is installed on a server that is secured by SSL.
6. If the policy manager server connects to the Internet through a proxy server, perform the following steps in the **Customer proxy server** section:
   a. Click **Enable proxy server for Policy manager only**.
   b. Enter the hostname or IP address and port number.
   c. Click **Proxy requires authentication** and enter the username and password to access the proxy server.
7. Test the connection and click **Apply**.

Enabling Dell Technologies connectivity services

Dell Technologies connectivity services is a secure support technology for the data center. Dell Technologies connectivity services connects to secure connect gateway, which is used for call home functionality and remote connectivity. You can enable Dell Technologies connectivity services as part of the initial configuration wizard. Alternatively, you can enable it later by adding it as a destination to a notification policy in Events and Alerts.

About this task

PowerFlex Manager provides support through integration with the secure connect gateway, ensuring better alignment with Dell Technologies services initiatives to enhance the user experience. Dell Technologies connectivity services is the functionality that enables this connection.

When you enable Dell Technologies connectivity services, you can take advantage of the following benefits, depending on the service agreement on your device:

- Automated issue detection - Dell Technologies connectivity services monitors your Dell Technologies devices and automatically detects hardware issues, both proactively and predictively.
- Automated case creation - When an issue is detected, Dell Technologies connectivity services automatically opens a support case with Dell Technologies Support.
- Automated diagnostic collection - Dell Technologies connectivity services automatically collects system state information from your devices and uploads it securely to Dell Technologies. Dell Technologies Support uses this information to troubleshoot the issue.
- Proactive contact - A Dell Technologies Support agent contacts you about the support case and helps you resolve the issue.

The first time that you access the initial configuration wizard, the connection status displays as not configured.

If you do not enable Dell Technologies connectivity services in the initial setup, you can add it later as a destination when adding a notification policy through **Settings** > **Events and Alerts** > **Notification Policies**.

Prerequisites

Ensure you have the details about your Dell Technologies connectivity services configuration.

Steps

1. Click **Enable Dell Technologies connectivity services**.
2. From the **Connection Type** tab, there are two options:

If you want to:	Do the following:
Connect directly to Dell Technologies.	Click **Connect Directly**.
Connect to a Dell Technologies connectivity services gateway server that connects to Dell Technologies by remote access.	Click **Connect via Gateway Server** and provide the following details: • The IP and port number that is assigned to the primary and secondary gateway servers. The port number is auto populated to **9443**. • Optionally, click **Add Gateway** to include an additional gateway server. The maximum number of gateways is eight. • If your environment has a proxy server, enter the IP address, port number, username, and password for the proxy server. • Enter the access key and PIN for authentication to Dell Technologies connectivity services. • Optionally, click **Test connection** to test the connection status. • Optionally, click **Send Test Alert** to send a test alert.

3. From the **Support Contacts** tab, identify the primary support contact by providing their first name, last name, email, and phone details. Optionally, you can add up to two additional support contacts.
4. From the **Device Registration** tab, register your device by completing the following steps:
   a. Select the type of device registration from the **Type** menu.
   b. Information does not need to be entered for PowerFlex software.
5. From the **Customize Connection to Dell Technologies connectivity services** tab, you can:
   a. Click **Connect to CloudIQ** to enable Dell Technologies to send telemetry data, alerts, and analytics through Dell Technologies connectivity services.
   b. Click **Enable Remote Support** to enable authorized Dell Technologies support engineers to troubleshoot your system remotely.
6. From the **Re-authenticate Dell Technologies connectivity services** tab, click **request a New Access Key and PIN** to generate an access key and pin. The **AccessKey Portal** opens.
   a. From the menu, select **SERIALNUMBER**.
   b. Enter the serial number of your device or PowerFlex software and click **Submit**.
   c. Click **Generate New Access key**.
7. From the **Re-authenticate Dell Technologies connectivity services** tab, enter the access key and pin.
8. Click **Next**.

Next steps

After you complete the initial configuration, go to the **Settings** page and verify that Dell Technologies connectivity services has been turned on. You may need to configure the Dell Technologies connectivity services policy explicitly on the **Settings** > **Events and Alerts** > **Notification Policies** page.

Getting started

The **Getting Started** page guides you through the common configurations that are required to prepare a new PowerFlex Manager environment. A green check mark on a step indicates that you have completed the step. Only superusers have access to the **Getting Started** page.

If you would like to use the legacy PowerFlex Installer to provide an installation topology file, click **Deploy With Installation File**. For more information, see the *Dell PowerFlex and PowerFlex Manager Software Install Guide*.

The following table lists the configuration options available in the **Getting Started** page. Not all configuration options are required for all system types as noted.

Table 25. Common configurations options

Step	Description
Add compliance file	Provide compliance file location and authentication information for use within PowerFlex Manager. The compliance file (`Software_Only_Complete_4.6.x_xx.zip`) defines the specific hardware components and software version combinations that are tested and

Table 25. Common configurations options (continued)

Step	Description
	certified by Dell for hyperconverged infrastructure and other Dell products. This step enables you to choose a default compliance version for compliance or add new compliance versions. You can also click **Settings** > **Repositories** > **Compliance Versions**. Select **Default Version** and choose a version from the drop-down list to set it as the default for compliance. Before you make the software catalog the default compliance version, upload a suitable compatibility matrix file (`cm-xxxxxxxx.gpg`), under **Settings** > **Repositories** > **Compatibility Management**. If you have more than one compliance version, set one of them as default. For more information, see the *Add a compliance file* topic in related information.
Define networks	Enter detailed information about the available networks in the environment. This information is used later during deployments to configure nodes to have the right network connectivity. PowerFlex Manager uses the defined networks in templates to specify the networks or VLANs that are configured on PowerFlex nodes for your resource groups. This step is enabled immediately after you perform an initial configuration for PowerFlex Manager. You can also click **Settings** > **Networking** > **Networks**. **NOTE:** This step is not applicable for co-resident deployments.
Discover resources	Grant PowerFlex Manager access to resources (nodes) in the environment by providing the management IP address and credential for the resources to be discovered. You can also click **Resources** > **Discover Resources**. **NOTE:** This step is not applicable for co-resident deployments.
Manage deployed resources (Optional)	Add an existing resource group for a cluster that is already deployed and manage the resources within PowerFlex Manager. This step is not enabled until you define your networks. You can also click **Lifecycle** > **Resource Groups** > **Add Existing Resource Group**. **NOTE:** This step is not applicable for co-resident deployments.
Deploy resources	Create a template with requirements that must be followed during a deployment. Templates enable you to automate the process of configuring and deploying infrastructure and workloads. For most environments, you can clone one of the sample templates that are provided with PowerFlex Manager and make modifications as needed. Choose the sample template that is most appropriate for your environment. For example, for a PowerFlex hyperconverged deployment, clone one of the PowerFlex hyperconverged templates. For a PowerFlex two-layer deployment, clone the compute-only templates. Then clone one of the storage templates. This step is not enabled until you define your networks. You can also click **Lifecycle** > **Templates**. **NOTE:** This step is not applicable for co-resident deployments.

Table 25. Common configurations options (continued)

Step	Description
Manage PowerFlex license	Configure licensing for PowerFlex. You can also click **Settings** > **License Management**. For information on obtaining and activating the PowerFlex capacity license, see Dell PowerFlex Family Licensing Guide.

To go to the **Getting Started** page, click **Getting Started** on the help menu.

Configuring license management

PowerFlex Manager comes by default with a trial or evaluation license for first 90 days. There is no requirement to upload an evaluation license. After 90 days, the license is enforced or it displays an error message and no operation can be performed.

The following licensing are supported:

- Dynamic licensing: Provides an ability to manage license allocations more easily across product registrations, enabling a flexible and keyless experience. It automates and simplifies license management activities such as, registrations, capacity expansions, and renewals. Dynamic licensing is applicable for all systems with PowerFlex Manager 5.1.x and later.

  The following methods are used to activate and use your products with dynamic licensing:
  - Connected method
  - Offline method

- Key-based format licensing: Using this licensing, you can upload a license in PowerFlex Manager. The license is a single file that includes both PowerFlex Manager and PowerFlex licensing information. Once the license capacity is exceeded, you must upload a new license with more capacity.

For information on registering or migrating licenses, see *Dell PowerFlex Family Licensing Guide*.

Add a compliance file

A compliance file must be in an available state for PowerFlex Manager to add the file.

Steps

1. From the **Getting Started** page, click **Upload Compliance File**.

   Alternatively, click **Settings** > **Repositories**. On the **Repositories** page, select the **Compliance Versions** tab, and click **Add**.

2. In the **Add Compliance File** dialog, select one of the following options:
   - **Download from Dell Technologies connectivity services (Recommended)**—Select this option to import the compliance file that contains the firmware bundles you need.
     - **Download from local network path**—Select this option to download the compliance file from a CIFS file share. This option is intended for sites that do not have Internet accessibility to Dell Technologies connectivity services.
3. If you selected **Download from Dell Technologies connectivity services (Recommended)**, click the **Available Compliance Files** drop-down list, and select the file.

   Before downloading a compliance file from Dell Technologies connectivity services, you must configure Dell Technologies connectivity services. To configure Dell Technologies connectivity services , you must enable Dell Technologies connectivity services in the **Initial Configuration** wizard.

   If you are downloading a compliance file from Dell Technologies connectivity services, the file type is a ZIP or TAR.GZ file.

   If you are downloading a compliance file from the Dell Technologies support site, the file will be called `Software_Only_Complete_4.5.x_xxx.zip`

   NOTE: The XML file type is not supported as a download option.

4. If you selected **Download from local network path**, perform the following:

- In the **File Path** box, enter the location of the compliance file. Use the following format:
  - CIFS share for ZIP file: **\\host\share\filename.zip**
  - CIFS share for TAR.GZ file: **\\host\share\filename.tar.gz**
- If you are using a CIFS share, enter the **User Name** and **Password**.
- Mark the target upgrade catalog as the default version for compliance checking.

Download the 3.6.x and the 4.5.x software catalogs. PowerFlex requires the catalog for the version that the system is on and the target version.

5. Click **Save**.

   PowerFlex Manager unpacks the compliance file as a repository that includes the firmware and software bundles that are required to perform any updates.

   If the compliance file contains an embedded ISO file for an operating system image (such as for a VMware ESXi or PowerFlex image), the download process unpacks the file and adds it to the operating system image repository.

   A compliance file can be large and take some time to download and unpack. To help you monitor the progress of the downloading and unpacking operations, PowerFlex Manager provides percentage complete information for these operations in the **State** column on the **Compliance Versions** tab.

Upload the compatibility matrix file

To facilitate upgrades, PowerFlex Manager uses information that is provided in the compatibility matrix file to determine which upgrade paths are valid and which are not.

About this task

When you attempt an upgrade, PowerFlex Manager warns you if the current version of the software is incompatible with the target version, or if the software catalog versions that are currently loaded are incompatible with the target compliance versions. To determine which paths are valid and which are not, PowerFlex Manager uses information that is provided in the compatibility matrix file. The compatibility matrix file maps all the known valid and invalid paths for all previous releases of the software.

When you first install PowerFlex Manager, the software does not have the compatibility matrix file, so you must upload the file before performing any updates. You must upload the latest compatibility matrix file to ensure that PowerFlex Manager has access to the latest upgrade information.

You can download the file from Dell Technologies connectivity services or upload it from a local directory path. The compatibility matrix file (`cm-xxxxxxxx.gpg`) can be found in the `Software_Only_Complete_4.5.x_xxx.zip` file. The file has a GPG extension and an associated compatibility version number.

Prerequisites

If you are uploading from a local directory path, ensure that you have access to a valid compatibility matrix file that has the GPG extension. If you are using Dell Technologies connectivity services, ensure that Dell Technologies connectivity services has been properly configured.

Steps

1. On the menu bar, click **Settings** and click **Compatibility Management**.
2. Click **Edit Settings**.
3. Click **Download from Dell Technologies connectivity services** if you are using Dell Technologies connectivity services.
4. Click **Upload from Local** to use a local file, then click **Choose File** to select the GPG file.

7

Deploying PowerFlex storage

The next stage of installation involves defining infrastructure and resources, and then deploying PowerFlex storage.

Deploy the PowerFlex data storage as a resource group by using one of the following methods:

- Deploying a PowerFlex cluster using a CSV topology file from PowerFlex Manager.
- Deploying a PowerFlex cluster using PowerFlex Manager templates and resource groups (for PowerFlex file deployments only).
- Deploy a PowerFlex cluster using REST API commands.

NOTE: This is an advanced deployment method for deploying block storage. If you use this method, ensure that you are familiar with setting up the CSV file, as described in detail in this guide.

Deployment design for co-residency management

The following deployment design principles are recommended:

- If you have a multirack system, place the MDM nodes and the PowerFlex management platform components on different racks.
- If the hardware requirements are met, any node running SDS can also optionally run an SDR, SDT.
- Ensure that the LIA is installed on every node, as it is required for maintenance purposes.

PowerFlex component layout

The following table details best practices in relation to component layout for a co-resident deployment, based on the number of storage nodes you are deploying:

Table 26. PowerFlex component layout

Number of nodes	Components installed on the node						
	Node 1	Node 2	Node 3	Node 4	Node 5	Node 6	Node 7
Four nodes	MDM, SDS	MDM, SDS, SDC, PowerFlex management platform	Tiebreaker, SDS, SDC, PowerFlex management platform	SDS, SDC, PowerFlex management platform	N/A	N/A	N/A
Five nodes	MDM, SDS	MDM, SDS	MDM, SDS, SDC, PowerFlex management platform	Tiebreaker, SDS, SDC, PowerFlex management platform	Tiebreaker, SDS, SDC, PowerFlex management platform	N/A	N/A
Six nodes	MDM, SDS	MDM, SDS	MDM, SDS	Tiebreaker, SDS, SDC, PowerFlex management platform	Tiebreaker, SDS, SDC, PowerFlex management platform	SDS, SDC, PowerFlex management platform	N/A
Seven nodes and greater	MDM, SDS	MDM, SDS	MDM, SDS	Tiebreaker, SDS	Tiebreaker, SDS, SDC, PowerFlex management platform	SDS, SDC, PowerFlex management platform	SDS, SDC, PowerFlex management platform

NOTE: For systems with additional storage nodes, separate the PowerFlex management platform services from the MDM roles, if possible.

The following table describes the PowerFlex MDM certificates that must be installed on the participating nodes for SUSE Linux Enterprise Server before starting a co-resident deployment.

Table 27. PowerFlex MDM certificates for SUSE Linux Enterprise Server

For PowerFlex MDM certificates	Location	Repository
pycparser-2.21-py2.py3 or higher	Third-party	Python Package Index
cffi-1.15.1 or higher	Third-party	Python Package Index
cryptography-40.0.2 or higher	Third-party	Python Package Index

Download the appropriate versions of pycparser, cffi and cryptography for your operating system release and Python version. SLES 15.4 and above comes with these modules installed.

The following table describes the PowerFlex MDM certificates that must be installed on the participating nodes for Red Hat Enterprise Linux before starting a co-resident deployment.

Table 28. PowerFlex MDM certificates for Red Hat Enterprise Linux

For PowerFlex MDM certificates	Location	Repository
pycparser-2.21-py2.py3 or higher	Third-party	Python Package Index
cffi-1.15.1 or higher	Third-party	Python Package Index
cryptography-40.0.2 or higher	Third-party	Python Package Index

Download the appropriate version of pip, pycparser, cffi and cryptography for your operating system release and Python version. Python v3 and Python3-cryptography includes the above mentioned modules and can be used to run the MDM certificate generation script. RedHat 9 and above will already include them.

Operating system requirements

The following table lists the required packages for each operating system for PowerFlex 4.6.x. A valid time source should be configured on all nodes before deploying PowerFlex.

For the most updated list of operating systems supported by PowerFlex, see *Dell PowerFlex 4.6.x Support Matrix* on the Dell Technologies support site for the supported versions.

Linux

Table 29. Linux requirements

Component or feature	Required packages
All components, all Linux types	• numactl • libaio • wget • apr • libapr • python2-rpm or python3-rpm • yum-utils
MDM	• bash-completion (for SCLI completion) • Latest version of Python 2.x or Python 3.x (for newer operating system such as Red Hat Enterprise Linux 8.x) • binutils • java-11-openjdk-headless
Storage data server	• smartmontools

Table 29. Linux requirements (continued)

Component or feature	Required packages
	• smartmontoolslibs • binutils • sg3_utils • hdparm • pciutils
Diagnostic tools	• sar • sysstat
Storage controllers	• DELL: perccli • LSI raid controllers: storcli • HP raid controllers: hpssacli
SDS nodes that support SDPMs or NVDIMMs	• ndctl • jq • daxio • libpmem
Secure authentication mode	Ensure that OpenSSL 64-bit v1.1 is installed on all servers in the system: • `libopenssl1_0_0-1.0.1g-0.40.1.x86_64.rpm` • `openssl1-1.0.1g-0.40.1.x86_64.rpm`
LDAP	Ensure that OpenLDAP 2.4 is installed on all servers
LIA	pciutils

VMware ESXi

PowerFlex only supports the SDC component VMware ESXi 7.x or 8.x for compute only nodes (SDC).

Microsoft Windows

PowerFlex only supports the SDC component on Microsoft Windows.

Table 30. Microsoft Windows requirements

Operating system	Requirement
Windows	• Only SDC is a supported component on Windows. • To install SDC, ensure that Security Update for x64-based Systems is installed. To use the secure authentication mode, ensure that these are installed on all servers in the system: • OpenSSL 64-bit v1.0.1 or later (v1.1, however, is not supported) • Visual C++ redistributable 2010 package, 64-bit The Microsoft Windows *Storage Spaces* feature (available from Windows 8.1 and later) is not supported in Microsoft failover cluster running on PowerFlex volumes.

Preparing the storage node

For large-scale systems with up to 2048 storage data client, use the following procedure to prepare the operating systems of servers that host PowerFlex metadata manager, storage data server, and storage data replication components.

Prepare Linux-based server operating systems for support of large-scale systems (up to 2048 SDCs)

For large-scale systems with up to 2048 storage data client, use this procedure to prepare the Linux-based operating systems of servers that host PowerFlex metadata manager, storage data server, and storage data replication components.

Prerequisites

Superuser permissions are required to configure the servers.

About this task

Configure the servers hosting the MDM, SDS, and SDR components to support large-scale systems with up to 2048 SDCs.

- A system setting must be validated and configured on servers hosting MDM, SDS, and SDR.
- For all servers hosting MDM, SDS, and SDR components, certain threshold parameters must be configured. This change enables a larger number of records in the Address Resolution Protocol (ARP) cache table of the operating system.

Steps

1. On servers hosting PowerFlex metadata manager, storage data server, and storage data replication components, do the following:
   a. Run this command:

      ```
      cat /proc/sys/fs/file-max
      ```

      The output for the command is the number of files that the operating system can open.
   b. If the number is 32768 or higher, go to step 2. If the number is less than 32768, go to the next substep.
   c. Open the file `/etc/sysctl.conf` using the vi editor.
   d. If the following line exists, edit it with the vi editor. If it does not exist, add the full line with the following setting:

      ```
      fs.file-max = 32768
      ```

2. On servers hosting MDM, SDS, and SDR components, if the following lines exist in the file `/etc/sysctl.conf`, edit them with the vi editor. If they do not exist, add the full lines with the following settings:

   ```
   net.ipv4.neigh.default.gc_thresh1 = 8192
   net.ipv4.neigh.default.gc_thresh2 = 16384
   net.ipv4.neigh.default.gc_thresh3 = 32768
   ```

3. Save the file.
4. Apply the changes, using the command:

   ```
   sysctl -p
   ```

5. Repeat the steps on the next server, until all servers are configured.

Prepare the DAX devices

If you are using NVDIMM devices for acceleration, perform the following procedures to prepare the NVDIMMs as DAX acceleration devices. This step is not required if you are deploying with PowerFlex Manager as a resource group.

NVDIMM or SDPM information table

Before you prepare the NVDIMM or SDPM as a DAX device, use the following NVDIMM or SDPM information table. Use it to record the information that you discover while preparing an NVDIMM or SDPM as a DAX device. You need the information to add the acceleration device to PowerFlex.

Table 31. NVDIMM or SDPM information table

Item	Information
DAX device name (chardev)	
Acceleration device path	

Prepare an NVDIMM or SDPM as a DAX (acceleration) device

Prepare a new NVDIMM or SDPM as a DAX or acceleration device before adding it to the SDS in a Linux system.

Steps

1. Using SSH, log in to the Linux server.
2. Run this command to create the namespace, and DAX on the new NVDIMM or SDPM. where <x> is 0 or 1. Create the namespace with region0 and then run the same command with region1.

   ```
   ndctl create-namespace --mode=devdax --region=region<X> --align=4k
   ```

   For example,

   ```
   ndctl create-namespace --mode=devdax --region=region1 --align=4k
   {
     "dev":"namespace1.0",
     "mode":"devdax",
     "map":"dev",
     "size":"15.75 GiB (16.91 GB)",
     "uuid":"e47c3ce0-8a98-4256-8fed-ff2bacead155",
     "daxregion":{
       "id":1,
       "size":"15.75 GiB (16.91 GB)",
       "align":4096,
       "devices":[
         {
           "chardev":"dax1.0",
           "size":"15.75 GiB (16.91 GB)"
         }
       ]
     },
     "align":4096
   }
   ```

3. Run this command again to validate the namespace was created. In this example, you can see namespace1.0 and dax1.0 was created for region1 NVDIMM or SDPM.

   ```
   ndctl list --regions --verbose
   {
     "regions":[
       {
         "dev":"region1",
         "size":17179869184,
         "available_size":17179869184,
         "max_available_extent":17179869184,
   ```

```
      "type":"pmem",
      "numa_node":1,
      "iset_id":-4649799281728009192,
      "persistence_domain":"unknown"
    },
  {
    "dev":"namespace1.0",
    "mode":"devdax",
    "map":"dev",
    "size":16909336576,
    "uuid":"e47c3ce0-8a98-4256-8fed-ff2bacead155",
    "chardev":"dax1.0",
    "align":4096
  },
    {
      "dev":"region0",
      "size":17179869184,
      "available_size":0,
      "max_available_extent":0,
      "type":"pmem",
      "numa_node":0,
      "iset_id":4335710697273129127,
      "persistence_domain":"unknown",
      "namespaces":[
        {
          "dev":"namespace0.0",
          "mode":"devdax",
          "map":"dev",
          "size":16909336576,
          "uuid":"265ef925-571d-4a0a-bb96-dca4a3873000",
          "raw_uuid":"ff7b05b9-510e-4604-8329-7f3dccf10300",
          "daxregion":{
            "id":0,
            "size":16909336576,
            "align":4096,
            "devices":[
              {
                "chardev":"dax0.0",
                "size":16909336576
              }
            ]
          },
          "align":4096,
        }
      ]
    }
```


4. Record the acceleration device name in the NVDIMM or SDPM information table.
5. Find the full acceleration device path:

   ```
   ls -ltr /dev/ | grep dax
   ```

   The following output should appear:

   ```
   /dev/daxX.X
   ```

   For example:

   ```
   /dev/dax0.0
   ```

6. Record the acceleration device path in the NVDIMM or SDPM information table.

Deploying PowerFlex using a non-root user

Install or expand PowerFlex in Linux environments using a nonroot sudo user in noninteractive mode. This option is used only when a CSV file is used to install PowerFlex.

Sudo is a program that allows you to run or install a program as the root user. Create a sudo user to install PowerFlex.

To deploy PowerFlex with a nonroot user, the nonroot user must meet the following conditions:

- The username in the CSV file must already exist.
- A nonroot user must be a sudo user.
- Nonroot user must be in noninteractive mode.
- The requirement for TTY must be disabled.

In the CSV file used for installation, you must indicate that you are intending to use a sudo nonroot username by appending the string **(sudo)** to the username in the **Username** field. For example, if you are using a nonroot user with the username *non_root*, enter the string **non_root(sudo)** in the username field of the CSV file.

Configure a non-root non-interactive sudo user

In Linux, you can install or extend PowerFlex with a nonroot user when installing PowerFlex using a CSV file. You must configure a non-root sudo user in non-interactive mode.

About this task

The following procedure describes one method for configuring a nonroot noninteractive sudo user. Perform the commands from the operating system console of the node where PowerFlex will be installed.

Steps

1. Create a user group named **admin**.

```
groupadd admin
```

2. Create a user name **non_root** and add it to the admin group.

```
useradd -G admin non_root
```

3. Change the password of the non-root user.

```
passwd non_root
```

When prompted, enter the new password and then confirm it by entering it again.

4. Open the sudoers `/etc/sudoers` file for editing.

```
vi /etc/sudoers
```

5. Search the sudoers file for **## Same thing without a password**.

```
:s/## Same thing without a password
```

6. In the line below the search result, add the text `%admin ALL=(ALL) NOPASSWD: ALL` to the file.
7. Search the sudoers file for **Defaults requiretty**, and replace it with `Defaults ! requiretty`.
8. Exit the vi editor by typing the following command to exit: `:wq!`
9. Create a hidden directory in the non_root user's home directory to store the SSH configuration.

```
mkdir /home/non_root/.ssh
```

10. Copy the SSH configuration from the root user to the non_root user's directory.

```
cp -rf /root/.ssh/* /home/non_root/.ssh/
```

Deploying a PowerFlex cluster using a CSV topology file

This section describes how to prepare a CSV topology file, upload the CSV topology file using REST API commands or PowerFlex Manager, and then deploy block storage using REST API commands or PowerFlex Manager.

If you are using the legacy PowerFlex to provide an installation topology file using PowerFlex Manager, click **Deploy With Installation File**.

NOTE: This is an advanced deployment method for deploying block storage. If you use this method, ensure that you are familiar with setting up the CSV file, as described in detail in this guide.

Consider the following:

- While deploying PowerFlex using PowerFlex installer, if you see an error indicating that the channel cannot be opened, then verify if PowerFlex is already deployed.
- While deploying additional servers with SDTs on an existing system with components such as SDTs, you may see errors listed in the **Events** tab. These errors are for the already existing components, such as, SDTs in the cluster.
- Ensure that you use the single SSH key for all nodes if you want to deploy a PowerFlex cluster using a CSV topology file.

Prepare the CSV topology file

Create a CSV topology file when a CSV file and REST API commands are used to deploy block storage. Follow the instructions that match your environment and the PowerFlex support matrix for your version. CSV file templates are provided as part of the software download.

Prerequisites

Ensure that the settings you enter comply with PowerFlex product limits. See the *Dell PowerFlex and PowerFlex Manager Technical Overview* for product limits.

Ensure that you have a copy of the following CSV templates. Select a template that is most suitable for your requirements. In this explanation, the CSV file is referred to and illustrated as a spreadsheet.

Table 32. CSV templates

CSV template	Description
`PowerFlex_config_example_all fields.csv` (All fields)	This spreadsheet template contains all available fields, both required and optional. In particular, it contains an optional section for storage pool configuration, which is relevant for multiple nodes.
`PowerFlex_basic_example_minimum fields.csv` (Minimum fields)	This spreadsheet template contains only the required fields. The optional fields (those that are in the all fields spreadsheet, but not in the minimum fields spreadsheet) will be assigned default values.

NOTE: During CSV file preparation, note that there are important differences between hyperconverged and two-layer topologies.

- In hyperconverged topologies, each row in the upper section represents a server that can host MDM, SDS, and SDC components.
- In two-layer topologies where front-end servers are Linux or Windows-based, SDCs must be represented by separate, additional rows. The values for SDC-specific rows are described in *CSV topology for two-layer deployment*. Alternatively, omit the SDCs from the CSV file, and install them manually after deploying the back-end.

NOTE: Save a copy of the CSV file for future use. It is recommended to save the file in a secure encrypted folder on your operating system, after deployment. When you add nodes to the system in the future, you need the CSV file that you used for initial deployment of the system.

You can edit a CSV file with Microsoft Excel or another file-editing software.

Steps

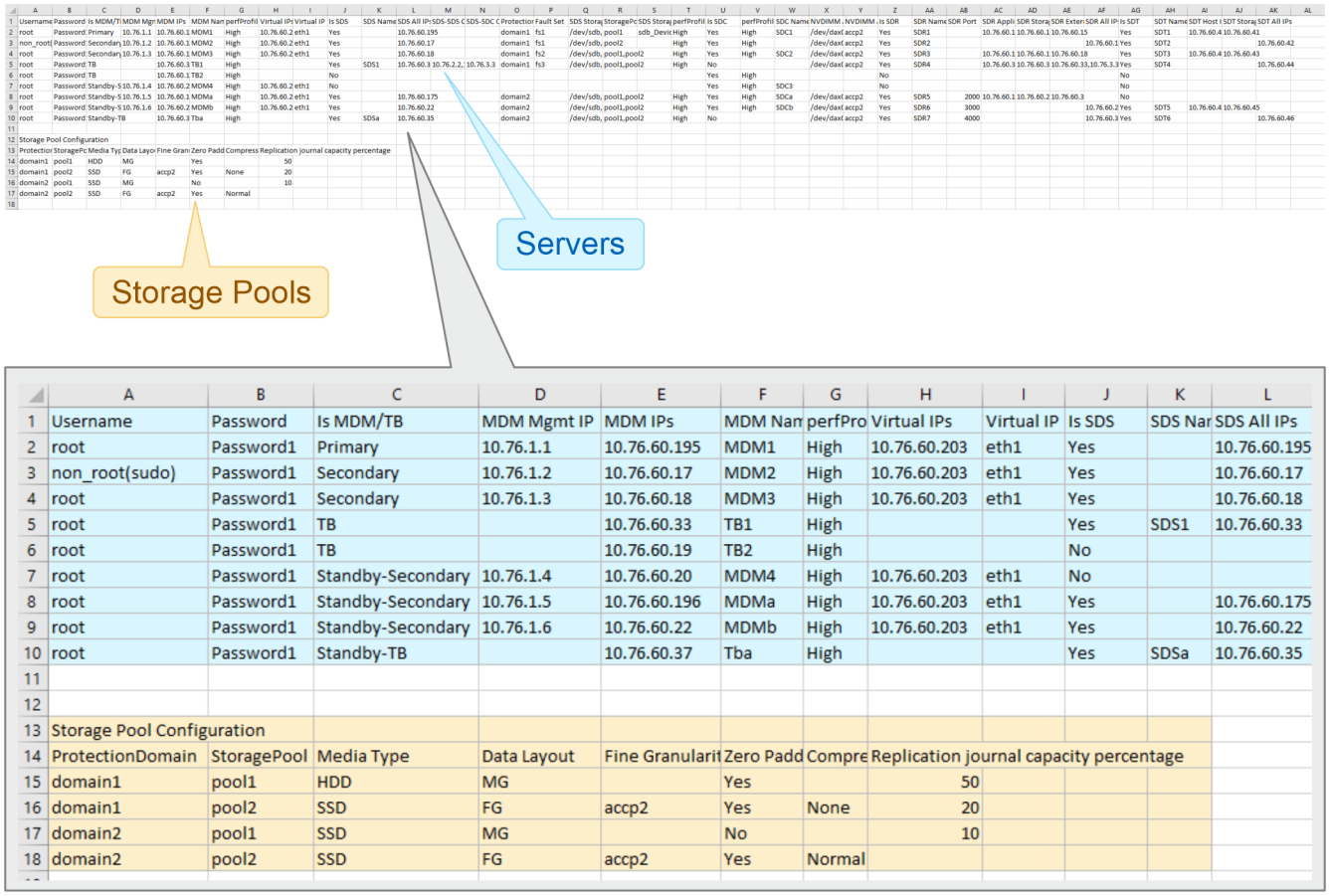
1. In the CSV file, enter your site-specific information in the appropriate places. You can overwrite the default information provided in the file.

   You must only use one spreadsheet for the installation, as follows:

   - To manually enter all configuration details, use the all fields spreadsheet.
   - To use default values for the non-mandatory fields, use the minimum fields spreadsheet.
   - To configure non-default values for columns that are not in the minimum fields spreadsheet, either use the all fields spreadsheet, or copy the column heading from there into the minimum fields spreadsheet, and enter your custom values into the minimum fields spreadsheet.

   NOTE: The examples provided here do not show all possible parameters that are configurable in a PowerFlex system.

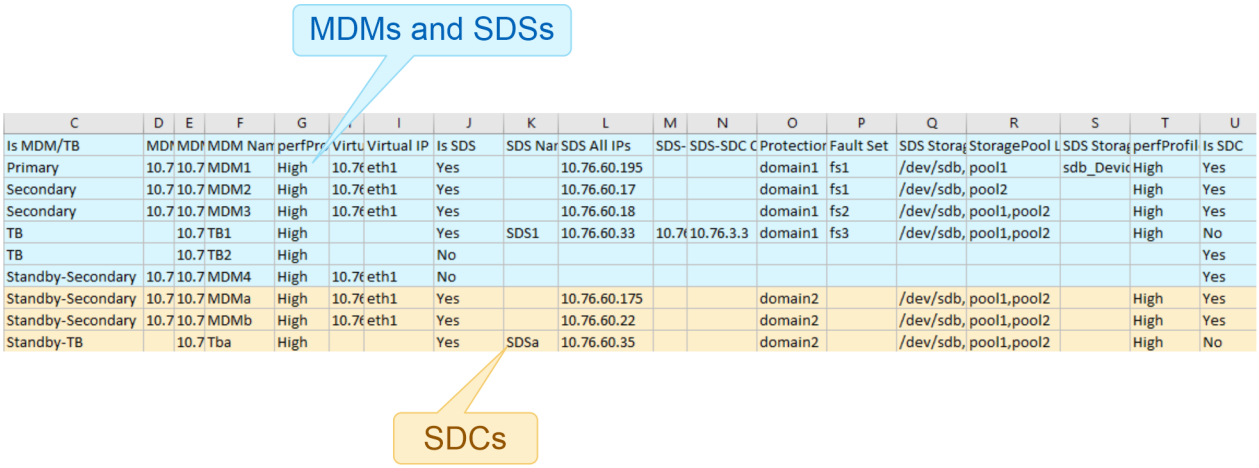
   The following figure shows part of a sample an all fields CSV file used to deploy Linux nodes in a hyperconverged system:



	A	B	C	D	E	F	G	H	I	J	K	L
1	Username	Password	Is MDM/TB	MDM Mgmt IP	MDM IPs	MDM Nam	perfPro	Virtual IPs	Virtual IP	Is SDS	SDS Nar	SDS All IPs
2	root	Password1	Primary	10.76.1.1	10.76.60.195	MDM1	High	10.76.60.203	eth1	Yes		10.76.60.195
3	non_root(sudo)	Password1	Secondary	10.76.1.2	10.76.60.17	MDM2	High	10.76.60.203	eth1	Yes		10.76.60.17
4	root	Password1	Secondary	10.76.1.3	10.76.60.18	MDM3	High	10.76.60.203	eth1	Yes		10.76.60.18
5	root	Password1	TB		10.76.60.33	TB1	High			Yes	SDS1	10.76.60.33
6	root	Password1	TB		10.76.60.19	TB2	High			No		
7	root	Password1	Standby-Secondary	10.76.1.4	10.76.60.20	MDM4	High	10.76.60.203	eth1	No		
8	root	Password1	Standby-Secondary	10.76.1.5	10.76.60.196	MDMa	High	10.76.60.203	eth1	Yes		10.76.60.175
9	root	Password1	Standby-Secondary	10.76.1.6	10.76.60.22	MDMb	High	10.76.60.203	eth1	Yes		10.76.60.22
10	root	Password1	Standby-TB		10.76.60.37	Tba	High			Yes	SDSa	10.76.60.35
11												
12												
13	Storage Pool Configuration											
14	ProtectionDomain	StoragePool	Media Type	Data Layout	Fine Granularit	Zero Padd	Compre	Replication journal capacity percentage				
15	domain1	pool1	HDD	MG		Yes		50				
16	domain1	pool2	SSD	FG	accp2	Yes	None	20				
17	domain2	pool1	SSD	MG		No		10				
18	domain2	pool2	SSD	FG	accp2	Yes	Normal					

Figure 3. Sample CSV file used to deploy Linux nodes in a hyperconverged system

The following figure shows a sample of all fields CSV file, where SDCs are defined below the rows for MDMs and SDSs. In a two-layer configuration. The Linux servers form the MDM cluster, and each of those servers will also be SDSs. SDCs can be added here, in separate rows, or added to the system later.



C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
Is MDM/TB	MDI	MDI	MDM Nan	perfPro	Virtu	Virtual IP	Is SDS	SDS Nar	SDS All IPs	SDS-	SDS-SDC C	Protectior	Fault Set	SDS Storag	StoragePool L	SDS Storag	perfProfil	Is SDC
Primary	10.7	10.7	MDM1	High	10.76	eth1	Yes		10.76.60.195			domain1	fs1	/dev/sdb,	pool1	sdb_Devic	High	Yes
Secondary	10.7	10.7	MDM2	High	10.76	eth1	Yes		10.76.60.17			domain1	fs1	/dev/sdb,	pool2		High	Yes
Secondary	10.7	10.7	MDM3	High	10.76	eth1	Yes		10.76.60.18			domain1	fs2	/dev/sdb,	pool1,pool2		High	Yes
TB		10.7	TB1	High			Yes	SDS1	10.76.60.33	10.76	10.76.3.3	domain1	fs3	/dev/sdb,	pool1,pool2		High	No
TB		10.7	TB2	High			No											Yes
Standby-Secondary	10.7	10.7	MDM4	High	10.76	eth1	No											Yes
Standby-Secondary	10.7	10.7	MDMa	High	10.76	eth1	Yes		10.76.60.175			domain2		/dev/sdb,	pool1,pool2		High	Yes
Standby-Secondary	10.7	10.7	MDMb	High	10.76	eth1	Yes		10.76.60.22			domain2		/dev/sdb,	pool1,pool2		High	Yes
Standby-TB		10.7	Tba	High			Yes	SDSa	10.76.60.35			domain2		/dev/sdb,	pool1,pool2		High	No

Figure 4. SDCs defined in a sample CSV file

2. For Linux nodes, the **Domain** column is not relevant. Leave the column blank or remove the column.

   The following table describes the fields in the spreadsheets. The required fields appear in both all fields and minimum fields spreadsheets. Field names are case-sensitive; the order of the columns is not significant.

Table 33. Fields in the CSV file

Field	Description	Required
Domain	If using a domain user, the name of the domain (not relevant for Linux)	-
Username	The name of the user. • Linux nodes: Enter either `root` or a non-root sudo user, indicated by appending **`(sudo)`** to the end of the username (default: `root`). For more information about deploying with a non-root user, see *Deploying PowerFlex using a non-root user*. • Windows nodes: Enter a user with administrator rights (default: `administrator`).	-
Password	Password used to log in to the node. This should be the password of the user entered in the **Username** column.	Yes
IPs	IP address to be used for multiple purposes. Use this field to designate one IP address that will be assigned to all the following: MDM IP, MDM Mgmt IP and SDS All IP. This option is provided for use cases where separate networks for data and management are not required.	-
Is MDM/TB	The MDM role to deploy on this node: `Primary`, `Secondary`, `TB`, `Standby-Secondary`, `Standby-TB`, or blank (if not an MDM). For more information, see the *Dell PowerFlex and PowerFlex Manager Technical Overview* for information about the MDM cluster.	Yes
MDM Mgmt IP	The IP address for the management-only network. The management IP address is not required for: tiebreaker, standby tiebreaker, and any host that is not an MDM. In such cases, leave this field blank.	-
MDM IPs	MDM IP addresses used to communicate with other PowerFlex components in the storage network. This is required for all MDMs, tiebreakers, and standbys. Leave this field blank for hosts that are not part of the MDM cluster. Maximum of eight addresses, comma-separated, no spaces.	-
MDM Name	The name to be assigned to an MDM node, up to 31 characters in length. MDM names must be unique and may not contain spaces. Ensure that each server on which MDM is installed has a unique hostname.	-

Table 33. Fields in the CSV file (continued)

Field	Description	Required
perfProfileForMDM	Optional performance profile to set for MDMs: `High` (default) or `Compact`. When this field is left empty, the default option is applied. High performance is required for the fine granularity data layout feature.	-
Virtual IPs	A virtual IP address (VIP) for each possible manager MDM. This virtual IP address can be used for communication between the MDM cluster and SDCs on the PowerFlex data networks (not applicable for the management network). Only one virtual IP address can be mapped to each NIC, with a maximum of four virtual IP addresses per system.	-
Virtual IP NICs	The NIC to which the virtual IP addresses are mapped. Maximum length of 11 characters.	-
Is SDS	Deploy the SDS component on this node: `Yes` or `No`	Yes
SDS Name	The name for the SDS node, up to 31 characters in length. The name must not contain spaces. Ensure that each server on which SDS is installed has a unique hostname.	-
SDS All IPs	SDS IP addresses to be used for communication among all nodes. Maximum of eight addresses, comma-separated, no spaces. **NOTE:** When using this option, all roles use the same IP addresses, and there is no need to fill the SDS\SDC only columns.	-
SDS-SDS Only IPs	SDS IP addresses to be used for communication among SDS nodes. When the replication feature is used, these addresses are also used for SDS-SDR and SDT-SDT communication. Maximum of eight addresses, comma-separated, no spaces.	-
SDS-SDC Only IPs	SDS IP addresses to be used for communication among SDS and SDC nodes only. Maximum of eight addresses, comma-separated, no spaces. For SDC-only nodes, enter the IP address in this column.	-
Protection Domain	The protection domain to which to assign this SDS or SDR. The name can be up to 31 characters in length.	-
Fault set	The fault set for an SDS. **NOTE:** When using fault sets, you must assign a minimum of three. See the *Dell PowerFlex and PowerFlex Manager Technical Overview* for guidelines on fault sets.	-

SDS Storage Device List	Storage devices to be added to an SDS. For more than one device, use a comma-separated list, with no spaces. Ensure that the device names and paths do not contain any special characters, such as: `?, ` `, !, @, #`
	See the *Dell PowerFlex and PowerFlex Manager Technical Overview* to configure direct attached storage (DAS).
	Device name format on Linux:
	`/dev/sdb,/dev/sdc`
	When specifying the SDS device path on a Linux node, use the path according to how it is listed in `cat/proc/partitions` (and not according to the output of `fdisk -l`). Use persistent device names.
	For example:
	`fdisk` output: `/dev/mapper/samah-lv1, /dev/sdb`
	`cat /proc/partitions` output: `dm-3, sdb`
	Use these values in the CSV file: `/dev/dm-3, /dev/sdb`
		-

Table 33. Fields in the CSV file (continued)

Field	Description	Required
	To enable volume creation, you must add (at least) one device to (at least) three SDSs, where each SDS is in a separate fault set. Each device has a minimum of 130 GB free storage capacity (an SDS which has not been assigned to a fault set is treated as a fault set of its own). You can do that using the CSV file, or at a later stage. For the maximum number of devices per SDS, see product limits in the *Dell PowerFlex and PowerFlex Manager Technical Overview*. Device data is erased when devices are added to an SDS. When adding a device to an SDS, PowerFlex checks that the device is clear before adding it. If the device is not clear, an error message is displayed for each device. PowerFlex might not perform optimally if there are large differences between the sizes of the devices in the storage pool—for example, if one device is as big as the rest of the devices. After adding devices, you can define how much of the device capacity is available to PowerFlex by using the SCLI command `modify_sds_device_capacity`. For optimal performance, try to balance the number of devices of a storage pool, and the capacity of those devices among the relevant SDSs.	
StoragePool List	Sets storage pool names, up to 31 characters in length. Use this option in one of the following ways to assign storage pools to the devices in the **SDS Storage Device List**: • A comma-separated list of storage pool names; the length of the list must be the same length as the list of devices in the **SDS Storage Device Names** list. In this case, the storage pools are mapped to the devices in that list, respectively. The list must be comma-separated, with no spaces. • If one storage pool is entered here, the same storage pool are mapped to all the devices in the **SDS Storage Device List**. • If no storage pool is listed, a storage pool is automatically created during installation, named `default`.	-
SDS Storage Device Names	Sets names for devices. Assigns device names to the devices in the **SDS Storage Device List**, respectively. The number of names must equal the number of devices. Comma-separated, no spaces.	-
perfProfileForSDS	Optional performance profile to set for SDSs: `High` (default) or `Compact` When this field is left empty, the default option is applied. High performance is required for the fine granularity data layout feature. For more information, see *Configure performance profile during deployment*.	-
Is SDC	`Yes` or `No`.	Yes
perfProfileForSDC	Optional performance profile to set for SDCs: `High` (default) or `Compact`. When this field is left empty, the default option is applied. High performance is required for the fine granularity data layout feature. For more information, see *Configure performance profile during deployment*.	-
SDC Name	Sets a name for the SDC, with a maximum length of 31 characters. Ensure that each server on which SDC is installed has a unique hostname.	-
NVDIMM Acceleration Device List	List of NVDIMM devices used to provide acceleration for storage pools using fine granularity data layout. Up to six devices, comma-separated, with no spaces. Ensure that the NVDIMM devices are configured for DAX mode.	-
NVDIMM acceleration pool List	Sets NVDIMM acceleration pool names. Assigns NVDIMM acceleration pools to the devices in the **NVDIMM Acceleration Device List**, respectively. The number of NVDIMM acceleration pools must equal the number of devices. Comma-separated, no spaces.	-
Is SDR	Deploy the SDR component on this node: `Yes` or `No` (default).	-

Table 33. Fields in the CSV file (continued)

Field	Description	Required
	A minimum of two SDRs per peer system must be deployed for the replication feature. SDRs must be deployed on nodes that also contain the SDS component.	
SDR Name	The name of the SDR node can be up to 31 characters in length. The name must not contain spaces. Default: empty field.	-
SDR Port	The port used by the SDR. Default: 11088	-
SDR Application IPs	The IP addresses through which the SDC communicates with the SDR. When an application writes data, it works with the SDR through these IP addresses. Maximum of eight addresses, comma-separated, no spaces. The same IP addresses can be used for Application, Storage, External. IPv6 is supported.	-
SDR Storage IPs	The IP addresses through which the SDR communicates with the MDM for server-side control communications. Maximum of eight addresses, comma-separated, no spaces. The same IP addresses can be used for application, storage, external. IPv6 is supported.	-
SDR External IPs	The IP addresses through which the SDR communicates with peer systems' SDRs. Maximum of eight addresses, comma-separated, no spaces. The same IP addresses can be used for Application, Storage, External. IPv6 is supported.	-
SDR All IPs	SDR IP addresses to be used for communication among all nodes (including all three roles) Maximum of eight addresses, comma-separated, no spaces. This is the default setting.	-
perfProfileForSDR	Optional performance profile to set for SDRs: `High` (default) or `Compact`. When this field is left empty, the default option is applied. High performance is required for the fine granularity data layout feature. See *Configuring performance profile during deployment* in Related information.	-
Storage pool Configuration section	Optional section of the CSV file that lets you configure properties for each of the storage pools to which the SDS devices are assigned.	-
ProtectionDomain	The protection domain to which the storage pool belongs.	-
StoragePool	The storage pool name	-
Media Type	The expected device media type in the storage pool: `HDD` or `SDD`	-
External Acceleration	External acceleration type, if used: • `Read`—All devices are accelerated by a non- PowerFlex read cache • `Write`—All devices are accelerated by a non- PowerFlex write cache • `ReadAndWrite`—All devices are accelerated by both non- PowerFlex read and write cache	-
Data Layout	Data layout type: • `FG`—Fine granularity layout • `MG`—Medium granularity layout	-
Fine granularity ACCP	The name of the fine granularity acceleration pool associated with the storage pool.	-

Table 33. Fields in the CSV file (continued)

Field	Description	Required
Zero Padding	Zero padding enabled on the storage pool (mandatory for fine granularity): • `Yes`—zero padding enabled • `No`—zero padding disabled	-
Compression Method	The compression method used in the fine granularity storage pool. Compression methods might affect performance. • `None`—no compression is used • `Normal`—compression is enabled	-
Replication journal capacity percentage	The percentage of capacity dedicated to the replication journal: a value between 1–100 The percentage is calculated from the net capacity, and not from the total capacity. Use the information provided in the system limits section of system documentation to calculate the percentage of capacity that is required. This calculation also applies to expansion use cases where SDR is being added to an existing system. Each protection domain that contains SDRs should have at least one storage pool with replication journal capacity higher than 0%. Each SDR needs at least 108 GB of replication journal capacity in order to work. Take this into consideration when calculating capacity requirements.	-
Is SDT	Deploy the SDT component on this node: `Yes` or `No` SDT is an NVMe target, which is a front-end component that translates NVMe over Fabric protocol into internal protocols.	-
SDT Name	The name for the SDT node, can be up to 31 characters in length. The name must not contain spaces.	-
SDT All IPs	SDT IP addresses used for both hosts communication and MDM communication (including both roles). Maximum of eight IP addresses, comma-separated, no spaces.	-

3. Save the CSV file and retain it for future use.

Prepare the CSV topology file for two-layer deployment

When deploying SDC nodes on a Linux server in a two-layer environment, add rows to the CSV file specifically for SDCs, that only contain the values listed in this table.

Table 34. CSV topology file for a two-layer deployment

Field	Description	Required
Domain	If using a domain user, the name of the domain (not relevant for Linux)	-
Username	The name of the user Linux nodes: Enter either `root` or a `non-root` sudo user, indicated by appending "(sudo)" to the end of the username (default: `root`). For more information about deploying with a `non-root` user, see related information.	-
Password	Password used to log in to the node. This should be the password of the user entered in the **Username** column.	Yes
Operating System	The server's operating system: `linux` or `windows`	Yes
SDS-SDC Only IPs	SDC IP addresses used for communication among SDS and SDC nodes only. Comma-separated, no spaces.	-

Table 34. CSV topology file for a two-layer deployment (continued)

Field	Description	Required
	NOTE: For SDC-only nodes, enter the IP address in this column.	
Is SDC	`Yes`	Yes
perfProfileForSDC	Optional performance profile to set for SDCs: `High` (default) or `Compact`. When this field is left empty, the default option is applied. High performance is required for the fine granularity data layout feature. For more information, see *Configuring performance profile during deployment* in Related information.	-
SDC Name	Name for the SDC.	-

CSV file examples for typical configurations

Refer to the examples provided in this section to see how the CSV file is used to configure various system topologies and features.

NOTE: These examples represent a subset of all possible topologies and features that can be configured.

Four data networks: two front-ends and two back-ends, no fault sets

This example shows a CSV topology file for four data networks, where two are front-end networks, and two are back-end networks. No fault sets are configured.

Four data networks are configured as follows:

- Two front-end networks for SDCs
- Two back-end networks for MDMs, SDSs, and SDRs (for replication only)
- Two external networks for replication (if applicable)
- No fault sets

To download the example CSV file, see the *Dell PowerFlex and PowerFlex Manager Example CSV files* available on the Dell Technologies support site.

The following table lists the parameters that must be configured for this use case. Other optional parameters in the CSV file should be configured as necessary. For information about default credentials, see the *Dell PowerFlex Family 4.x Security Configuration Guide*.

Four data networks: two front-end and two back-end, with fault sets

This example shows a CSV topology file for four data networks, where two are front-end networks, and two are back-end networks. Fault sets are also configured.

Four data networks are configured as follows:

- Two front-end networks for SDCs
- Two back-end networks for MDMs, SDSs, and SDRs (when replication is required)
- Two external networks for replication (when applicable)
- Fault sets

To download the example `BE-FE-6_node_FaultSet.xlxs` CSV file, see the *Downloading the example CSV files Reference Guide* available on the Dell Technologies support site.

The following table lists the parameters that must be configured for this use case. Other optional parameters in the CSV file should be configured as necessary. For information about default credentials, see the *Dell PowerFlex and PowerFlex Manager Security Configuration Guide*.

Table 35. CSV file parameters

Parameter	Example
Username	root

Table 35. CSV file parameters (continued)

Parameter	Example
Password	password
Is MDM/TB	Primary Other options: Secondary, tiebreaker (TB)
MDM Mgmt IP	192.168.150.11
MDM IPs	192.168.151.11,192.168.152.11
MDM Name	MDM1
perfProfileForMDM	high
Virtual IPs	192.168.153.10,192.168.154.10
Virtual IP NICs	ens153,ens154
Is SDS	Yes
SDS Name	SDS1
SDS-SDS Only IPs	192.168.151.11,192.168.152.11
SDS-SDC Only IPs	192.168.153.11,192.168.154.11
Protection Domain	PD01
Fault Set	FS01
SDS Storage Device List	/dev/sdb,/dev/sdc,/dev/sdd,/dev/sde
StoragePool List	SP01,SP01,SP02,SP02
perfProfileForSDS	high
Is SDC	Yes
perfProfileForSDC	high
SDC Name	SDC1
NVDIMM Acceleration Device List	/dev/dax0.0
NVDIMM Acceleration Pool List	accp01
Is SDR (optional—for replication only)	Yes
SDR Name (optional—for replication only)	SDR1
SDR Application IPs (optional—for replication only)	192.168.153.11,192.168.154.11
SDR Storage IPs (optional—for replication only)	192.168.151.11,192.168.152.11
SDR External IPs (optional—for replication only)	192.168.161.11,192.168.162.11
Storage Pool Configuration (section is used to configure data layout and replication)	
ProtectionDomain	PD01
StoragePool	SP01
Media Type	SSD
External Acceleration	None
Data Layout	FG
Fine Granularity ACCP	accp01

Table 35. CSV file parameters (continued)

Parameter	Example
Zero Padding	Yes
Compression Method	Normal
Replication journal capacity percentage	18

Four data networks: Front-end and back-end on all four networks, no fault sets

This example shows a CSV topology file for four data networks, where all four are both front-end and back-end networks. No fault sets are configured.

Four data networks are configured as follows:

- Four networks for both front-end and back-end: SDCs, MDMs, SDSs, and SDRs (for replication only)
- Two external networks for replication (if applicable)
- No fault sets

To download the example `SDS-SDC-Same-Network-5_node.xlsx` CSV file, see the *Dell PowerFlex and PowerFlex Manager Example CSV files* available on the Dell Technologies support site.

The following table lists the parameters that must be configured for this use case. Other optional parameters in the CSV file should be configured as necessary. For information about default credentials, see the *Dell PowerFlex Family 4.x Security Configuration Guide*.

Four data networks: front-end and back-end on all four networks, with fault sets

This example shows a CSV topology file for four data networks, where all four are both front-end and back-end networks. Fault sets are also configured.

Four data networks are configured as follows:

- Four networks for both front-end and back-end: SDCs, MDMs, SDSs, and SDRs (for replication only)
- Two external networks for replication (if applicable)
- Fault sets

To download the example `SDS-SDC-Same-Network-6_node_withFaultSet.xlsx` CSV file, see the *Downloading the example CSV files Reference Guide* available on the Dell Technologies support site.

The following table lists the parameters that must be configured for this use case. Other optional parameters in the CSV file should be configured as necessary. For information about default credentials, see the *Dell PowerFlex and PowerFlex Manager Security Configuration Guide* .

Table 36. CSV file parameters

Parameter	Example
Username	root
Password	password
Is MDM/TB	Primary Other options: Secondary, TB
MDM Mgmt IP	192.168.150.11
MDM IPs	192.168.151.11,192.168.152.11,192.168.153.11,192.168.154.11
MDM Name	MDM1
perfProfileForMDM	high
Virtual IPs	192.168.151.10,192.168.152.10,192.168.153.10,192.168.154.10

Table 36. CSV file parameters (continued)

Parameter	Example
Virtual IP NICs	ens151,ens152,ens153,ens154
Is SDS	Yes
SDS Name	SDS1
SDS All IPs	192.168.151.11,192.168.152.11,192.168.153.11,192.168.154.11
Protection Domain	PD01
Fault Set	FS01
SDS Storage Device List	/dev/sdb,/dev/sdc,/dev/sdd,/dev/sde
StoragePool List	SP01,SP01,SP02,SP02
perfProfileForSDS	high
Is SDC	No
perfProfileForSDC	high
SDC Name	SDC1
NVDIMM Acceleration Device List	/dev/dax0.0
NVDIMM Acceleration Pool List	accp01
Is SDR (optional—for replication only)	Yes
SDR Name (optional—for replication only)	SDR1
SDR Application IPs (optional—for replication only)	192.168.151.11,192.168.152.11,192.168.153.11,192.168.154.11
SDR Storage IPs (optional—for replication only)	192.168.151.11,192.168.152.11,192.168.153.11,192.168.154.11
SDR External IPs (optional—for replication only)	192.168.161.11,192.168.162.11
Storage Pool Configuration (this section is used to configure data layout and replication)	
ProtectionDomain	PD01
StoragePool	SP01
Media Type	SSD
External Acceleration	None
Data Layout	FG
Fine Granularity ACCP	accp01
Zero Padding	Yes
Compression Method	Normal
Replication journal capacity percentage	18

Configuring performance profile during deployment

During deployment, system performance is enhanced by default implementation of the high performance profile for the MDM, SDC, SDS, and SDR components. Most servers support the requirements for this configuration and can benefit considerably from its implementation.

The high performance profile is mandatory for fine granularity data layout. However, for medium granularity data layout, you can choose to use the compact performance profile instead.

High performance profile should be enabled on servers equipped with processors based on Haswell-equivalent or newer architectures and 48 GB of RAM. Other configurations may benefit from this setting as well.

The performance profile is set to High by default, but if you want to change the profile to Compact in the CSV file, use the `perfProfileFor<COMPONENT>` columns.

Adding new components or features by editing the existing CSV file

To add components or new features to your system, use the existing CSV topology file for the PowerFlex system.

The CSV file that is used to deploy a system can be used later to add features or components to an existing PowerFlex system. The following scenarios are when you can use the CSV file at a later time:

- Add SDSs, with or without new storage and acceleration devices
- Add MDMs or tiebreakers. The PowerFlex installer adds them to the MDM cluster and configures them automatically.
- Add SDRs for the replication feature
- Add SDCs
- Add SDTs
- Configure storage pools
- Configure PowerFlex system features which can be configured using the CSV file, such as performance profiles, acceleration, and so on.

NOTE: You must use the saved copy of the CSV file that was used for the initial deployment to add nodes to the system.

To add and configure components and features, make a copy of your system's latest CSV topology file. Add rows or columns, or edit existing rows and columns. For new features, you can copy columns from the latest complete CSV template, and paste them into the copy of your system's CSV file. Do the following to download example CSV files:

1. Log in to PowerFlex Manager.
2. Left-click the question mark (?) icon in the upper right.
3. Select **Getting Started**.
4. On the **Getting Started** page, click **Deploy With Installation File**.
5. In the **PowerFlex Installer** window, click **Next**.

   Under **Installation Topology**, you can download example CSV files.
6. To download the example CSV files, click the **Minimal** link (minimum fields) or the **Complete** link (all fields).

The following examples show how you can edit the CSV topology file which can be edited for various system modification scenarios.

Add storage and acceleration

CSV file of topology currently in use

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Username	Passwor	Operat	Is MDM/TI	MDM Nan	MDM Mgn	MDM IPs	Is SDS	SDS Name	SDS All IPs	Prote	Fault Set	SDS Storage Device L	StoragePool
2	root	Scaleio1	linux	Primary	MDM_114	10.55.137.	10.55.137	Yes	sds_idd_1	10.55.137.114	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1
3	root	Scaleio1	linux	Secondar	MDM_115	10.55.137.	10.55.137	Yes	sds_idd_2	10.55.137.115	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1
4	root	Scaleio1	linux	Secondar	MDM_116	10.55.137.	10.55.137	Yes	sds_idd_4	10.55.137.116	pd1		/dev/sdb,/dev/sdc	sp1
5	root	Scaleio1	linux	TB	MDM_117	10.55.137.	10.55.137	Yes	sds_idd_5	10.55.137.117	pd1		/dev/sdb,/dev/sdc	sp1
6	root	Scaleio1	linux	TB	MDM_119	10.55.137.	10.55.137	Yes	sds_idd_3	10.55.137.119	pd2		/dev/sdb	sp2
7														
8														

Edited CSV file—new rows for added SDSs

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Username	Password	Is MDM/TI	MDM Mgn	MDM IPs	MDM Nan	perfProfil	Virtual IPs	Virtual IP	Is SDS	SDS Name	SDS All IPs	SDS-SDS C	SDS-SDC C	Protectior	Fault Set	SDS Stora	StoragePc
2	root	Password	Primary	10.76.1.1	10.76.60.1	MDM1	High	10.76.60.2	eth1	Yes		10.76.60.195			domain1	fs1	/dev/sdb,	pool1
3	non_root(	Password	Secondar	10.76.1.2	10.76.60.1	MDM2	High	10.76.60.2	eth1	Yes		10.76.60.17			domain1	fs1	/dev/sdb,	pool2
4	root	Password	Secondar	10.76.1.3	10.76.60.1	MDM3	High	10.76.60.2	eth1	Yes		10.76.60.18			domain1	fs2	/dev/sdb,	pool1,poo
5	root	Password	TB		10.76.60.3	TB1	High			Yes	SDS1	10.76.60.3	10.76.2.2,:	10.76.3.3	domain1	fs3	/dev/sdb,	pool1,poo
6	root	Password	TB		10.76.60.1	TB2	High			No								
7	root	Password	Standby-S	10.76.1.4	10.76.60.2	MDM4	High	10.76.60.2	eth1	No								
8	root	Password	Standby-S	10.76.1.5	10.76.60.1	MDMa	High	10.76.60.2	eth1	Yes		10.76.60.175			domain2		/dev/sdb,	pool1,poo
9	root	Password	Standby-S	10.76.1.6	10.76.60.2	MDMb	High	10.76.60.2	eth1	Yes		10.76.60.22			domain2		/dev/sdb,	pool1,poo
10	root	Password	Standby-TB		10.76.60.3	Tba	High			Yes	SDSa	10.76.60.35			domain2		/dev/sdb,	pool1,poo
11																		

Figure 5. Add storage and acceleration to the CSV topology file

Add an MDM and a tiebreaker

CSV file of topology currently in use

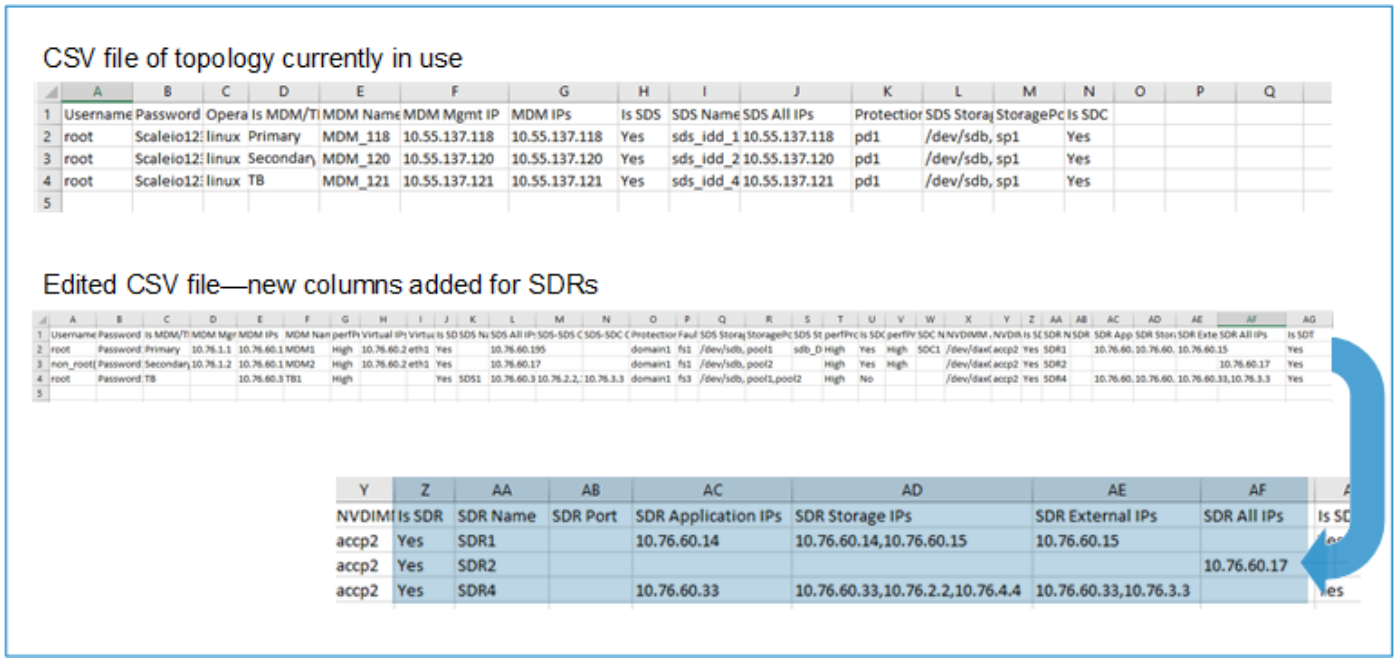
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Username	Password	Operating	Is MDM/TI	MDM Nan	MDM Mgn	MDM IPs	Is SDS	SDS Name	SDS All IPs	Protectior	SDS Stora	StoragePc	Is SDC	Is SDR
2	root	Scaleio12	linux	Primary	MDM_118	10.55.137.	10.55.137.	Yes	sds_idd_1	10.55.137.	pd1	/dev/sdb,	sp1	Yes	Yes
3	root	Scaleio12	linux	Secondary	MDM_120	10.55.137.	10.55.137.	Yes	sds_idd_2	10.55.137.	pd1	/dev/sdb,	sp1	Yes	Yes
4	root	Scaleio12	linux	TB	MDM_121	10.55.137.	10.55.137.	Yes	sds_idd_4	10.55.137.	pd1	/dev/sdb,	sp1	Yes	Yes
5															

Edited CSV file—new rows added for an additional MDM and TB

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Username	Password	Is MDM/TI	MDM Mgn	MDM IPs	MDM Nan	perfProfil	Virtual IPs	Virtual IP	Is SDS	SDS Name	SDS All IPs	SDS-SDS C	SDS-SDC C	Protectior	Fault Set
2	root	Password	Primary	10.76.1.1	10.76.60.1	MDM1	High	10.76.60.2	eth1	Yes		10.76.60.195			domain1	fs1
3	non_root(	Password	Secondar	10.76.1.2	10.76.60.1	MDM2	High	10.76.60.2	eth1	Yes		10.76.60.17			domain1	fs1
4	root	Password	Secondar	10.76.1.3	10.76.60.1	MDM3	High	10.76.60.2	eth1	Yes		10.76.60.18			domain1	fs2
5	root	Password	TB		10.76.60.3	TB1	High			Yes	SDS1	10.76.60.3	10.76.2.2,:	10.76.3.3	domain1	fs3
6	root	Password	TB		10.76.60.1	TB2	High			No						
7	root	Password	Standby-S	10.76.1.4	10.76.60.2	MDM4	High	10.76.60.2	eth1	No						
8	root	Password	Standby-S	10.76.1.5	10.76.60.1	MDMa	High	10.76.60.2	eth1	Yes		10.76.60.175			domain2	
9	root	Password	Standby-S	10.76.1.6	10.76.60.2	MDMb	High	10.76.60.2	eth1	Yes		10.76.60.22			domain2	
10	root	Password	Standby-TB		10.76.60.3	Tba	High			Yes	SDSa	10.76.60.35			domain2	
11																

Figure 6. Add an MDM and a tiebreaker to the CSV topology file

Add storage data replicators (SDRs) for replication



Y	Z	AA	AB	AC	AD	AE	AF
NVDIMM	Is SDR	SDR Name	SDR Port	SDR Application IPs	SDR Storage IPs	SDR External IPs	SDR All IPs
accp2	Yes	SDR1		10.76.60.14	10.76.60.14,10.76.60.15	10.76.60.15	
accp2	Yes	SDR2					10.76.60.17
accp2	Yes	SDR4		10.76.60.33	10.76.60.33,10.76.2.2,10.76.4.4	10.76.60.33,10.76.3.3	

Figure 7. Add storage data replicators to the CSV topology file

Add storage data clients

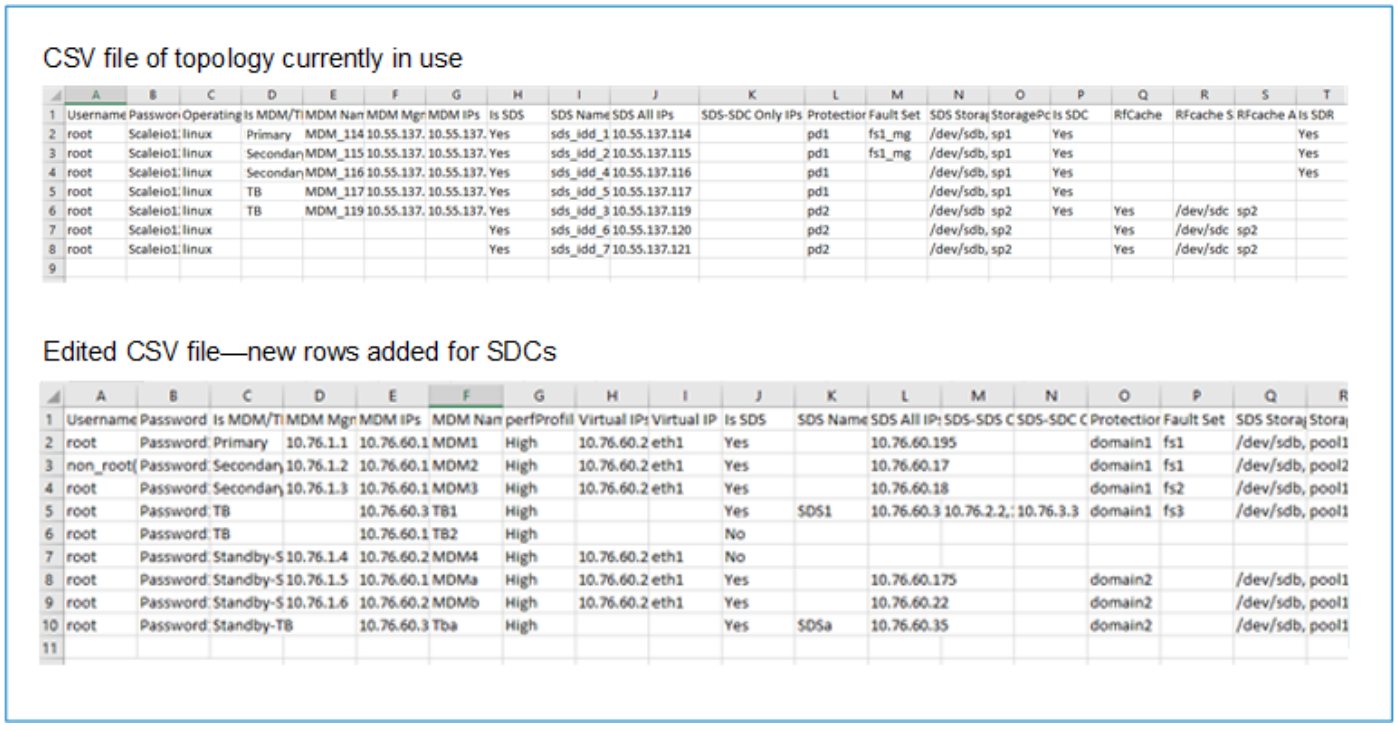


Figure 8. Add storage data clients (SDCs) to the CSV topology file

Configure storage pools

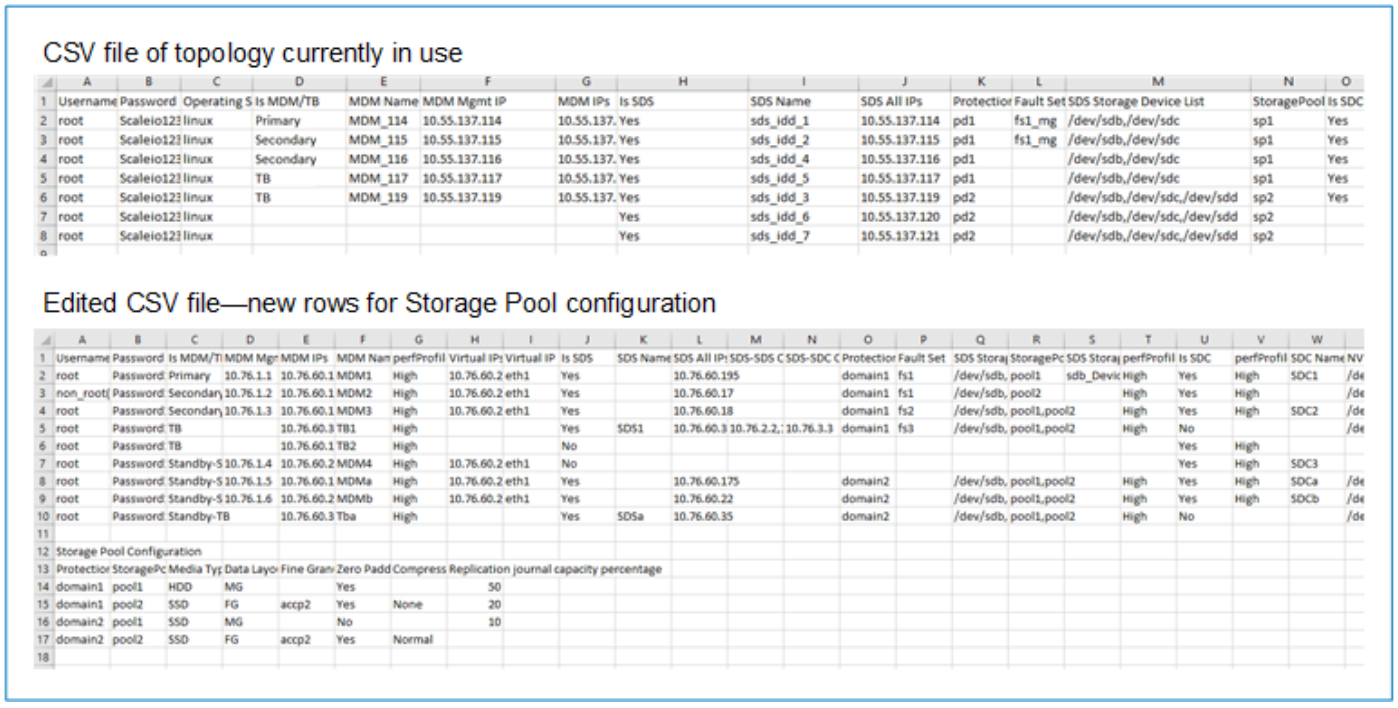


Figure 9. Configure storage pools in the CSV topology file

Deploy a PowerFlex cluster using a CSV topology file from PowerFlex Manager

Use the following procedure to deploy a PowerFlex cluster using a CSV topology file.

Steps

1. Log in to PowerFlex Manager.
2. On the **Getting Started** page, click **Deploy With Installation File** to provide an installation topology file for deployment of nodes.
3. In the **PowerFlex Installer** window, click **Browse** to upload the appropriate artifacts (RPMs) according to the operating system of your PowerFlex operating system machines.
4. Click **Upload Files** and click **Next** when the RPMs have finished uploading.
5. Click **Browse** to upload an installation CSV topology file that describes the desired topology of PowerFlex.

   To view sample files, click the links to the following two example files: **PowerFlex_config_example_all fields.csv (all fields)** and **PowerFlex_basic_example_minimum fields.csv (minimum fields).**
6. Click **Upload Installation CSV**.
7. Select the **I accept the terms of the End User License Agreement (mandatory)** check box.
8. Click **Next**.
9. In the **Advanced Settings** tab, you can select how you would like to install PowerFlex 4.6.x. You can skip options or force an overwrite.
10. Select **Force overwrite of existing data on devices used for PowerFlex storage**.

    PowerFlex can be installed on new or clean disks (storage node-DGWT devices). If the installation is done on a previously deployed PowerFlex cluster, or used disks, you must tell the installer to overwrite the older data.

    NOTE: If you select this option, the data is erased.
11. Click **Next**.
12. In the **Summary** tab, verify all the configurations and click **Finish**.

13. Click the **Go to Events Page** button to track the installation progress.

Next steps

In case of failure or to check the progress, see the following log files:

- PowerFlex management platform installer logs: `/opt/dell/pfmp/PFMP_Installer/logs`
- Platform installer logs: `/opt/dell/pfmp/atlantic/logs/bedrock.log`

If the PowerFlex management platform deployment fails, collect the PowerFlex management platform support data and provide the support bundle files to Dell Technologies Support. For information about collecting PowerFlex management platform data, see *(Optional) Collect PowerFlex management platform support data*.

Deploy or expand a cluster using REST API commands

After you have prepared the CSV topology file, use the REST API to parse it and upload the configuration to PowerFlex Manager, and then deploy block storage resources using REST API commands. This procedure is only recommended for advanced users familiar with working with the REST API.

Prerequisites

- Ensure that you have prepared the CSV topology file.
- Ensure that you know the load balancer IP address and user credentials of PowerFlex Manager. All the steps performed in this procedure must be performed using the load balancer IP address. The load balancer IP address is the same IP address you are using to access PowerFlex Manager.
- Ensure that you have the software packages of the PowerFlex core components that you want to deploy, and the ActiveMQ package.
- See the Dell Technologies Developer portal for information about working with the REST API.
- Log in using the REST API, and make a note of the Keycloak token that was returned from `/auth/login` or from `/api/gatewayLogin`.

About this task

The following procedure provides examples using cURL.

Steps

1. Get the configuration from the CSV file, by parsing it using the PowerFlex REST API. Use the command `POST /im/types/Configuration/instances/actions/parseFromCSV`, where the body is the CSV file reference. For example:

   ```
   curl -v -k -X POST -H "Content-Type:application/json" -H "Authorization: Bearer
   ${token}" -F "file=<PATH_TO_CSV_FILE>" https://<IP_ADDRESS>/im/types/Configuration/
   instances/actions/parseFromCSV
   ```

   Where: *{token}* is the Keycloak token, *<PATH_TO_CSV_FILE>* is the path to the CSV topology file, and *<IP_ADDRESS>* is the PowerFlex Manager IP address.
   The command returns a JSON file containing the required cluster configuration. The installation command requires this file in order to install resources. Save this file.
2. Add packages:
   a. Copy the installation packages of the required core components (including ActiveMQ RPM) to: `"/usr/local/tomcat/temp/scaleio"`.
   b. Open the shell, and run the following command:

      ```
      kubectl get pods -n powerflex | grep block-legacy
      ```

   c. Make a note of the block-legacy-gateway ID that is returned in the first line of the command output. For example, in this sample output, the ID is `76594f9459-mqgxj`:

      ```
      block-legacy-gateway-76594f9459-mqgxj    1/1    Running    0    24h
      block-legacy-gateway-mds-6696f6dd4c-s9qp8    1/1    Running    0    24h
      ```

d. In the shell, run the following command:

```
kubectl exec --stdin --tty <ID> -n powerflex -- /bin/sh
```

Where *<ID>* is the block-legacy-gateway ID obtained in the previous step.

e. In the shell, run the following commands:

```
apt-get update
apt-get install openssh-client
scp root@<RPM_MACHINE_IPS>:<RPMS_LOCATION>* temp/scaleio
```

Where *<RPM_MACHINE_IPS>* is the IP address where the packages are saved, and *<RPMS_LOCATION>* is the path to the location of the packages.

3. Install block storage resources using the PowerFlex REST API. Use the command `POST /im/types/Configuration/actions/install`, where the body is the configuration that was returned from parsing the CSV file.

   If the parameter *systemId* (type string) is added to the configuration, the cluster is created with the given ID. Otherwise, the gateway generates an ID.

   Optional parameters:

   - noUpload—Skip upload package phase (true or false, default if false)
   - noInstall—Skip install package phase (true or false, default if false)
   - noConfigure—Skip configure phase and repository files (true or false, default if false)
   - noSecurityBootstrap—Skip security bootstrap (part of install phase) (true or false, default if true)
   - noLinuxDevValidation (true or false, default if true)
   - sdsInstances (number, optional)
   - extend—extend an existing system (true or false, default if false)
   - globalZeroPadPolicy (true or false, default if true)
   - forceDeviceTakeover—Clean devices before installing SDS (without it, installation will fail when it runs after uninstall; must not be used in production) (true or false, optional)

   For example:

   ```
   curl -v -s -k -i -X POST -H "Content-Type:application/json" -H
   "Authorization: Bearer ${token}" "https://<IP_ADDRESS>/im/types/Configuration/actions/
   install?noSecurityBootstrap=false&forceDeviceTakeover=true" -d @<PATH>/config.json
   ```

   Where: *{token}* is the Keycloak token, *<IP_ADDRESS>* is the PowerFlex Manager IP address, and *<PATH>* is the path to the JSON configuration file that was output in step 1.

4. Installation is asynchronous and consists of four stages: query, upload, install, and configure. Manually monitor and move from one phase to the next one, using the commands `queryPhaseState` and `moveToNextPhase`. You can also monitor the progress of each stage by viewing events in PowerFlex Manager.
   The following is an example of an installation event:

   ```
   COMMAND_STARTED_ON_OPERATION_INSTALL_AND_PHASE_QUERY
   ```

5. After installation is complete, clear the gateway state. Use the command `POST im/types/Command/instances/actions/clear`.

   For example:

   ```
   curl -v -s -k -i -X POST -H "Content-Type:application/json" -H "Authorization: Bearer
   ${token}" "https://<IP_ADDRESS>/im/types/Command/instances/actions/clear"
   ```

   Where: *{token}* is the Keycloak token, and *<IP_ADDRESS>* is the PowerFlex Manager IP address.

Deployment design for co-residency management

The following deployment design principles are recommended:

- If you have a multirack system, place the MDM nodes and the PowerFlex management platform components on different racks.

- If the hardware requirements are met, any node running SDS can also optionally run an SDR, SDT.
- Ensure that the LIA is installed on every node, as it is required for maintenance purposes.

8

Post-installation activities

Perform the procedures listed in the post-installation checklist after system installation to get your PowerFlex storage up and running in an optimal manner.

See the post-installation checklist to determine the procedures that are relevant for your system. Some procedures are performed using PowerFlex Manager, and some are performed using command line. PowerFlex Manager activities are described in detail in the online help system.

Post-installation checklist

The following table lists the configuration tasks that are performed after installing PowerFlex software.

The table also indicates whether tasks are required or optional. It also indicates where you can find documentation for each task.

Table 37. Post-installation configuration tasks

Item	Required or optional?	Follow the procedures in this post-installation section	Follow PowerFlex Manager online assist procedures
Enable OpenSSL FIPS compliance	Optional	X	-
Check port usage	Required	X	-
Configure SDC driver for security-enhanced Linux (SELinux)	Optional	X	-
Mount a PowerFlex volume	Optional	X	-
Configure NVMe targets and hosts for block storage if they were not configured by the installation template.	Required	-	X
Add block storage if it is not configured by the installation template, in this order: Add protection domains > Add storage pools > Add acceleration pools (for fine granularity data layout only) > Add devices to SDS (including mapping them to storage pools) > Add volumes > Map volumes to hosts	Required	-	X
Create and map volumes for block storage if not configured by the installation template	Required	-	X
Associate PowerFlex volumes with physical disks	Optional	X	-
Configure local protection for block storage (local snapshots)	Optional		X
Configure remote protection for block storage (asynchronous replication to a peer computer) if not configured by the installation template, in this order: Install the remote PowerFlex computer > Extract certificates on both systems and copy to peers > Configure journal capacity > ensure	Optional		X

Table 37. Post-installation configuration tasks (continued)

Item	Required or optional?	Follow the procedures in this post-installation section	Follow PowerFlex Manager online assist procedures
SDR is deployed and added > Define peer systems > Create and activate RCG			
Configure the SDS for optimal memory usage	Required	X	-
Configure the SDS for NVDIMM health awareness	Required	X	-
Configure local and LDAP users	Required	-	X
Fine tune block storage computer performance	Optional	X	-
Configure event and alert notifications to external systems	Optional	-	X
Install the computer license	Required	-	X
Configure SDC access to the MDM	Optional	X	-

Configuring NVMe initiators as hosts connecting to PowerFlex system

Use the following procedures to configure the NVMe initiators on hosts for Linux and VMware ESXi-based systems.

The procedure may vary depending on the operating system versions. For more information, see the operating system vendor documentation.

NOTE: NVMe over TCP initiator host-side support is only available for VMware ESXi and SUSE. All other Linux platforms are considered as technology preview.

Configure NVMe initiators on hosts for Linux-based systems

For systems using NVMe over TCP technology, configure the NVMe initiator on all hosts that access PowerFlex storage. The following procedure is provided as an example only. Consult your operating system's documentation for detailed instructions about configuring NVMe initiators for NVMe over TCP topologies.

Prerequisites

Ensure that you have the login credentials for each host.

About this task

In an erasure coding environment, when NVMe over TCP technology is used, PowerFlex software is not installed on the compute nodes. Instead, configure the NVMe initiators in the operating system on the PowerFlex compute-only node, and configure the NVMe targets in PowerFlex.

In a mirroring environment, when NVMe over TCP technology is used, PowerFlex software is not installed on the compute nodes. Instead, configure the NVMe initiators in the operating system on the PowerFlex compute-only node or PowerFlex hyperconverged node, and configure the NVMe targets in PowerFlex.

NOTE: Ensure that the hosts are running Linux operating system versions that are supported by this version of PowerFlex. For more information, see the system requirements in the *Dell PowerFlex and PowerFlex Manager Technical Overview*.

The following example shows one way of configuring the NVMe initiator. See your operating system's documentation for details and other configuration options.

Steps

1. Ensure that the NVMe modules are loaded on the NVMe initiator:
   a. Log in to the host.
   b. In the command line, run the command:

      ```
      lsmod | grep nvme
      ```

   c. Ensure that `nvme_tcp` and `nvme_fabrics` modules are loaded.
      If they are loaded, go to step 2. If they are not loaded, continue to the next substep.
   d. Perform one of the following procedures:
      - For manual loading of modules (perform after every reboot), run:

        ```
        modprobe nvme_tcp
        modprobe nvme_fabrics
        ```

      - For automatic loading of NVMe modules, run:

        ```
        echo nvme_tcp >> /etc/modules-load.d/nvme_tcp.conf
        echo nvme_fabrics >> /etc/modules-load.d/nvme_tcp.conf
        ```

2. Ensure that the NVMe CLI is installed on the initiator:
   a. Run the following command:

      ```
      nvme
      ```

      If the CLI responds, go to step 3. If it does not, go to the next substep.
   b. Run the command:

      ```
      dnf install nvme-cli
      ```

3. Ensure that a unique host NQN and unique host ID are configured in the initiator.
   a. To verify that a host NQN is present, run the command:

      ```
      nvme show-hostnqn
      ```

      Example of a host NQN:

      ```
      nqn.2014-08.org.nvmexpress:uuid:4c4c4544-0051-5610-8046-b5c04f484b32
      ```

   b. If no host NQN exists, generate one by running the command:

      ```
      nvme gen-hostnqn > /etc/nvme/hostnqn
      ```

      NOTE: Make a note of each PowerFlex node's host NQN. You need this information when configuring the NVMe targets with PowerFlex Manager.

   c. To verify that a host ID is present, run the command:

      ```
      cat /etc/nvme/hostid
      ```

      Example of a host ID:

      ```
      4c4c4544-0051-5610-8046-b5c04f484b32
      ```

      Last 36 characters of a nvme gen-hostqn name can be used. The format must be precise, otherwise the connection fails.
4. Perform the following actions for each required protection domain:
   a. Add an NVMe host to PowerFlex.
   b. Create volume.
   c. Map volume

For more information, see the *Dell PowerFlex and PowerFlex Manager Software Administration Guide* and *Dell PowerFlex and PowerFlex Manager CLI Reference Guide*.

5. Discover and connect to the target PowerFlex system.
6. Configure the type of multipath used:

   Multipath type: Native NVMe Multipath

   Use the following procedure:

   - Check if the native NVMe multipath is enabled using the following command:

     ```
     cat /sys/module/nvme_core/parameters/multipath
     ```

   - If `Y` is returned, native NVMe multipath is already enabled.
   - To enable the NVMe multipath, run the following command:

     ```
     grubby --update-kernel=ALL --args="nvme_core.multipath=Y"
     grub2-mkconfig -o /boot/grub2/grub.cfg
     reboot
     ```

   - To enable active/active protocol which remains persistent after reboots, run the following command:

     ```
     echo 'ACTION=="add", SUBSYSTEM=="nvme-subsystem", ATTR{iopolicy}="round-robin"' >
     /lib/udev/rules.d/71-nvme-iopolicy.rules
     ```

7. Configure a persistent initiator. This initiator automatically forms NVMe connections after reboots.
   a. Create a service that runs `nvme connect-all` on reboots:

      ```
      vi /etc/systemd/system/nvme_fabrics_persistent.service
      [Unit]
      Description=NVMf auto discovery service
      Requires=network.target
      After=systemd-modules-load.service network.target
      [Service]
      Type=oneshot
      ExecStart=/usr/sbin/nvme connect-all
      StandardOutput=journal
      [Install]
      WantedBy=multi-user.target timers.target
      ```

   b. Start a new service, and enable it:

      ```
      systemctl start nvme_fabrics_persistent.service
      systemctl enable nvme_fabrics_persistent.service
      ```

8. Run the following command to update the udev rule to make the recovery delay 15 seconds:

   ```
   echo ACTION!=\"remove\", KERNELS==\"ctl\", SUBSYSTEMS==\"nvme-fabrics\",
   ATTR{transport}==\"tcp\", ATTR{model}==\"powerflex\", ATTR{recovery_delay}=\"15\" >>
   /etc/udev/rules.d/71-nvme-recovery-delay.rules
   ```

   NOTE: This step is required only for SUSE Linux Enterprise Server (SLES) 15.5 version.

Discover and connect to the target PowerFlex system

Use the following procedure to obtain SDT IP addresses for discovering the target PowerFlex system.

Prerequisites

- Ensure that a volume is mapped to the host to be able to connect the SDT paths.
- Obtain SDT IP addresses.

Steps

1. From PowerFlex Manager, click **Block** > **Storage systems** > **NVMe Targets**.
2. Select any one of the listed SDTs.
3. Record the IP addresses and discovery port in the lower right corner.
4. To perform the discovery to check that there is an NVMe connection, use the following command:

   ```
   #nvme discover --transport=tcp --traddr=192.168.163.17 --host-traddr=192.168.163.251 --
   trsvcid=8009
   ```

   192.168.163.17 is the NVMe over TCP controller IP address (data 1 VLAN) , and **192.168.163.251** is the NVMe over TCP host IP address (data 1 VLAN).

   ```
   Discovery Log Number of Records 4, Generation counter 2
   =====Discovery Log Entry 0======
   trtype:  tcp
   adrfam:  ipv4
   subtype: nvme subsystem
   treq:    not specified, sq flow control disable supported
   portid:  1
   trsvcid: 4420
   subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
   traddr:  192.168.163.19
   eflags:  not specified
   sectype: none
   =====Discovery Log Entry 1======
   trtype:  tcp
   adrfam:  ipv4
   subtype: nvme subsystem
   treq:    not specified, sq flow control disable supported
   portid:  17
   trsvcid: 4420
   subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
   traddr:  192.168.163.18
   eflags:  not specified
   sectype: none
   =====Discovery Log Entry 2======
   trtype:  tcp
   adrfam:  ipv4
   subtype: nvme subsystem
   treq:    not specified, sq flow control disable supported
   portid:  32
   trsvcid: 4420
   subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
   traddr:  192.168.161.20
   eflags:  not specified
   sectype: none
   =====Discovery Log Entry 3======
   trtype:  tcp
   adrfam:  ipv4
   subtype: nvme subsystem
   treq:    not specified, sq flow control disable supported
   portid:  49
   trsvcid: 4420
   subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
   traddr:  192.168.163.17
   eflags:  not specified
   sectype: non
   ```

5. On the NVMe over TCP client, connect the client to the PowerFlex NVMe over TCP target system.
6. Configure the `/etc/nvme/discovery.conf file` to add the parameters used in the NVMe discover command and use `-p` at the end of the command for persistent controller discovery.

   Persistent discovery controller will:

   a. Notify a host of fabric changes after initial discovery.
   b. The host retains connection to discovery controller.

NOTE: Add an entry for all the SDT IP addresses discovered in the previous step:

```
echo "--transport=tcp --traddr=192.168.163.17 --host-traddr=192.168.163.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
echo "--transport=tcp --traddr=192.168.163.18 --host-traddr=192.168.163.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
echo "--transport=tcp --traddr=192.168.163.19 --host-traddr=192.168.163.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
echo "--transport=tcp --traddr=192.168.161.20 --host-traddr=192.168.161.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
```

Where

- `--traddr=192.168.161.xx`— Data 1 IP address configured in SDT.
- `--traddr=192.168.163.xx`—Data 2 IP address configured in SDT.
- `--host-traddr=192.168.163.251`—Data 1 IP address configured in the NVMe host.
- `--host-traddr=192.168.161.251`—Data 2 IP address configured in the NVMe host.

7. Connect the NVMe over TCP host to the controller system.
   `# nvme connect-all`
8. Verify that the NVMe over TCP host can access the namespace:
   `# nvme list-subsys`

Configure NVMe initiators on hosts for VMware ESXi-based systems

For systems using NVMe over TCP technology, configure the NVMe initiator on all hosts that access PowerFlex storage. This procedure is provided as an example only. See VMware documentation for detailed instructions about configuring NVMe initiators for NVMe over TCP topologies.

About this task

When NVMe over TCP technology is used, PowerFlex software is not installed on the compute nodes. Instead, configure the NVMe initiators in the operating system on the PowerFlex compute-only node or PowerFlex hyperconverged node, and configure the NVMe targets in PowerFlex.

NOTE: Ensure that the hosts are running VMware ESXi operating system versions that are supported by this version of PowerFlex. For more information, see the system requirements in the *Dell PowerFlex and PowerFlex Manager Technical Overview*.

The following example shows one way of configuring the NVMe initiator. See your operating system's documentation for details and other configuration options.

Steps

1. Configure at least three vSwitches: one for management, and two for data (for example: data1 and data2)
2. Configure the VMkernel on each data vSwitch:
   a. In the command line, add the **NVMeTCP** tag to the data VMkernels. For example:

   ```
   esxcli network ip interface tag add -t NVMeTCP -i vmk1
   esxcli network ip interface tag add -t NVMeTCP -i vmk2
   ```

   b. Verify that the **NVMeTCP** tag was added:

   ```
   esxcli network ip interface tag get -i vmk1
   ```

   The returned output should include the following:

   ```
   Tags: NVMeTCP
   ```

c. Enable NVMeTCP on the data vSwitch uplink network adapters:

```
esxcli nvme fabrics enable -p TCP -d vmnic4
esxcli nvme fabrics enable -p TCP -d vmnic5
```

d. Verify that NVMeTCP was enabled on the network adapters:

```
esxcli nvme adapter list
```

Output similar to the following should be returned:

```
Adapter  Adapter Qualified Name           Transport Type  Driver   Associated
Devices

-------  -------------------------------  --------------  -------
------------------

vmhba64  aqn:nvmetcp:ec-f4-bb-d4-95-18-T  TCP             nvmetcp  vmnic4
vmhba65  aqn:nvmetcp:ec-f4-bb-d4-95-18-T  TCP             nvmetcp  vmnic5
```

3. Get the ESXi NQN of each ESXi node, and make a note of each one. You need this information when configuring the NVMe targets with PowerFlex Manager.

```
esxcli nvme info get
```

Output similar to the following should be returned:

```
Host NQN:nqn.2014-08.com.xyz.lab.sio:nvme:sio-dcoe-116AW-13
```

4. Perform the following actions:
   a. Add NVMe host to PowerFlex.
   b. Create volume.
   c. Map volume.

   For more information, see the *Dell PowerFlex and PowerFlex Manager Administration Guide* and *Dell PowerFlex and PowerFlex Manager CLI Reference Guide*.

Discover and connect to the target PowerFlex system

Use the following procedure to obtain SDT IP addresses for discovering the target PowerFlex system.

Prerequisites

- Ensure that a volume is mapped to the host to be able to connect the SDT paths.
- Obtain SDT IP addresses.

Steps

1. From PowerFlex Manager, click **Block** > **Storage systems** > **NVMe Targets**.
2. Select any one of the listed SDTs.
3. Record the IP addresses and discovery port in the lower right corner.
4. To discover and connect the NVMe over TCP Target PowerFlex system, perform the following:

   NOTE: Use the esxcli command to perform the operation, there is issue with persistent discovery when perform the steps using UI.

   a. Log in to VMware ESXi server using SSH.
   b. Run a discovery query on each adapter:

```
#esxcli nvme fabrics discover -a vmhba6x -i 192.168.x.x -p 8009
#esxcli nvme fabrics discover -a vmhba6y -i 192.168.y.y -p 8009
```

Where 6x is the first NVMe over TCP software adapter, and 6y is the second NVMe over TCP software adapter.

Where 192.168.x.x is the first IP address and 192.168.y.y is the second IP address of any one of the SDTs.

c. Connect to the PowerFlex system (append **-c** on the discovery query command).

```
#esxcli nvme fabrics discover -a vmhba6x -i 192.168.x.x -p 8009 -c
#esxcli nvme fabrics discover -a vmhba6y -i 192.168.y.y -p 8009 -c
```

d. Get the controller list. Verify the connected controllers:

```
#esxcli nvme controller list
```

5. Perform a rescan of the storage.
   a. In the vSphere Client, browse to a host, a cluster, a data center, or a folder that contains hosts.
   b. From the right-click menu, select **Storage** > **Rescan Storage**.
   c. Select **Scan for new Storage Devices** and click **OK**.
6. Create a new VMFS datastore.
   a. Log in to vCenter
   b. Select **Home/Inventory** and select the storage icon.
   c. In the left right, click the PowerFlex compute-only cluster, and select **Storage/New Datastore**.
   d. Select **VMFS** and click **NEXT**.
   e. Enter the name of the datastore and select a host from the list.
      You should see a new disk called NVMe TCP Disk (eui.######)
   f. Select the disk and click **NEXT**.
   g. Select **VMFS 6** and click **NEXT**.
   h. Leave the default configuration unless instructed by the customer and click **NEXT**.
   i. Review your selections and click **FINISH**.

Automate SDC driver updates due to OS kernel patching

Use the following procedure to deploy SDC-only on SLES, Oracle Linux, or Ubuntu servers.

Ensure that the PowerFlex SDC is compatible with the Linux kernel.

NOTE: All references to Ubuntu in this section apply to supported versions of SLES and Oracle Linux.

To ensure that a client node always has an SDC kernel driver that matches the client kernel, the SDC startup script uses `driver_sync.sh`, a synchronization utility that fetches SDC kernel drivers from a configured repository. This repository can be either the Dell Technologies repository, or a local repository in your private network.

For Linux nodes with PowerFlex SDC drivers:

- The `driver_sync.sh` script is located under `<base_dir>`.
- In the repository, a driver package tarball is located at `<base_dir>/driver_cache/<distribution>/<PowerFlex_version>/<kernel_version>/<driver>.tar`.

Where:

- `<base_dir>` is `/bin/emc/scaleio/<driver>_sync` by default.
- `<distribution>` is one of: CentosStream8, CentosStream9, CoreOS, RH, RHEL8, RHEL9, SLES, SLES15.3, Ubuntu, Ubuntu16.04, Ubuntu18.04, Ubuntu20.04, or Ubuntu22.04
- `<kernel_version>` is the output of `uname -r`.
- `<driver>` is scini for SDC or xcache for RF.

It may be necessary to manually restart the driver/s and services on hosts.

In a normal work flow, an SDC driver is loaded on boot. If there is a mismatch between the version of the running kernel and the kernel against which the driver was compiled, the `driver_sync.sh` script is run automatically to fetch the appropriate driver. If the script fails for any reason (for example, due to incorrect configuration) after fixing the issue with the `driver_sync.sh` script, the driver services must be restarted manually, as described below. Root permission is required to run the commands.

To manually restart driver and services, run this command:

```
service scini restart
```

Ensure that a driver is available for the running kernel

Ensure that the kernel version is correct by performing the following steps.

Steps

1. Ensure that your kernel version matches a version in the repository, by performing the following:

   NOTE: If all servers have the same kernel, you can run this command on a single server.

   a. Run this command on every Ubuntu server:

   ```
   uname -r
   ```

   Output similar to the following is displayed:

   ```
   3.16.0-62-generic
   ```

   b. Copy the following text into a browser, and compare the output to the Ubuntu kernel version in the PowerFlex repository:

   ```
   sftp://mft.dell.com/
   ```

   NOTE: You can use Hypertext Transfer Protocol Secure (HTTPS) while checking driver availability manually.

2. To use a mirror repository, you must ensure that the SSH public and SSH private keys exist in all system nodes in the same path.

   The private key should have directory-level permission only (chmod 600 <private_key_path>).

3. The GPG key must be located in all system nodes in the same path. This key is required for using a mirror or Dell Technologies repository.

Next steps

The configuration of `driver_sync.sh` is specified in `driver_sync.conf` file, which can be created using the following method:

- Create the configuration file manually

NOTE: The SDC package now supports automated installation on Linux operating systems with SELinux. For supported operating systems, see the support matrix. To ensure that SELinux is activated and that SDCs are deployed as expected, verify that the relevant pre-requisite packages are installed.

Parameters to add to the driver_sync file

Add the following parameters to the driver synchronization utility.

Steps

1. Validate that the repository contains the kernel your operating system requires.

   ```
   sftp://mft.dell.com/
   ```

   Or

   ```
   https://mft.dell.com/
   ```

NOTE: You can use Hypertext Transfer Protocol Secure (HTTPS) while checking driver availability manually.

2. Download `sftp_keys_pkb.tar` from `<location needed>` and extract the contents to your SDC/RF host, as user root.
3. Place scini_repo_key.pub at the location specified in repo_public_rsa_key, `/etc/emc/scaleio/<driver>_sync/scini_repo_key.pub` by default.
4. Place scini_key at the location specified in user_private_rsa_key, `/etc/emc/scaleio/<driver>_sync/scini_key` by default. Where `<driver>` is scini for the SDC.
5. Change the permissions of a private key file.

```
chmod 600 /etc/emc/scaleio/scini_sync/scini_key
```

6. Edit /etc/emc/scaleio/scini_sync/driver_sync.confsimilar to the example below:

```
repo_address = sftp://mft.dell.com
repo_user = PowerFlexRepo
user_private_rsa_key = /etc/emc/scaleio/scini_sync/scini_key
repo_public_rsa_key = /etc/emc/scaleio/scini_sync/scini_repo_key.pub
```

NOTE: The `repo_password` line is not used with SFTP and can be left with the default value. Use the real password supplied above to check driver availability.

7. Restart the Storage Data Client RFCache driver:
   - Option1: Run from the operating system command prompt:

     ```
     systemctl restart <driver>
     ```

     NOTE: You can restart the Storage Data Client from the operating system only if the operating system is not using any volumes.
   - Option2: If needed, plan for a reboot cycle of the operating system for this setting to take effect.
8. Check that the `<driver>` is running:

```
systemctl status <driver>
```

9. Once you are finished, perform an operating system update and reboot to load the new kernel. The Storage Data Client driver sync automatically downloads the kernel object (KO) file, if available.

Create the configuration file manually

The `driver_sync.conf` file can be created or edited manually using a text editor. This topic is relevant for deployments on Linux flavors other than RedHat.

About this task

The file exists in the `etc` folder, with dummy values, and you must edit that file, and then copy it to the `etc` folder on every SDC server in the system. The `etc` folder is located in:

- SDC

Table 38. SDC host and path

Host	Path
Ubuntu/OL	`/etc/emc/scaleio/scini_sync/driver_sync.conf`

The configuration of the `driver_sync` script is read from `driver_sync.conf` (an example appears below), which contains the following parameters:

Table 39. driver_sync.conf parameters

Parameter	Description
repo_address	Encodes the chosen method and address.
repo_user	Identifies the username that is used when logging in to the drivers server. Required for SFTP and FTP methods.
repo_password	The FTP server login password (for user repo_user). Required for FTP method
user_private_rsa_key	Contains the path of the private RSA key file that is used to SFTP connection. This key file (provided by PowerFlex on installation) enables SFTP connection without a password.
local_dir	The local directory in which the downloaded SDC/RF drivers are stored. For SDC/RF clients, this value is set up during installation and does not require a change.
repo_public_rsa_key	The path of the public RSA key of the repository server machine (also known as the host public key).
module_sigcheck	Determines whether to verify (1) or not to verify (0) the downloaded kernel drivers.
emc_public_gpg_key	A key that is used to verify signatures of downloaded drivers. Enter the path to the public GPG signing key of PowerFlex (like the one that comes with PowerFlex LIA). This is needed only if module_sigcheck is set to 1.
The following parameters are used when driver_sync.sh is used to synchronize a local repository against a remote one.	
sync_pattern	A regular expression that can be used to fetch several drivers at once. This parameter is not needed, and becomes relevant only when your own private repository uses driver_sync.sh to fetch from a remote (For example: Dell) repository.
is_local_repo	Set to 0 (default) or omit, for SDC/RF clients. Set to 1 when driver_sync.sh is used to synchronize local repository against a remote repository. Setting to 1 causes the fetched tar files not to be extracted.

For example:

```
################################################
#driver_sync Configuration file
#Everything after a '#' until the end of the line is ignored
################################################
#Repository address, prefixed by protocol
repo_address = sftp://localhost/path/to/repo/dir
#repo_address = ftp://localhost/path/to/repo/dir
#repo_address = file://local/path/to/repo/dir
# Repository user (valid for ftp/sftp protocol)
repo_user = scini
# Repository password (valid for ftp protocol)
repo_password = scini
# Local directory for modules
local_dir = /bin/emc/scaleio/scini_cache/
# User's RSA private key file (sftp protocol)
user_private_rsa_key = /bin/emc/scaleio/scini_key
# Repository host public key (sftp protocol)
repo_public_rsa_key = /bin/emc/scaleio/scini_repo_key.pub
# Should the fetched modules' signatures be checked [0, 1]
module_sigcheck = 1
# EMC public signature key (needed when module_sigcheck is 1)
emc_public_gpg_key = /bin/emc/scaleio/emc_key.pub
# Sync pattern (regular expression) for massive retrieve
sync_pattern = .*
```

Perform the following steps for updating the driver_sync file on the storage data client:

Steps

1. Validate that the SFTP contains the kernel your operating system requires.

```
sftp://mft.dell.com/
```

Or

```
https://mft.dell.com/
```

NOTE: You can use Hypertext Transfer Protocol Secure (HTTPS) while checking driver availability manually.

2. Download and extract the attached sftp_keys to your SDC host, as user root, to /etc/emc/scaleio/scini_sync. Find the file that is attached to the article: sftp_keys_pkb.tar.
3. Change the permissions of a private key file.

```
chmod 600 /etc/emc/scaleio/scini_sync/scini_key
```

4. Edit /etc/emc/scaleio/scini_sync/driver_sync.confsimilar to the example below:

```
repo_address = sftp://mft.dell.com
                                    repo_user = PowerFlexRepo
                                    user_private_rsa_key = /etc/emc/scaleio/
scini_sync/scini_key
                                    repo_public_rsa_key = /etc/emc/scaleio/scini_sync/
scini_repo_key.pub
```

NOTE: The repo_password line is not used with SFTP and can be left with the default value. Use the real password supplied above to check driver availability.

5. Restart the storage data client driver:
   - Option1: Run from the operating system command prompt:

     ```
     systemctl restart scini
     ```

     NOTE: This is possible only if no volumes are in use by the operating system.
   - Option2: If needed, plan for a reboot cycle of the operating system for this setting to take effect.
6. Check that the `sciniservice` is running:

```
systemctl status scini
```

7. Once you are finished, perform an operating system update and reboot to load the new kernel. The storage data client driver sync automatically downloads the kernel object (KO) file, if available.

Update the PowerFlex signature key

If the PowerFlex package-signing key expires, you must update the configuration for each SDC.

About this task

When a PowerFlex package-signing key expires, a new public key is issued. You must update `driver_sync.conf` to use the new public key.

Steps

1. Download the new public key from the Dell Technologies Support site.
2. On each SDC node, edit `driver_sync.conf` so that the value of `emc_public_gpg_key` is the path to the new key file.

   The expired public key will still be in the gpg keyring, and older packages that were signed with the old key can still be verified.

   Location of `driver_sync.conf`:

- SDC

Table 40. SDC host and path

Host	Path
Ubuntu/OL	`/etc/emc/scaleio/scini_sync/driver_sync.conf`

Create a mirror repository

For Ubuntu or Oracle Linux (OL, also known as OEL), set up a local mirror repository (optional).

This section describes how to set up a local repository. This is necessary for scenarios when connecting all SDC nodes to the PowerFlex repository is not possible, perhaps due to security considerations or other limitations.

A common workflow would be to configure your mirror repository to synchronize with the PowerFlex repository, while your SDC nodes will fetch drivers (synchronize) from the mirror repository.

Installing the SDC manually

If a host of the SDC type was not installed on front-end servers during initial installation activities, follow the appropriate instructions for your front-end servers' operating system.

See the *Dell PowerFlex and PowerFlex Manager Support Matrix* to ensure that the intended SDC operating system of the host is compatible for use with your PowerFlex system.

You can use any of the following methods to install the SDC manually:

- VMware ESXi
- Linux
- Windows

Install the SDC on a VMware ESXi 8.x server

Install SDC on a VMware ESXi 8.x server using command-line esxcli commands.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.

  NOTE: Obtain and include the IP addresses of as many potential manager MDMs as possible when performing this task, to make it easier to switch MDMs in the future. You can add a total of eight IP addresses.

- Login credentials for the intended SDC host.
- The required installation software package for your operating system of SDC (available from the zipped software packages that can be downloaded from the Dell Technologies support site) .
- A GUID string for the SDC. These strings can be generated by tools that are freely available online. The GUID must conform to OSF DCE 1.1. The expected format is xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx where each x can be a digit (0–9) or a letter (a–f).

About this task

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system during initial deployment activities.

NOTE: This procedure requires two servers reboots.

Steps

1. On the VMware ESXi-based server on which you are installing the SDC, set the acceptance level:

```
esxcli software acceptance set --level PartnerSupported
```

2. Install the SDC:
   a. Change the directory where you have copied SDC.ZIP file.
   b. Unzip the SDC.ZIP file, run the following command: **`unzip <filename.zip>`**.
   c. Install the SDC package, run the following command: **`esxcli software vib install -v <path_to_vib>`**.

   Where **`<path_to_vib>`** is the path to SDC.VIB file which is unzipped in the above command.

3. Get permission to reboot the VMware ESXi-based server, and perform a graceful reboot cycle to load the SDC driver on the server.

   In a first-time install, the SDC will not automatically boot up at this point. Further configuration and an additional reboot are required, as described in this procedure.

4. Set the IP address of the MDM:

```
esxcli system module parameters set -m scini -p "IoctlIniGuidStr=<XXXXXX>
IoctlMdmIPStr=<LIST_VIP_MDM_IPS> IoctlMdmPasswordStr=<ip>-<password>"
```

   Where

   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses, or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
     - Separate the IP addresses of the same MDM cluster with a `,` symbol.
     - Separate multiple MDM clusters with the **+** symbol.

       In the following example, there are two MDM clusters. The first has two IP addresses and the second has only one:

       `10.20.30.40,50.60.70.80+11.22.33.44`
   - *<XXXXXX>* is the user-generated GUID string. Example GUID: **`12345678-90AB-CDEF-1234-567890ABCDEF`**
   - Optional: `IoctlMdmPasswordStr` is used for SDC authentication to the MDM for access to the PowerFlex system.

   CAUTION: A few applications require an advanced settings in ESXi for specific vital product data (VPD) pages. For more information about changing `blkScsiUnitSerialLenByStandard` parameter on an ESXi host for PowerFlex volumes, see Procedure to change parameter blkScsiUnitSerialLenByStandard on ESXi for PowerFlex volumes. Perform this procedure only if your applications require this setting. This is not a common setting.

5. Back up the parameters:

```
/bin/auto-backup.sh
```

6. Perform another graceful reboot cycle.
7. Ensure that the driver is loaded:

```
vmkload_mod -l | grep scini
```

   Output similar to the following indicates that the driver is loaded:

```
scini 2 808
```

   The SDC component is installed on the VMware ESXi-based server.

Results

The SDC is installed on the VMware ESXi-based server and is connected to PowerFlex.

Install the SDC on a VMware VCF/ESXi 9.x based server

Use this procedure to install the SDC on a VMware VCF/ESXi 9.x based server.

About this task

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system during initial deployment activities.

NOTE: This procedure requires multiple servers reboots.

Prerequisites

Ensure that you have the following:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.

  NOTE: Obtain and include the IP addresses of as many potential manager MDMs as possible when performing this procedure to make it easier to switch MDMs in the future. You can add a total of eight IP addresses.

- Login credentials for the intended SDC host.
- The required installation software package for your operating system of SDC (available from the zipped software packages that can be downloaded from the Dell Technologies support site).
- A GUID string for the SDC. These strings can be generated by tools that are available online, for example GUID generator or UUID generator. The GUID must conform to OSF DCE 1.1. The expected format is xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx where each x can be a digit (0–9) or a letter (a–f).

Steps

1. On the VMware ESXi-based server on which you are installing the SDC, set the acceptance level, type the following: **`esxcli software acceptance set --level PartnerSupported`**
2. Put the VMware ESXi node into maintenance mode: **`esxcli system maintenanceMode set --enable true`**
3. Install the SDC agent package: **`esxcli software component apply -d /<PATH>/scaleio-daemon-esx_5.x.x.xxx-x.zip`**

   Where **`<PATH>`** is the directory containing the SDC agent package (.zip file).

4. Install the SDC driver package: **`esxcli software component apply -d /<PATH>/scaleio-sdc-esx9x_5.x.x.xxx-xOEM.xxx.x.xxxxxxxx.zip`**.
5. Get permission to reboot the VMware ESXi-based server, and perform a graceful reboot cycle to load the SDC driver: **`reboot`**
6. Ensure that the driver is loaded: **`vmkload_mod -l | grep scini`**.

   Output similar to the following indicates that the driver is loaded:

   ```
   scini 2 808
   ```

7. Set the module parameters:

   ```
   esxcli system module parameters set -m scini -p "IoctlIniGuidStr=<XXXXXX>
   IoctlMdmIPStr=<LIST_VIP_MDM_IPS> IoctlMdmPasswordStr=<ip>-<password>"
   ```

   Where

   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the virtual IP addresses, or the MDM IP address. Do not include tiebreaker addresses here.
     - Separate the IP addresses of the same MDM cluster with a `,` symbol.
     - Separate multiple MDM clusters with the + symbol.

       In the following example, there are two MDM clusters. The first has two IP addresses and the second has only one, use the following format for the `IoctlMdmIPStr` parameter: `10.20.30.40,50.60.70.80+11.22.33.44`
   - `<XXXXXX>` is the user-generated GUID string. Example GUID: `12345678-90AB-CDEF-1234-567890ABCDEF`
   - Optional: `IoctlMdmPasswordStr` is used for SDC authentication to the MDM for access to the PowerFlex system.

For example:

```
esxcli system module parameters set -m scini -p "IoctlIniGuidStr=a9745569-
eed1-43fe-82fd-b28199dd7ffb  IoctlMdmIPStr=192.168.211.93,172.168.211.108"
```

8. Reboot the VMware ESXi host to set the parameters.
9. Exit the VMware ESXi host from maintenance mode: **esxcli system maintenanceMode set --enable false**
10. Back up the parameters: **/bin/auto-backup.sh**.
11. Log in to the primary MDM or PowerFlex Manager to verify that the SDC is connected, and proceed with volume mapping.

Install the SDC on a Linux-based server

Install the storage data client on a Linux-based server with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more storage data clients to an existing system, and for adding storage data clients to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node.
- The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the Dell Technologies support site).
- Secure boot is not supported; ensure that secure boot is disabled in the BIOS.

About this task

This section describes how to manually install an external SDC on a Linux-based server using the command line.

The steps in the procedure apply to Citrix Hypervisor(XenServer)/Red Hat Enterprise Linux/Oracle Linux operating systems.

Steps

1. Install the GPG key on every server on which SDC is installed. From the PowerFlex installation folder, run the following command on every server:

   ```
   rpm --import RPM-GPG-KEY-ScaleIO
   ```

2. From the command line, install the SDC:
   - Red Hat Enterprise Linux/Oracle Linux/SLES

     ```
     MDM_IP=<LIST_VIP_MDM_IPS> rpm -i <SDC_PATH>.rpm
     ```

   - Ubuntu (Extract these files before use, as described above.)

     ```
     MDM_IP=<LIST_VIP_MDM_IPS> dpkg -i <SDC_PATH>.deb
     ```

   Where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
   - *<SDC_PATH>* is the path where the SDC installation package is located
3. Verify that the driver is loaded using one of the following commands:
   - SLES 15:

     ```
     systemctl status scini
     ```

   - All other Linux:

     ```
     service scini status
     ```

Results

The SDC is installed on the Linux-based server, and is connected to PowerFlex.

Install the SDC on an AIX-based server

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node.
- The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the Dell Technologies support site).

About this task

The following procedure explains how to manually install an external SDC on an AIX-based server.

Steps

1. From the command line, install the SDC:

   ```
   MDM_IP=<LIST_VIP_MDM_IPS> rpm -i <SDC_PATH>.rpm
   ```

   Where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
   - *<SDC_PATH>* is the path where the SDC installation package is located.

     The SDC package is in a format similar to this: `EMC-ScaleIO-sdc-.aix7.aix7.2.ppc.rpm`

2. Optional: For SDC authentication to the MDM for access, enter the following command:

   ```
   chdev -l scini -a "mdm_password=<ip>-<password>"
   ```

Results

The SDC is installed on the AIX-based server and is connected to PowerFlex.

Install SDC on a Windows-based server

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node.
- The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the Dell Technologies support site).

About this task

The following procedure explains how to manually install an external SDC on a Windows-based server.

NOTE: This procedure requires a server reboot.

Steps

1. On the Windows-based server on which you are installing the SDC, run the following command:

   ```
   msiexec /i <SDC_PATH>.msi MDM_IP=<LIST_VIP_MDM_IPS>
   ```

   where
   - *<SDC_PATH>* is the path where the SDC installation package is located.
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses, or the virtual IP address of the MDM. Do not add tiebreaker addresses here.
2. Get permission to reboot the Windows-based server, and perform a reboot to load the SDC driver on the server.

Results

The SDC is installed on the Windows-based server, and is connected to PowerFlex.

Run the storage data client driver on Red Hat Enterprise Linux or SUSE Enterprise Linux Enterprise Server with multipath enabled

Run the storage data client driver on Red Hat Enterprise Linux or SUSE Enterprise Linux Enterprise Server with multipath enabled

Steps

1. On the machine where the storage data client is running, open `/etc/udev/rules.d/20-scini.rules` for editing.
2. In the first line at the end add:

   ```
   ENV{ID_WWN}="%c"
   ```

   The line should read like this:

   ```
   KERNEL=="scini*[!0-9]", SUBSYSTEM=="block", PROGRAM="/bin/emc/scaleio/drv_cfg --
   query_block_device_id --block_device $tempnode", SYMLINK+="disk/by-id/emc-vol-%c",
   ENV{ID_BUS}="scsi", ENV{ID_SERIAL}="%c", ENV{ID_WWN}="%c"
   ```

3. Create or open `/etc/multipath.conf` and add the following additional settings to the configuration file:

   ```
   defaults {
   retain_attached_hw_handler "no"
   find_multipaths no
   }
   ```

   ```
   blacklist_exceptions {
   devnode "scini"
   }
   ```

   NOTE: From the command line, run `dracut -f.` This includes the changes in initrd.
   `[retain_attached_hw_handler no]` is the default setting in Red Hat Enterprise Linux 7.x or 8.x.
4. Restart the PowerFlex service and multipathd service:

   ```
   systemctl restart scini
   ```

   ```
   systemctl restart multipathd.service
   ```

NOTE: If LUNs are in use and a restart is not possible, restart the node.

Enable OpenSSL FIPS compliance on the storage node, MDM, LIA

Enable the implementation of OpenSSL Federal Information Processing Standards (FIPS) compliance on the storage node, MDM, LIA for communication between the external components, including PowerFlex Manager, PowerFlex gateway, and CLI, to the MDM. It is also enabled for any other usage of the OpenSSL library.

Prerequisites

The MDM must be hosted on Linux with the OpenSSL package installed.

Steps

1. On each host running PowerFlex, open the configuration file of each component with a text editor.
   The configuration file is `/opt/emc/scaleio/<COMPONENT>/cfg/conf.txt`, where *<COMPONENT>* is the lowercase name of the component (for example, pds).
2. Add the parameter `security_enable_fips=1` to the file.
3. Save and close the file.
4. Open the SCLI configuration file with a text editor:
   The configuration file is at: `~/.scli/conf.txt`.
5. Add the parameter `security_enable_fips=1` to the file.
6. Save and close the file.
7. On each host, restart each component's service:

```
service pds restart
service dgwt restart
```

8. Verify that OpenSSL FIPS compliance has been enabled by running:

```
cat  /proc/sys/crypto/fips_enabled
```

   If it has been enabled correctly, the output should be `1`. If the output is not `1`, enable OpenSSL FIPS at the operating system level.

Check port usage

If you are using a firewall to protect your network, and need to enable ports and IP addresses, see the *Dell PowerFlex Family 4.x Security Configuration Guide*. This guide contains information about port usage for all components, and what to modify in case port numbers need to be changed.

Configure the SDC-only (scini) driver for SELinux

SELinux may prevent the scini driver from starting on Red Hat Enterprise Linux-based SDC nodes. This procedure prevents this issue.

About this task

If SELinux is enabled and enforced, it can prevent the SDC-only driver (scini) from starting on Red Hat Enterprise Linux-based nodes.

NOTE: To remove the SELinux module, run: `semodule -r scini`

Steps

1. On the SDC, use a text editor to create two text files in any directory:
   a. Create a file called `scini.fe`, and add the following text:

```
/usr/bin/emc/scaleio/(.*).ko  system_u:object_r:modules_object_t:s0
/bin/emc/scaleio/(.*).ko  system_u:object_r:modules_object_t:s0
```

   b. Create a file called `scini.te`, and add the following text:

```
module scini 1.0;

require {
        type insmod_t;
        type modules_object_t;
        class system module_load;
}
```

2. Build and install the SELinux module, by running the following commands:

```
checkmodule -M -m scini.te -o scini.mod
semodule_package -o scini.pp -m scini.mod -f scini.fe
semodule -i scini.pp
restorecon -R -v /bin/emc/scaleio/
```

3. To check the current configuration, run the command:

```
sestatus
```

   Output similar to the following should be displayed, showing that *SELinux status* is enabled, and *Current mode* is enforcing:

```
SELinus status: enabled
SELinuxfs mount: /sys/fs/selinux
SELinux root directory: /etc/selinux
Loaded policy name: targeted
Current mode: enforcing
Mode from config file: enforcing
Policy MLS status: enabled
Policy deny_unknown status: allowed
Memory protection checking: actual (secure)
Max kernel policy version: 31
```

4. To check that the scini module is configured in SELinux, run the command:

```
semodule --list=full | grep scini
```

   Output similar to the following should be displayed:

```
400    scini                pp
```

5. Start the scini service:

```
service scini start
```

Mount a PowerFlex volume on SDC-only

The exposed PowerFlex volumes are connected to the servers by the network. To configure the mounting options of PowerFlex devices, follow the instructions for your specific Linux-based operating system, below.

About this task

Use persistent device names since the `/dev/sciniX` names are not guaranteed to persist between reboots.

Steps

1. Determine the `/dev/disk/by-id` correlation to `/dev/sciniX`:

   ```
   ls -l /dev/disk/by-id/ |grep scini
   ```

   Output similar to the following appears:

   ```
   lrwxrwxrwx 1 root root 12 Mar 2 05:35 emc-vol-7ec27ef55b8f2108-85a0f0330000000a ->
   ../../scinia
   lrwxrwxrwx 1 root root 12 Mar 2 05:35 emc-vol-7ec27ef55b8f2108-85a0f03200000009 ->
   ../../scinib
   lrwxrwxrwx 1 root root 12 Mar 2 05:35 emc-vol-7ec27ef55b8f2108-85a0f02c00000003 ->
   ../../scinic
   ```

2. Run the mount command:

   ```
   /dev/disk/by-id/<EMC-vol-id> <mount point>
   ```

   Example:

   ```
   mount /dev/disk/by-id/emc-vol-7ec27ef55b8f2108-85a0f0330000000a /mnt_scinia
   ```

3. To make the mount command persistent, edit the `/etc/fstab` file according to the instructions for your operating system:
   - Red Hat Enterprise Linux 7.x:

     In `/etc/fstab`, use a text editor to add `_netdev` to the PowerFlex mount lines.

     Example:

     ```
     /dev/disk/by-id/emc-vol-7ec27ef55b8f2108-85a0f0330000000a /mnt_scinia xfs
     defaults,_netdev 0 0
     ```

     Ensure that you comply with the `netdev` and syntax rules for your file system, as described in the `man` page.

   - SLES:

     In `/etc/fstab`, use a text editor to add `nofail` to the mount lines.

     Example:

     ```
     /dev/disk/by-id/emc-vol-7ec27ef55b8f2108-85a0f0330000000a /mnt_scinia xfs nofail 0 0
     ```

     Ensure that you comply with the `nofail` and syntax rules for your file system, as described in the `man` page.

Associate PowerFlex volumes with physical disks on MDM-only

If necessary, you can associate PowerFlex volumes with physical disks.

To get PowerFlex volume information, run the CLI command `scli --query_all_volumes` (or `--query_all` or `--query_volume`).

Output similar to the following appears:

```
Query-all-volumes returned 10 volumes
Protection Domain 0728185d00000000 Name: pd1
Storage Pool ad99eaab00000000 Name: default
<No volumes defined>

Storage Pool ad99eaac00000001 Name: sp1
Volume ID: fac22a6300000000 Name: vol0 Size: 152.0 GB (155648 MB) Mapped to 1 SDC Thick-provisioned
Volume ID: fac22a6400000001 Name: vol1 Size: 400.0 GB (409600 MB) Mapped to 1 SDC Thin-provisioned
Volume ID: fac22a6500000002 Name: vol2 Size: 80.0 GB (81920 MB) Mapped to 1 SDC Thick-provisioned
Volume ID: fac22a6600000003 Name: vol3 Size: 392.0 GB (401408 MB) Mapped to 1 SDC Thick-provisioned
Volume ID: fac22a6700000004 Name: vol4 Size: 96.0 GB (98304 MB) Mapped to 1 SDC Thin-provisioned
Volume ID: fac22a6800000005 Name: vol5 Size: 112.0 GB (114688 MB) Mapped to 1 SDC Thick-provisioned
Volume ID: fac22a6900000006 Name: vol6 Size: 96.0 GB (98304 MB) Mapped to 1 SDC Thin-provisioned
Volume ID: fac22a6a00000007 Name: vol7 Size: 176.0 GB (180224 MB) Mapped to 1 SDC Thick-provisioned
Volume ID: fac22a6b00000008 Name: vol8 Size: 272.0 GB (278528 MB) Mapped to 1 SDC Thick-provisioned
Volume ID: fac22a6c00000009 Name: vol9 Size: 360.0 GB (368640 MB) Mapped to 1 SDC Thin-provisioned
```

This output shows the volume ID and name, and other volume information.

Retrieve volume information for Linux from SDC-only

Retrieve volume information from the SDC host.

On the SDC host, run the following command to get the operating system volume information that correlates to the PowerFlex scini device name:

```
ls -l /dev/disk/by-id/ |grep scini
```

Output similar to the following appears:

```
lrwxrwxrwx 1 root root 12 Aug 25 19:40 emc-vol-62c093a52d14aec7-fac22a6300000000 -> ../../scinia
lrwxrwxrwx 1 root root 12 Aug 25 19:40 emc-vol-62c093a52d14aec7-fac22a6400000001 -> ../../scinic
lrwxrwxrwx 1 root root 12 Aug 25 19:40 emc-vol-62c093a52d14aec7-fac22a6500000002 -> ../../scinib
lrwxrwxrwx 1 root root 12 Aug 25 19:41 emc-vol-62c093a52d14aec7-fac22a6600000003 -> ../../scinie
lrwxrwxrwx 1 root root 12 Aug 25 19:41 emc-vol-62c093a52d14aec7-fac22a6700000004 -> ../../scinid
lrwxrwxrwx 1 root root 12 Aug 25 19:42 emc-vol-62c093a52d14aec7-fac22a6800000005 -> ../../scinif
lrwxrwxrwx 1 root root 12 Aug 25 19:42 emc-vol-62c093a52d14aec7-fac22a6900000006 -> ../../scinig
lrwxrwxrwx 1 root root 12 Aug 25 19:42 emc-vol-62c093a52d14aec7-fac22a6a00000007 -> ../../scinii
lrwxrwxrwx 1 root root 12 Aug 25 19:42 emc-vol-62c093a52d14aec7-fac22a6b00000008 -> ../../scinih
lrwxrwxrwx 1 root root 12 Aug 25 19:43 emc-vol-62c093a52d14aec7-fac22a6c00000009 -> ../../scinij
```

This output shows the scini volume name and the volume ID.

By matching the volume ID in both outputs, you can match the operating system name, sciniX, with the PowerFlex volume name.

For example:

- scinia = fac22a6300000000 = vol0
- scinic = fac22a6400000001 = vol1

Alternatively, run the `sg_inq /dev/sciniX` SCSI query command (which requires that the sg3_utils be installed on the Linux host). The result of this command includes the EMC volume ID at the bottom of the output, as illustrated:

```
Vendor identification: EMC
Product identification: ScaleIO
Product revision level: 1.3
Unit serial number: EMC-62c093a52d14aec7-fac22a6300000000
```

NOTE: The product identification remains as ScaleIO (not PowerFlex).

Add PowerFlex volumes to Linux LVM physical volumes on SDC-only

Add PowerFlex volumes to Linux LVM physical volumes.

Steps

1. In the `/etc/lvm/lvm.conf` file add the following lines:

```
types = [ "scini", 16 ]
```

2. After configured, the `lvmdiskscan` command should yield results similar to the following:

```
/dev/scinia  [      96.00 GiB] LVM physical volume
/dev/scinib  [     320.00 GiB] LVM physical volume
/dev/scinic1 [      56.00 GiB]
/dev/scinid  [      32.00 GiB]
1 disk
1 partition
2 LVM physical volume whole disks
0 LVM physical volumes
```

3. Continue with normal LVM steps.

Retrieve volume information for Windows from MDM-only

You can retrieve volume information from the MDM.

About this task

The `sg_inq.exe` file was added to the MSI installation and can be found at `C:\Program Files\EMC\ScaleIO\SDC\diag\`.

Steps

1. Run the `sg_inq HardiskX` SCSI query command.
   The result of this command includes the EMC volume ID at the bottom of the output.
2. On the MDM, get the PowerFlex volume information:

```
C:\Program Files\emc\scaleio\sdc\bin\drv_cfg --query_vol
```

   Output similar to the following is displayed:

```
Retrieved 5 volume(s)
VOL-ID 6acb988100000000 MDM-ID 0b246c9a755ca3dd
VOL-ID 6acb988200000001 MDM-ID 0b246c9a755ca3dd
VOL-ID 6acb988300000002 MDM-ID 0b246c9a755ca3dd
VOL-ID 6acb988400000003 MDM-ID 0b246c9a755ca3dd
VOL-ID 6acb988500000004 MDM-ID 0b246c9a755ca3dd
```

3. From the Windows command prompt, run this command:

```
wmic diskdrive get deviceid,serialnumber | findstr "EMC"
```

   Output similar to the following is displayed:

   `\\.\PHYSICALDRIVE13 EMC-0b246c9a755ca3dd-6acb988500000004`

   The first part of the output is the disk name. In our example: `PHYSICALDRIVE13`

   The second part is the disk serial number. The last set of the second part (after the dash) is the PowerFlex volume ID. In our example: `6acb988500000004`

Retrieve volume information for AIX from SDC-only

On AIX servers, associate the PowerFlex volume ID with the AIX physical device.

Steps

1. On the SDC host, run the following command to get the operating system volume information:

```
#odmget -q "name like scinid* and attribute=vol_id" CuAt
```

Output similar to the following is displayed:

```
CuAt:
name = "scinid0"
attribute = "vol_id"
value = "e120a92d00000000"
type = "R"
generic = "D"
rep = "s"
nls_index = 22
[root@cnode02 /]#odmget -q "name like scinid* and attribute=vol_id" CuAt

CuAt:
name = "scinid2"
attribute = "vol_id"
value = "e120a92f00000002"
type = "R"
generic = "D"
rep = "s"
nls_index = 22

CuAt:
name = "scinid8"
attribute = "vol_id"
value = "e120a93500000008"
type = "R"
generic = "D"
rep = "s"
nls_index = 22

CuAt:
name = "scinid0"
attribute = "vol_id"
value = "e120a92d00000000"
type = "R"
generic = "D"
rep = "s"
nls_index = 22
```

You can get information for a single volume by using this command:

```
#odmget -q "name=scinid0 and attribute=vol_id" CuAt
```

2. Match the value of the `value` field with the PowerFlex volume ID.

Configure the SDS for optimal memory usage

For certain system configurations, the SDS must be configured to be able to use more than one nonuniform memory access (NUMA) domain.

About this task

Often, memory is split into a few NUMA domains. For example, in a two-CPU-socket server configuration, each CPU is in a different NUMA domain. The SDS is configured by default to have affinity to socket 0 in a server or VM, and therefore, by default, the SDS is connected only to NUMA 0, and only has access to the memory in NUMA 0.

In VMware ESXi hyperconverged infrastructure configurations, the NUMA definition requires special attention. When configuring the SVM, there are cases where the SVM includes two vNUMAs. For example, if the number of vCPUs required for the SVM is larger than the number of physical cores in the CPU, VMware ESXi configures the SVM with two vNUMAs. Other cases are VMware ESXi hyperconverged infrastructure with replication, where the SVM includes both the SDR and the SDS. The SDS is configured on vNUMA 0 and the SDR on vNUMA 1. The total memory is split equally between the two vNUMAs, and therefore, the SDS may not have enough memory on NUMA 0.

The memory size required for the SDS depends on the configuration, and on other factors. In order to determine the required size for the SDS, and whether NUMA configuration is required, follow the size recommendations provided in the *Dell PowerFlex 4.6.x Technical Overview*.

If the memory supported by NUMA 0 (usually half the total memory) is less than the amount of memory required by the SDS, you must allow the SDS access to the memory in the other NUMA. For example, if a node has 96 GiB of physical memory, divided into 48 GiB on each CPU, and the SDS requires 58 GiB of memory, this NUMA modification is required, because 48 GiB is not sufficient for the SDS.

This operation has a slight impact to performance (less than 10%), and for this reason, is not the default setting.

NOTE: This setting can also be modified on operational SDSs that are expected to use more memory in the future, for example, due to adding SDS devices. This procedure requires a graceful SDS restart in order to take effect.

Steps

1. Use one of the following options to edit the NUMA configuration:

Option	Description
Use the echo command.	On the node where the SDS is deployed, issue the command: `echo "numa_memory_affinity=0" >> /opt/emc/scaleio/sds/cfg/conf.txt`
Use a text editor.	a. On the node where the SDS is deployed, navigate to the file: `/opt/emc/scaleio/sds/cfg/conf.txt` b. Open the file with a text editor, and add the following lines: *numa_memory_affinity=0* c. Save the file.

2. Gracefully restart the SDS:
   a. Use one of the PowerFlex management interfaces to place the SDS in instant maintenance mode.
   b. Restart the SDS process:

```
/opt/emc/scaleio/sds/bin/delete_service.sh
/opt/emc/scaleio/sds/bin/create_service.sh
```

   c. Exit instant maintenance mode.
   d. Wait for the rebuild to finish before performing other tasks.

Next steps

Repeat this procedure on each SDS that requires this modification.

Configure the SDS-only for NVDIMM or SDPM health awareness

Optionally, configure the SDS to enable periodic monitoring of bad blocks in NVDIMM or SDPM devices.

Prerequisites

Root access permissions are required for this procedure.

About this task

To enable this feature in the PowerFlex system, perform this procedure on every SDS that uses NVDIMM or SDPM acceleration devices. This feature is disabled by default.

Steps

1. In the command line, run the command:

```
echo "acc_dev_ndctl_error_detection_enable=1" >> /opt/emc/scaleio/sds/cfg/conf.txt
```

2. Restart the SDS:

```
/opt/emc/scaleio/sds/bin/delete_service.sh
```

```
/opt/emc/scaleio/sds/bin/create_service.sh
```

3. Wait until system rebuild or rebalance is complete, and then repeat steps 1 and 2 on the next SDS, until all SDSs are configured.

Performance tuning post-installation

This section describes steps to take after completing a successful installation to enhance performance.

You can improve the PowerFlex system performance in terms of IOPS, latency, and bandwidth, by performing environment-specific fine-tuning on the operating system, network, and PowerFlex components.

NOTE: Performance tuning is case-specific. To prevent undesirable effects, it is highly recommended to thoroughly test all changes. For further assistance, contact Dell Technologies Support.

Fine granularity layout performance considerations on SDS-only

Fine granularity layout enables 4 KB granularity and compression.

Table 41. Layout and performance considerations

Fine granularity	Medium granularity
The benefit of the more efficient fine granularity data layout in storage pools that allows 4 KB granularity plus compression.	In contrast to 1 MB granularity in medium granularity data layout, comes at some performance cost.
In most cases fine granularity is slower than medium granularity. When compression is enabled, reads are about 20% slower than in cases where compression is disabled. There are exceptions to these rules. For example, a small I/O write response time is similar in fine granularity and medium granularity.	Faster than fine granularity.
Snapshots have no impact on the performance of applications and workloads running during the snapshot creation	Snapshots might have an impact on performance in some cases. In cases that are very sensitive to performance and do not require snapshots, medium granularity would be a better choice. It would also be a good option for cases where the data cannot be compressed, as it may already be compressed, or if the data is encrypted by the application.

In cases that are very sensitive to performance and do not require snapshots, medium granularity would be a better choice. It would also be a good option for cases where the data cannot be compressed, as it may already be compressed, or if the data is encrypted by the application.

See the *Increase SDS threads* topic for details about increasing SDS threads for better performance in related information. For best performance, it is recommended to increase the number of SDS threads from eight to twelve when the CPUs have twelve or more physical cores.

NOTE: Increasing the SDS threads is expected to improve performance in cases where the CPU is bound; for example: small I/Os or fine granularity layout. In cases where performance is limited by drive performance (too few drives) or by the network (usually large I/Os), changing the number of SDS threads does not help.

Fine granularity metadata cache on MDM-only

Fine granularity MD_Cache helps improve the performance of the system when using the fine granularity layout.

When using the fine granularity layout, the fine granularity MD_Cache feature provides the following performance benefits:

- 20-40% increase in read IOPS
- Reductions in response times for reads

Fine granularity MD_Cache allocates a small portion of system memory to each SDS in the protection domain, for caching the metadata. The following formula below shows the amount of memory needed for fine granularity MD_Cache per SDS:

```
FGMC_RAM_in_GiB=(Total_Drive_Capacity_in_TiB/
2)*4%*Compression_Factor*Percent_of_Metadata_to_Cache
```

It is recommended to allocate 2 for compression factor and 2% for metadata to cache. Use the sizer tool to assess the amount of memory to allocate for metadata to cache. The maximum amount of memory that can be allocated per SDS is 32 GiB.

1. Run the following command to allocate the memory used by fine granularity MD_Cache on each SDS in the protection domain.

   ```
   scli --set_default_fgl_metadata_cache_size (--protection_domain_id <ID> | --
   protection_domain_name <NAME>) --metadata_cache_size_mb <SIZE>
   ```

2. Run the following command to enable fine granularity MD_Cache:

   ```
   scli --enable_fgl_metadata_cache (--protection_domain_id <ID> | --
   protection_domain_name <NAME>)
   ```

3. Run the following command to disable fine granularity MD_Cache:

   ```
   scli --disable_fgl_metadata_cache (--protection_domain_id <ID> | --
   protection_domain_name <NAME>)
   ```

Tuning considerations

In version 3.x and later, high_performance profile is the default setting. Medium Granularity configuration can be changed to the compact profile, which would require less memory and CPU resources, but may have some impact on the performance of the setup. With fine granularity configuration, and systems with a combination of fine and medium granularity, only high_performance profile is allowed.

The main difference between the high_performance (default) and compact profiles is the amount of server resources (CPU and memory) that are consumed. A high_performance profile (or configuration) will always consume more resources.

Users achieve optimum performance by always setting the performance profile to high_performance.

PowerFlex system changes for PowerFlex components using MDM

This section describes the various system changes available for enhancing PowerFlex performance.

Using the set_performance_parameters utility for MDM and storage nodes

The following table describes the commands for specific tasks:

Table 42. set_performance_parameters commands

Action	Command
View current settings.	`scli --query_performance_parameters`
By default is High_performance. Run the command to change to High_performance if the value is not set to High_performance.	`scli --set_performance_parameters --all_storage_node --all_sdc --apply_to_mdm --profile high_performance`
Change the profile back to compact.	`scli --set_performance_parameters --all_storage_node --all_sdc --apply_to_mdm --profile compact`
View full parameter settings for an MDM.	`scli --query_performance_parameters --print_all`
View the full parameter settings of a specific storage node (this also shows the MDM settings).	`scli --query_performance_parameters --storage_node_name<NAME> --print_all`
View the full parameter settings of a specific SDC (this also shows the MDM settings).	`scli --query_performance_parameters --sdc_name<NAME> --print_all`

Read RAM cache settings for SDS-only using MDM

You should keep read RAM cache disabled for SSD/flash pools. For HDD pools, read RAM cache can help increase performance. For PowerFlex storage-only nodes, turn on read RAM, the recommendation is to turn on read RAM cache for HDD pools and use as much of the server DRAM as possible.

In a configuration where PowerFlex shares the node with other applications, depending on the available DRAM resources, it may also help to turn on read RAM cache for HDD pools and increase the cache size from the default.

The following table describes the commands that are used for enabling read RAM cache and increasing the amount of read RAM cache.

NOTE: The read RAM cache may be enabled/disabled on the Protection Domain, or for each SDS in the cluster.

Table 43. Read RAM cache settings for SDS

Action	Command
Enable read RAM cache	Run the following commands: • `scli --set_rmcache_usage --protection_domain_name <domain NAME> --storage_pool_name <pool NAME> --`

Table 43. Read RAM cache settings for SDS (continued)

Action	Command
	use_rmcache [--dont_use_rmcache] • scli --enable_sds_rmcache [--disable_sds_rmcache] --sds_name <NAME> **NOTE:** These commands are required for every SDS in the cluster.

Increase the amount of read RAM cache	Run the following command:
	scli --set_sds_rmcache_size ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>]) | (--protection_domain_id <ID> | --protection_domain_name <NAME>)) --rmcache_size_mb <SIZE> [--i_am_sure]
	Where --rmcache_size_mb is the size of read RAM cache in MB, and the range is between 128MB—300GB.
	Ensure that read RAM cache is enabled at all levels (protection domain, storage pool, and SDS). When rmcache is properly enabled, the query output will look like this:
	`# scli --query_sds --sds_name SDS1`
	`SDS 60b4bdf300000003 Name: SDS1 Version: 4.5.2000`
	`Protection Domain: a5c9906f00000000, Name: PD1`
	`Medium granularity DRL mode: Volatile`
	`Authentication error: NONE`
	`IP information (total 2 IP Addresses):`
	`1: 192.168.160.237 Role: All (SDS and SDC)`
	`2: 192.168.152.237 Role: All (SDS and SDC)`
	`Port: 7072`
	`I/O flow control is enabled`
	`Fine granularity number of concurrent writes: 1000`
	`RAM Read Cache information:`
	`128.0 MB (131072 KB) total size`
	`Cache is enabled`
	`RAM Read Cache memory allocation state is successful`

Jumbo frames and the potential impact on performance for storage nodes and SDC-only

Enabling jumbo frames can result in approximately 10% improvement in performance if all network components fully support jumbo frames. If some network components do not fully support jumbo frames, it is recommended to use the default setting; MTU 1,500.

Before activating MTU settings on the logical level, set jumbo frames = MTU 9000 on the physical switch ports that are connected to the node. Failure to do so may lead to network disconnects and packet drops.

See your relevant vendor guidelines on how to configure jumbo frame support.

Configure jumbo frames for Linux

Configure jumbo frames for NIC cards in the Linux-based PowerFlex nodes.

Steps

1. Run the `ifconfig` command to get the NIC information.
2. Depending on the operating system, run the command:

Operating system	Command
Red Hat Enterprise Linux	Edit `/etc/sysconfig/network-scripts/ifcfg-` *<NIC_NAME>*
SLES	Edit `/etc/sysconfig/network/ifcfg-`*<NIC_NAME>*

3. Add the parameter `MTU=9000` to the file.
4. To apply the changes, type: **`service network restart`**
5. Run `ifconfig` again to verify that the settings have been changed.
6. To test the command, type: **`ping -M do -s 8972`***<DESTINATION_IP_ADDRESS>*
   Output should look similar to the following test ping to ensure MTU setting:

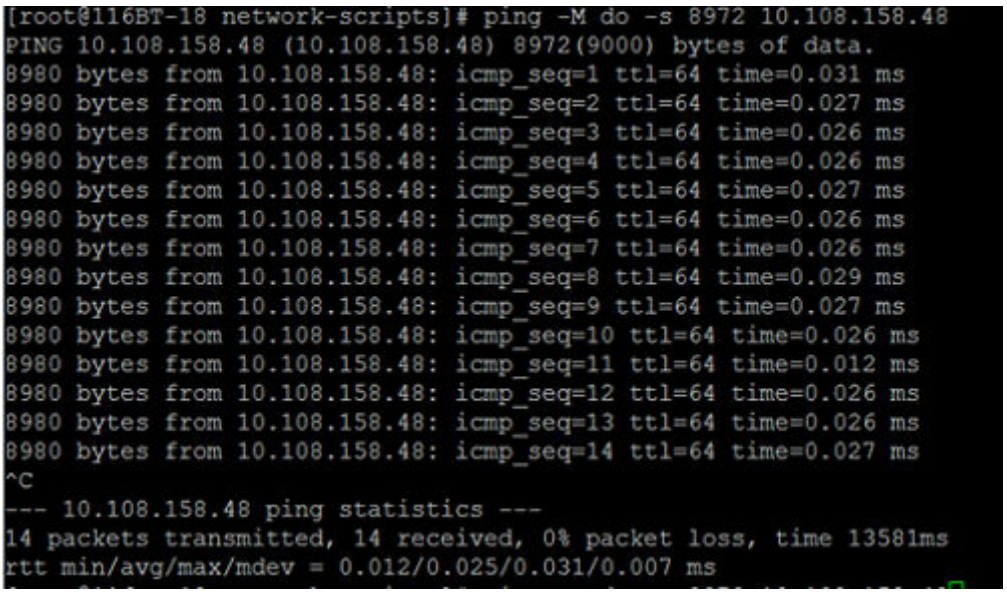


Figure 10. Test ping of MTU setting

Configure jumbo frames for Windows

Configure jumbo frames for NIC cards in the Windows-based PowerFlex nodes.

Steps

1. Change the `Maximum Transmission Unit (mtu)` setting to `9,000`, or the highest value that is supported by the switch and the connected nodes.
2. Determine the appropriate NIC name by typing the command:

```
netsh interface ipv4 show interface
```

   Sample output:

```
Idx     Met         MTU          State                Name
---  ----------  ----------  ------------  ---------------------------
  1          75  4294967295  connected     Loopback Pseudo-Interface 1
 10           5        1500  disconnected  Ethernet
  3          25        1500  connected     mgmt
  6           5        1500  disconnected  Ethernet 3
 12           5        1500  disconnected  Ethernet 4
  7          15        1500  connected     Data1
  9          15        9028  connected     Data2
```

   In this example, index 7 is the appropriate network.
3. Type the command:

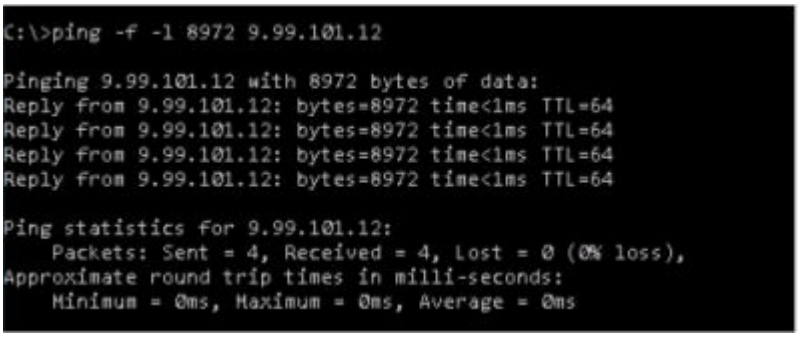
```
netsh interface ipv4 set subinterface <network_ID> mtu=9000 store=persistent
```

   Where *network_ID* is the ID from the output in the previous step; in this case, the ID is 17.

4. In the **Advanced** tab of the **Adapter Properties** dialog box for your vendor and driver, change the value of `Jumbo Packet` to *9000*.
5. Click **OK**.
   The network connection may disconnect briefly during this phase.
6. Verify that the configuration is working, by typing the command:

```
ping -f -l 8972 <Destination_IP_Address>
```

   Output similar to the following should display:



NOTE: Ensure that the switch supports 10 GB Ethernet.

Configure jumbo frames for VMware ESXi

Configure jumbo frames for NIC cards in the VMware ESXi-based PowerFlex nodes.

Steps

Change the `Maximum Transmission Unit (mtu)` setting to `9,000` on the vSwitches and on the SVM (be sure to make the change in `/etc/sysconfig/network/ifcfg-ethX`):

a. Type the command:

```
esxcfg-vswitch -m 9000 <vSwitch>
```

b. Create a VMkernel with jumbo frames support by typing the following commands:

i.
```
esxcfg-vswitch -d
```

ii.
```
esxcfg-vswitch -A vmkernel# vSwitch#
```

iii.
```
esxcfg-vmknic -a -i <ip address> -n <netmask> -m 9000 <portgroup name>
```

NOTE: Changing jumbo frames on a vSwitch will not change the VMkernel MTU size. For older vCenter versions, check to ensure the MTU setting has been changed. If it was not successful, users may need to delete and re-create the VMkernel. For newer vCenter versions, modify the MTU of VMkernel using the vSphere web client.

Configure jumbo frame for storage virtual machine

Configure jumbo frames for NIC cards in the storage virtual machine (SVM)-based PowerFlex nodes.

Steps

1. Edit the file `/etc/sysconfig/network/ifcfg-<NIC_NAME>`.
2. Add the parameter `mtu=9000` to the file.

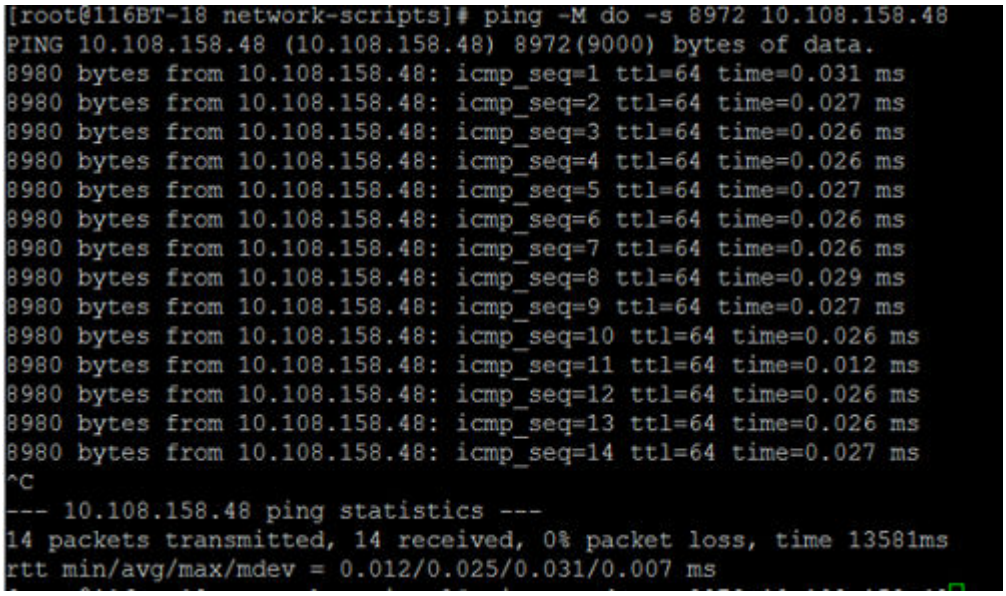
3. To apply the changes type:

```
service network restart
```

4. Run the `ifconfig` command again to confirm that the settings have been changed.
5. To test the command type:

```
ping -M do -s 8972 <DESTINATION_IP_ADDRESS>
```

Output should look similar to the following:



Increase SDS-only threads using MDM

The default number of SDS threads under a high-performance profile is eight. For best performance, Dell Technologies recommends increasing the number of SDS threads from eight to twelve when the CPUs have twelve or more physical cores.

Run the following command to view the current setting:

```
scli --query_performance_parameters --all_sds --print_all
```

The parameter `Active profile: high_performance:` shows the number of SDS threads currently being used:

- SDS_NUMBER_OS_THREADS
  - **Current value**: 8
  - **Profile value**: 8

The number of SDS threads is logged per SDS, and should be changed on all SDS nodes in the system.

Run the following command to change the SDS threads from eight to twelve:

```
scli --set_performance_parameters --all_sds --tech --sds_number_os_threads 12
```

NOTE: Increasing the SDS threads is expected to improve the performance in cases where the CPU is bound, for example: Small I/Os or fine granularity layout. In cases where the performance is limited by the drive performance (too few drives) or the network (large I/Os), changing the number of SDS threads does not help.

Maximize replication throughput over WAN on SDS and SDC using MDM

Use the following procedure to maximize replication throughput over WAN.

Run the following to maximize the use of WAN bandwidth by replication:

- On each node in the local or remote cluster, append the following lines to the file `/etc/sysctl.conf`:

```
net.core.rmem_max=100000000
net.core.wmem_max=100000000
net.ipv4.tcp_rmem=5000000 5000000 100000000
net.ipv4.tcp_wmem=10000000 10000000 100000000
```

  Run the command:

```
sysctl -p
```

- On the primary MDM of the local or remote clusters, run:

```
scli --set_performance_parameters --all_libsdc_remote --
sdc_number_sockets_per_sds_ip=9 -tech
```

A mandatory restart is required for all the components on each of the nodes.

Optimize Linux on SDS-only

When using the SSD devices, Dell Technologies recommends that the I/O scheduler of the devices be modified.

1. Change the disk scheduler in grub by appending "elevator=noop" to the variable `GRUB_CMDLINE_LINUX_DEFAULT`.

   For example, in `/etc/default/grub`, `GRUB_CMDLINE_LINUX_DEFAULT="quiet splash elevator=noop"`

2. Recompile the grub.cfg file

   For example, `grub2-mkconfig -o /boot/grub2/grub.cfg`

3. Reboot the Linux system.

When CPUs have twelve or more physical cores, and more performance is required from the node (for example: When CloudLink is used for software encryption, or when fine granularity is used and more performance per node is required), type the following:

```
scli --set_performance_parameters --all_sds --tech --sds_number_os_threads 12
```

NOTE: To make these changes persistent after reboot, either create a script that runs on boot, or change the kernel default scheduler using kernel command line.

Change the GRUB template for CPUs on storage nodes and SDC nodes

For CPU throughput on PowerFlex nodes running Red Hat Enterprise Linux 7.2 or later, change the GRUB template on every node to ensure that the CPUs maintain good latency performance.

Steps

1. Open the GRUB template for editing:

```
vim /etc/default/grub
```

2. Find the `GRUB_CMDLINE_LINUX` configuration option and append the following to the line:

```
intel_idle.max_cstate=1 intel_pstate=disable
```

For example:

```
GRUB_CMDLINE_LINUX="crashkernel=auto rd.lvm.lv=rhel/root
rd.lvm.lv=rhel/swap rhgb intel_idle.max_cstate=1
intel_pstate=disable quiet"
```

3. Compile the new GRUB:

```
grub2-mkconfig -o /boot/efi/EFI/redhat/grub.cfg
```

4. Stop and then disable `tuned`:

```
systemctl stop tuned
systemctl disable tuned
```

5. Reboot the node:

```
reboot
```

6. Ping the node to ensure that the GRUB change was implemented.
   The ping time should be in the 0.03 ms range.
7. Repeat the procedure on every relevant node.

Optimize VMware ESXi on storage nodes and SDC nodes

To improve I/O concurrency, you can increase the per device queue length value on a per data store basis. Per device queue length is seen as `No of outstanding IOs with competing worlds` in the display output.

Use the following command to increase the queue length:

```
esxcli storage core device set -d <DEVICE_ID> -O <Outstanding IOs>
```

Where *<Outstanding IOs>* can be a number ranging from 32–16384 (default=32). Dell Technologies recommends increasing queue length to 256.

Example:

```
esxcli storage core device set -d eui.16bb852c56d3b93e3888003b00000000 -O 256
```

Optimize Windows on storage nodes and SDC

The delayed acknowledgment feature causes high latencies for low I/O rates. Dell Technologies recommends that you disable the delayed ack feature on every network interface.

To change each SDC interface, excluding the management IP address, add the following registry entries:

- `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Tcpip\Parameters\Interfaces\<SAN interface GUID>`
- Entries: **`TcpAckFrequency TcpNoDelay`**
- Value type: **`REG_DWORD`**, number
- Value to disable: **`1`**

NOTE: Depending on the Windows version, the method for changing delayed ack varies. See your relevant vendor guidelines for specific details.

Ensure that the `customize power plan` is set to high performance, as shown in the following figure:

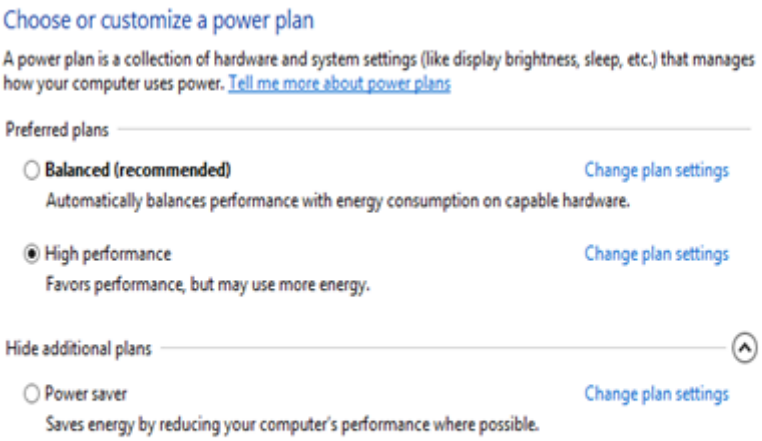


Figure 11. Customize power plan

Optimizing VM guests

Select the VM guest to optimize.

Modify the I/O scheduler

When using SSD devices, Dell Technologies recommends that you modify the devices' I/O scheduler.

1. Change the disk scheduler in grub by appending "elevator=noop" to the variable `GRUB_CMDLINE_LINUX_DEFAULT`.

   For example, in `/etc/default/grub`, `GRUB_CMDLINE_LINUX_DEFAULT="quiet splash elevator=noop"`

2. Recompile the grub.cfg file

   For example, `grub2-mkconfig -o /boot/grub2/grub.cfg`

3. Reboot the Linux system.

NOTE: For most Linux distributions, NOOP is not the default. Different Linux versions have different default values. For Red Hat Enterprise Linux 7 and SLES 11/12, the default value is deadline, but for older versions, the default is CFQ.

Configure the para virtual SCSI controller

Use the para virtual SCSI (PVSCSI) controller on guest VMs for high performance. It is important that you choose the correct PVSCSI controller, because choosing the wrong controller can adversely affect performance.

Current PVSCSI queue depth default values are 64 for the device and 254 for the adapter. Users can increase the PVSCSI queue depth to 256 for the device and 1024 for the adapter inside a Windows virtual machine.

Windows virtual machine:

1. From the command-line run:

```
REG ADD HKLM\SYSTEM\CurrentControlSet\services\pvscsi\Parameters\Device /v
DriverParameter /t REG_SZ /d "RequestRingPages=32,MaxQueueDepth=254"
```

2. Reboot the virtual machine.
3. Verify the successful creation of a registry entry:
   a. Open the Registry Editor by running the REGEDIT command from the command line.
   b. Browse to `HKLM\SYSTEM\CurrentControlSet\services\pvscsi\Parameters\Device`.
   c. Verify that the DriverParameter key exists with a value of RequestRingPages=32, MaxQueueDepth=254.

Linux guests:

1. Edit the `vmw_pvscsi.conf` file:

```
echo "options vmw_pvscsi cmd_per_lun=254 ring_pages=32" > /etc/modprobe.d/
vmw_pvscsi.conf
```

2. Rerun `vmware-config-tools.pl`:

```
vmware-config-tools.pl
```

You can add up to four PVSCSI controllers per guest. Allocating different PowerFlex volumes to different PVSCSI controllers can help realize the maximum potential of guest performance.

Dell Technologies strongly recommends that you review this Boardcom Knowledge Base article (article 2053145) so that you can make educated decisions regarding PVSCSI values.

Optimize Hyper-V

To improve Hyper-V performance and to minimize unnecessary resource consumption, review the following requirements:

- Disable NUMA spanning on Hyper-V hosts.
- Install SDC on the root partition.
- Dell Technologies recommends that you do not create virtual switches on data networks. Enable jumbo frames on virtual switches.

Optimize AIX

The following options optimize the AIX operating system:

- To turn off kernel tracker, run the following command:

```
bosedbug
```

- To reboot an AIX node, run the following command:

```
shutdown -r
```

- To review settings as viewed in the table, run the following command:

```
bosdebug
```

Table 44. AIX settings

AIX settings	Description
Memory debugger	off
Memory sizes	0
Network memory sizes	0
Kernel debugger	off
Real-time kernel	off
Backtracking fault log	off
Kernext memory tracking	off

Configure RAID controller virtual disk settings

Configure RAID controller virtual disk settings and optimal policies regarding each type of disk.

For SSDs in use by a RAID controller:

1. Create the virtual disk as a RAID 0 device.
2. Disable the PD cache for the hard drive.
3. Set the VD caching attributes as NRWTD (no read ahead, write through, direct).

NOTE: For VxFlex Ready Node-based installations, see the VxFlex Ready Node documentation on the Dell Technologies support site.

Apply performance profiles to system components

You can use PowerFlex Manager to apply performance profiles to system components.

About this task

The default setting configures a predefined set of parameters for high-performance use cases. When a container is provided in the command (system or protection domain or fault set), all the objects currently in the container are configured.

The profiles are applied separately to:

- SDSs
- SDCs
- SDRs

NOTE: After changing the performance profile of an SDS (on an SVM), you must perform manual memory allocation on the SVM, as described in this guide.

Steps

1. Depending on the system component that you want to configure, from the PowerFlex Manager menu bar, click the relevant menu options:
   - For SDS: **Block** > **SDSs**
   - For SDC: **Block** > **Hosts**
   - For SDR: **Protection** > **SDRs**
2. From the list of objects that is displayed, select the desired object, and then click **Modify** > **Modify Performance Profile**.
3. In the **Modify Performance Profile** dialog box, select a profile and click **Apply**.
4. Verify that the operation has finished and was successful, and then click **Dismiss**.

Configure SDC access to the MDM

It is possible to restrict SDC access to the MDM, pending approval of the SDC by the system. The default system setting is full access (restricted SDC mode is disabled).

When the restricted SDC mode is enabled, volumes can only be mapped to approved SDCs. Set a restricted SDC mode using the CLI. Approval is obtained by issuing the `--add_sdc` command for each SDC. You can set a restricted mode before or after SDCs have been added to your network. You can also approve SDCs using PowerFlex Manager.

The following CLI commands manage this feature:

Table 45. CLI commands

Action	Command
Enable or disable restricted SDC mode.	`set_restricted_sdc_mode command`
Add an SDC to the approved list, when restricted SDC mode is enabled.	`--add_sdc`

For more information, see the *Dell PowerFlex 4.6.x CLI Reference Guide*.

9

Optional deployment tasks

Deploying and configuring the PowerFlex management platform installer virtual machine

Add Entrust root certificate to the VMware vCenter certificate store

If you want to use third-party certificates in your environment, you must add a third-party trusted root certificate to the VMware vCenter certificate store.

Steps

1. Manually download the OVCS2 certificate (OVCS2-CSBR1-crosscert) from the Entrust website: Entrust Certificate Services Subordinate CAs.

   Search for the keyword **OVCS2**. Click **CRT** to download the certificate.
2. Upload the Entrust root certificate to the VMware vCenter certificate store:
   a. Log in to the vCenter Server.
   b. Click **Administration** > **Certificates** > **Certificate Management**.
   c. Next to **Trusted Root Store**, click **Add**.
   d. Browse and select the downloaded Entrust root certificate. The certificate is added in a panel under **Trusted Root Certificates**.
3. Once the certificates are installed, OVA can be verified and deployed.

Deploy the PowerFlex management platform installer VM

Use this procedure to deploy and configure the PowerFlex management platform installer VM in a VMware environment.

Prerequisites

- For a multi-VLAN or multi-subnet configuration, pfmc-vm-mgmt-<vlanid> (130) must be selected instead of the flex-node-mgmt-<vlanid> (105).
- For standard configurations without multi-networking, select either pfmc-vm-mgmt-<vlanid> (130) or flex-node-mgmt-<vlanid> (105).

Steps

1. Log in to the VMware vCSA.
2. Click **Menu** > **Shortcuts** > **Hosts and Clusters**.
3. Right-click the **ESXi Host** > **Select Deploy OVF Template**.
4. Select **Local File** > **Upload Files** > **Browse to the PowerFlex management platform OVA Template**
5. Click **Open** > **Next**.
6. Enter **pfmp-installer** for the VM name.
7. Click **Next**.
8. Verify that there are no compatibility warnings and click **Next**.
9. Click **Next**.

10. Select the **PFMP datastore**.
11. Select **Virtual disk format** > **Thin provision** > **Next**.
12. Select **flex-node-mgmt-<vlanid>** > **OK** > **Next**.

    NOTE:
    - For a multi-VLAN configuration, more than one port-group for pfmc-sds-mgmt-<vlanid> exists. Select the first port-group for VM deployment.
    - For a multi-VLAN or multi-subnet configuration, the management interface of the MVMs and installer VM must be configured on VLAN 130 instead of VLAN 105.

13. Click **Finish**.
14. Right-click the VM and select **Power** > **Power On**.

Configure the PowerFlex management platform installer networking interface

Use this procedure to configure the networking for the PowerFlex management platform installer VM.

About this task

The following network adapter needs to be configured on the Powerflex management platform installer:

VLAN 105: flex-node-mgmt (eth0) or VLAN 130: pfmc-mgmt (eth0)

For a multi-VLAN or multi-subnet configuration, the PowerFlex management cluster management is configured instead of the VMware ESXi management. For a standard deployment without multi-networking the network adapter can be configured on either VMware ESXi management (VLAN 105) or PowerFlex management cluster management (VLAN 130).

Steps

1. Launch the web console from vCSA and log in as **delladmin**.
2. To configure the <flex-node-mgmt> interface, type **sudo vi /etc/sysconfig/network/ifcfg-eth0**

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
```

Example:

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.105.10
NETMASK=255.255.255.0
```

a. To configure the default route, type:
   sudo vi /etc/sysconfig/network/routes
   default <gateway ip> - <interface>
   Example: **default 10.10.105.1 - eth0**
b. To configure DNS search and DNS servers, type **sudo vi /etc/sysconfig/network/config** and modify the following: **NETCONFIG_DNS_STATIC_SEARCHLIST="<search domain>" NETCONFIG_DNS_STATIC_SERVERS="< dns_ip1 dns_ip2>"**
   Example: **NETCONFIG_DNS_STATIC_SEARCHLIST="example.com" NETCONFIG_DNS_STATIC_SERVERS="10.10.105.240 10.10.105.241"**

Configure the NTP service

Configure the NTP services for the PowerFlex management platform installer virtual machine (VM) and the PowerFlex management platform cluster VMs.

Steps

1. To configure the NTP client, enter the following:
   server <ntp ip address> iburst
   For example: **server 10.10.105.240 iburst**
2. To start the chronyd service, enter:
   sudo systemctl start chronyd
3. To enable chronyd, enter:
   sudo systemctl enable chronyd
4. To restart chronyd, enter:
   sudo systemctl restart chronyd
5. To verify that the network time protocol service is running, enter:
   sudo systemctl status chronyd.service
6. To check that the server is synced to NTP, enter:
   chronyc tracking
   NOTE: The time may take a few minutes to sync. The following are examples of chronyd tracking status:

   Successful chronyc tracking status
   - A configured NTP server should be listed under `Reference ID`,
   - `Stratum` should be a value greater than zero.
   - `System Time` should report less than 1 s time difference:

```
PFMP2-450-747 :~ # chronyc tracking
Reference ID : 179628F2 (23.150.40.242)
Stratum : 3
Ref time (UTC) : Tue Mar 12 21:24:20 2024
System time : 0.000531569 seconds slow of NTP time
Last offset : -0.000336191 seconds
RMS offset :  0.000201320 seconds
Frequency : 6.809 ppm slow
Skew : 0.029 ppm
Root delay : 0.037072252 seconds
Root dispersion : 0.004111907 seconds
Update interval : 7242.7 seconds
Leap status : Normal
PFMP2-450-747 :~ #
```

   Unsuccessful chronyc tracking status
   - `Stratum` is 0 (not synced).
   - Leap status says `Normal`:

```
node1:~ # chronyc tracking
Reference ID : 00000000 ()
Stratum : 0
Ref time (UTC) : Thu Jan 01 00:00:00 1970
System time : 0.000000000 seconds fast of NTP time
Last offset : +0.000000000 seconds
RMS offset : 0.000000000 seconds
Frequency : 0.000 ppm slow
Residual freq : +0.000 ppm
Skew : 0.000 ppm
```

   The following image describes an unsuccessful chronyc tracking status.

```
Last login: Thu May  4 15:08:10 on tty1
pfmc-k8s-20230413-1:~ # chronyc tracking
Reference ID    : 00000000 ()
Stratum         : 0
Ref time (UTC)  : Thu Jan 01 00:00:00 1970
System time     : 0.000000013 seconds slow of NTP time
Last offset     : +0.000000000 seconds
RMS offset      : 0.000000000 seconds
Frequency       : 33.272 ppm slow
Residual freq   : +0.000 ppm
Skew            : 0.000 ppm
Root delay      : 1.000000000 seconds
Root dispersion : 1.000000000 seconds
Update interval : 0.0 seconds
Leap status     : Not synchronised
pfmc-k8s-20230413-1:~ #
```

7. Change the hostname, type the following: **`sudo hostnamectl set-hostname <mvm hostname>`**.

Deploying and configuring the PowerFlex management platform VMs

Use the following procedure to deploy the PowerFlex management platform cluster VMs.

Prerequisites

The PowerFlex management platform cluster requires three VMs to be deployed.

Steps

1. Log in to the VMware vCSA.
2. Click **Menu** > **Shortcuts** > **Hosts and Clusters**.
3. Right-click the **ESXi Host** > **Select Deploy OVF Template**.
4. Select **Local File** > **Upload Files** > **Browse to the PowerFlex management platform OVA**.
5. Click **Open** > **Next**.
6. Enter **`pfmp-mvm-<number>`** for the VM name.
7. Click **Next**.
8. Verify that there are no compatibility warnings and click **Next**.
9. Click **Next**.
10. Review details and click **Next**.
11. Select the datastore.
12. Select **Virtual disk format** > **Thin provision** > **Next**.
13. Select **Desired Destination Network** > **OK** > **Next**.
14. Click **Finish**.
15. Repeat all the steps for the three management VMs.

Configure the PowerFlex management platform networking interfaces

Use the following procedure to configure the networking files for the PowerFlex management platform cluster VMs.

Prerequisites

Ensure the following:

- In a multi-VLAN or multi-subnet configuration, the controller management must be configured to use PowerFlex management controller management (PFMC-MGMT) (VLAN 130).
- In standard builds without a multi-VLAN or multi-subnet configuration, the controller management can be either VMware ESXi management (VLAN 105) or PowerFlex management controller management (PFMC-MGMT) (VLAN 130).

NOTE: Based on your requirement, configure only the required interfaces on the management VMs. If you have unused interfaces on the deployed management VMs, then ensure you disconnect those interfaces within the vCenter, for example, an out-of-band interface. If the unused interfaces are connected and unconfigured, then PowerFlex Manager may display unexpected errors.

Steps

1. Launch the web console from the Virtual Machine Manager, log in as **delladmin**.
2. Configure the flex-node-mgmt-<vlanid> eth0 interface:
   a. Enter: **sudo vi /etc/sysconfig/network/ifcfg-eth0**

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
```

   For example:

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.105.30
NETMASK=255.255.255.0
```

   b. Configure the default route.
   Enter **sudo vi /etc/sysconfig/network/routes**
   default <gateway ip> - <interface>
   For example:
   default 10.10.105.1 - eth0
   c. Configure DNS search and DNS servers.
   Enter **sudo vi /etc/sysconfig/network/config** and modify the following:
   NETCONFIG_DNS_STATIC_SEARCHLIST="<search domain>"
   NETCONFIG_DNS_STATIC_SERVERS="< dns_ip1 dns_ip2>"
   For example:
   NETCONFIG_DNS_STATIC_SEARCHLIST="example.com"
   NETCONFIG_DNS_STATIC_SERVERS="10.10.105.240 10.10.105.241"
   d. To save and close, press **ESC** + **:wq**.
3. Configure the flex-data1-<vlanid> eth1 interface:
   a. Enter **sudo vi /etc/sysconfig/network/ifcfg-eth1**.

```
DEVICE=eth1
NAME=eth1
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
```

   For example:

```
DEVICE=eth1
NAME=eth1
STARTMODE=auto
BOOTPROTO=static
IPADDR=192.168.151.30
NETMASK=255.255.255.0
```

4. Configure the flex-data1-<vlanid> eth2 interface:

a. Enter: **sudo vi /etc/sysconfig/network/ifcfg-eth2**.

```
DEVICE=eth2
NAME=eth2
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
```

For example:

```
DEVICE=eth2
NAME=eth2
STARTMODE=auto
BOOTPROTO=static
IPADDR=192.168.152.30
NETMASK=255.255.255.0
```

5. Run the following command to update the network configuration:
   `sudo systemctl restart network`
6. Run the following command to change the delladmin account password:
   `passwd delladmin`
7. Change the root password with the following command:
   `sudo passwd root`
8. Repeat all steps for all PowerFlex management platform VMs.

Configure the NTP service

Configure the NTP services for the PowerFlex management platform installer virtual machine (VM) and the PowerFlex management platform cluster VMs.

Steps

1. To configure the NTP client, enter the following:
   server <ntp ip address> iburst
   For example: **server 10.10.105.240 iburst**
2. To start the chronyd service, enter:
   sudo systemctl start chronyd
3. To enable chronyd, enter:
   sudo systemctl enable chronyd
4. To restart chronyd, enter:
   sudo systemctl restart chronyd
5. To verify that the network time protocol service is running, enter:
   sudo systemctl status chronyd.service
6. To check that the server is synced to NTP, enter:
   chronyc tracking

   NOTE: The time may take a few minutes to sync. The following are examples of chronyd tracking status:

   Successful chronyc tracking status
   - A configured NTP server should be listed under `Reference ID`,
   - `Stratum` should be a value greater than zero.
   - `System Time` should report less than 1 s time difference:

```
PFMP2-450-747 :~ # chronyc tracking
Reference ID : 179628F2 (23.150.40.242)
Stratum : 3
Ref time (UTC) : Tue Mar 12 21:24:20 2024
System time : 0.000531569 seconds slow of NTP time
```

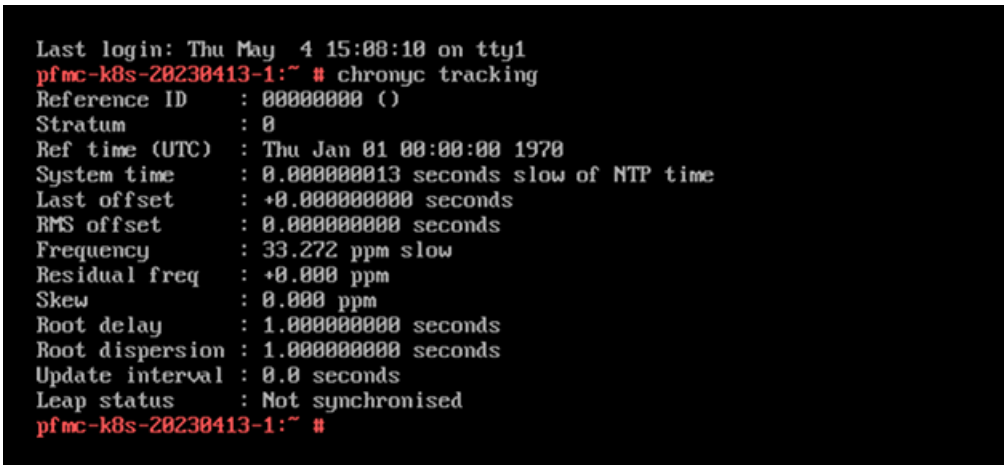
```
Last offset : -0.000336191 seconds
RMS offset :  0.000201320 seconds
Frequency : 6.809 ppm slow
Skew : 0.029 ppm
Root delay : 0.037072252 seconds
Root dispersion : 0.004111907 seconds
Update interval : 7242.7 seconds
Leap status : Normal
PFMP2-450-747 :~ #
```

Unsuccessful chronyc tracking status

- `Stratum` is 0 (not synced).
- Leap status says `Normal`:

```
node1:~ # chronyc tracking
Reference ID : 00000000 ()
Stratum : 0
Ref time (UTC) : Thu Jan 01 00:00:00 1970
System time : 0.000000000 seconds fast of NTP time
Last offset : +0.000000000 seconds
RMS offset : 0.000000000 seconds
Frequency : 0.000 ppm slow
Residual freq : +0.000 ppm
Skew : 0.000 ppm
```

The following image describes an unsuccessful chronyc tracking status.



7. Change the hostname, type the following: **`sudo hostnamectl set-hostname <mvm hostname>`**.

Deploy the PowerFlex management platform cluster

Use the following procedure to deploy the PowerFlex management platform.

Steps

1. Using SSH, connect as a non-root user (**`delladmin`**) to the PowerFlex management platform installer VM and run the following command:

   `ssh delladmin@<pfmp installer ip>`
2. Access the `config` directory:

   `cd /opt/dell/pfmp/PFMP_Installer/config`
3. To configure the `PFMP_Config.json`, run **`sudo vi PFMP_Config.json`** and update the configuration parameters.

   For example:

```
{
"Nodes":
[
    {
     "hostname": "pfmp-mvm-01",
```

```
     "ipaddress": "10.10.105.30"
    },
    {
     "hostname": "pfmp-mvm-02",
     "ipaddress": "10.10.105.31"
    },
    {
     "hostname": "pfmp-mvm-03",
     "ipaddress": "10.10.105.32"
    }
],
  "ClusterReservedIPPoolCIDR" : "10.42.0.0/23",
  "ServiceReservedIPPoolCIDR" : "10.43.0.0/23",
  "RoutableIPPoolCIDR" : [{"flex-node-mgmt-105":"10.10.105.20-10.10.105.24"},
                          {"flex-node-oob-101":"10.10.101.20-10.10.101.24"},
                          {"flex-data1-151":"192.168.151.20-192.168.151.24"},
                          {"flex-data2-152":"192.168.152.20-192.168.152.24"}],
  "PFMPHostname" : "dellpowerflex.com",
  "PFMPHostIP" : "10.10.105.20"
}
```

NOTE:

- **"RoutableIPPoolCIDR"** requires a pool range of at least five IP addresses. By default, the first IP address in the range becomes the PowerFlex Manager user interface address.
- PMFPHostIP has to be in the range of **"RoutableIPPoolCIDR" :{"flex-node-mgmt-105":"10.10.105.20-10.10.105.24"}**.
- The **"PFMPHostname" : "dellpowerflex.com"** and **"PFMPHostIP"** are user-defined and must have the appropriate DNS and PTR records created before installation.

Routed networks must be accessible through the interfaces on the hosts or VMs. If PowerFlex Manager requires access to non-routed networks, add the required interfaces to the host or VM and add an additional line to the RoutableIPPoolCIDR section for that interface.

4. Access the scripts folder in the working directory:
   cd /opt/dell/pfmp/PFMP_Installer/scripts
5. Set up the installer:
   sudo ./setup_installer.sh

   The **setup_installer.sh** starts a docker container with the appropriate network settings.
6. Install the PowerFlex management platform:
   sudo ./install_PFMP.sh

   The **install_PFMP.sh** uses the docker installation to deploy PowerFlex Manager management cluster across the three virtual machines or hosts.
7. When prompted, do the following:
   - Enter **No** to indicate that no SSH keys are used for authentication connection to the cluster nodes. If SSH keys are used, enter **Yes**.
   - Enter the non-default SSH username for the nodes specified in `PFMP_Config.json`.
   - Enter **Yes** to indicate that the passwords are the same for all cluster nodes.
   - Enter the non-default SSH password for the nodes specified in `PFMP_Config.json`.
   - Enter **No** to indicate that the nodes used for PowerFlex management platform are not integrated PowerFlex management (non-KVM) nodes.

   NOTE: If you want to deploy a PowerFlex cluster using a CSV topology file, ensure that you use the single SSH key for all nodes.

   View the progress of the installation in the following logs that are in the following directory:

   - `/opt/dell/pfmp/PFMP_Installer/logs/`
   - `/opt/dell/pfmp/atlantic/logs/`

NOTE: Installation may take more than two hours. When the deployment is complete, output similar to the following should appear:

```
UI can be accessed at: powerflex_manager.fqdn.com which needs to be resolved to
<PFMPHostIP>
```

- If the PowerFlex management platform deployment fails, collect the PowerFlex management platform support data and provide the support bundle files to Dell Technologies Support. For information about collecting PowerFlex management platform support data, see the *Dell PowerFlex and PowerFlex Manager Software Install Guide*.
- For more information about troubleshooting, see the *Dell PowerFlex and PowerFlex Manager Software Install Guide*.